

Introduction to Quantum Mechanics

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1 The Wave Function

Problem 1.1

(a)

$$N(14) = 1, N(15) = 1, N(16) = 2, N(22) = 2, N(24) = 2, N(25) = 5$$

$$\langle x \rangle = \sum xP(x) = 21 \implies \langle x \rangle^2 = 441 \quad \left(\because P(x) = \frac{N(x)}{N} \right)$$

$$\langle x^2 \rangle = \sum x^2 P(x) = 459.57$$

(b)

$$\Delta(14) = 7, \Delta(15) = 6, \Delta(16) = 5, \Delta(22) = 1, \Delta(24) = 2, \Delta(25) = 4$$

$$\sigma^2 = \langle (\Delta x)^2 \rangle = \sum (\Delta x)^2 P(x) = 18.57$$

(c)

$$\text{However, } \sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = 459.57 - 441$$

$$\sigma^2 = 18.57 \implies \text{Both results from (a) and (b) are valid}$$

Problem 1.2

(a)

$$\text{From the example, } \langle x \rangle = \frac{h}{3}, \rho(x) = \frac{1}{2\sqrt{hx}} (0 \leq x \leq h)$$

$$\langle x^2 \rangle = \int_0^h x^2 \rho(x) dx = \int_0^h \frac{x^{\frac{3}{2}}}{2\sqrt{h}} dx = \frac{1}{5} \sqrt{\frac{x^5}{h}} \Big|_0^h = \frac{h^2}{5}$$

$$\sigma = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{h^2}{5} - \frac{h^2}{9}} = \frac{2h}{2\sqrt{5}} \text{ metres}$$

(b) The probability that a photograph selected at random would give a value of x greater than $\langle x \rangle + \sigma$ would be,

$$P(x > \langle x \rangle + \sigma) = \int_{\langle x \rangle + \sigma}^h \rho(x) dx = \int_{\langle x \rangle + \sigma}^h \frac{1}{2\sqrt{hx}} dx$$

$$P(x > \langle x \rangle + \sigma) = \sqrt{\frac{x}{h}} \Big|_{\langle x \rangle + \sigma}^h = \sqrt{\frac{h - \frac{h}{3} - \frac{2h}{3\sqrt{5}}}{h}} = 0.607$$

Problem 1.3

(a)

$$\text{Given } \rho(x) = Ae^{-\lambda(x-a)^2}$$

$$\text{We know that } \int_{-\infty}^{+\infty} \rho(x) dx = 1 \implies \int_{-\infty}^{+\infty} Ae^{-\lambda(x-a)^2} dx = 1$$

$$\int_{-\infty}^{+\infty} Ae^{-\lambda(x-a)^2} dx = A\sqrt{\frac{\pi}{\lambda}} = 1 \implies A = \sqrt{\frac{\lambda}{\pi}}$$

(b)

$$\langle x \rangle = \int_{-\infty}^{+\infty} \sqrt{\frac{\pi}{\lambda}} x e^{-\lambda(x-a)^2} dx \xrightarrow{x \rightarrow x-a} \sqrt{\frac{\pi}{\lambda}} \int_{-\infty}^{+\infty} (u+a) e^{-\lambda u^2} du$$

$$\langle x \rangle = \sqrt{\frac{\pi}{\lambda}} \left(\int_{-\infty}^{+\infty} u e^{-\lambda u^2} du + a \int_{-\infty}^{+\infty} e^{-\lambda u^2} du \right) = \sqrt{\frac{\pi}{\lambda}} \left(0 + a\sqrt{\frac{\pi}{\lambda}} \right) = a$$

$$\therefore \langle x \rangle = a \implies \langle x \rangle^2 = a^2$$

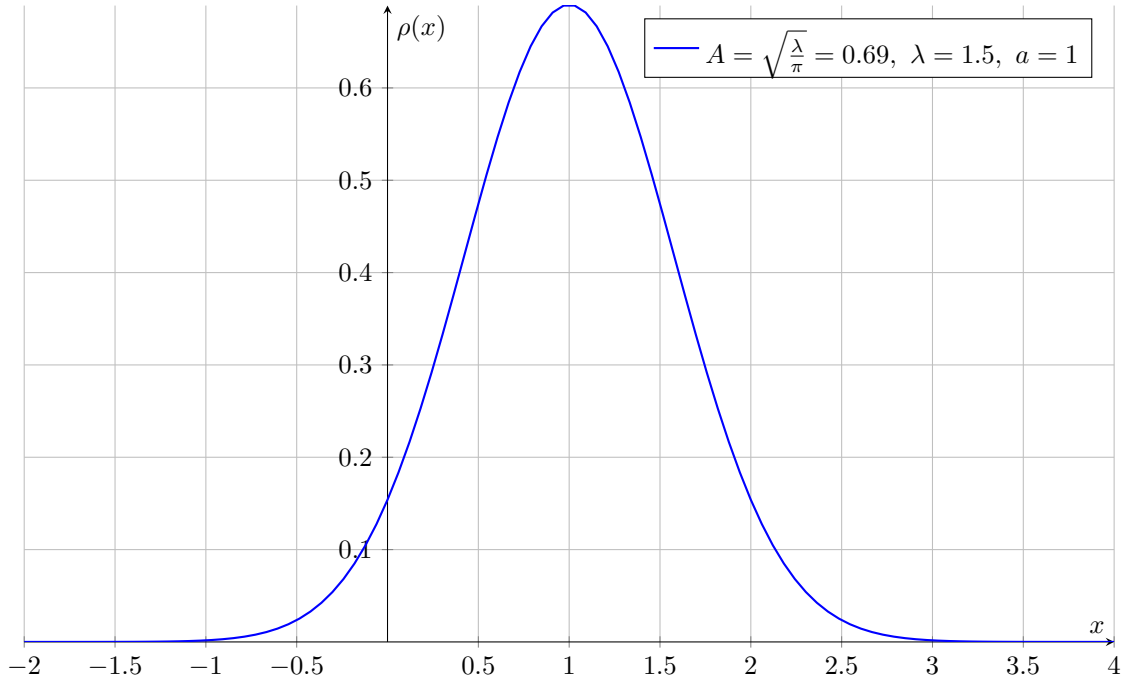
$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} \sqrt{\frac{\pi}{\lambda}} x^2 e^{-\lambda(x-a)^2} dx \xrightarrow{x \rightarrow x-a} \sqrt{\frac{\pi}{\lambda}} \int_{-\infty}^{+\infty} (x+a)^2 e^{-\lambda x^2} dx$$

$$\langle x^2 \rangle = \sqrt{\frac{\pi}{\lambda}} \left(\int_{-\infty}^{+\infty} x^2 e^{-\lambda x^2} dx + a \int_{-\infty}^{+\infty} x e^{-\lambda x^2} dx + a^2 \int_{-\infty}^{+\infty} e^{-\lambda x^2} dx \right) = \sqrt{\frac{\pi}{\lambda}} \left(a^2 \sqrt{\frac{\lambda}{\pi}} + 0 + \frac{1}{2\lambda} \sqrt{\frac{\lambda}{\pi}} \right)$$

$$\therefore \langle x^2 \rangle = a^2 + \frac{1}{2\lambda}$$

$$\therefore \sigma = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \frac{1}{\sqrt{2\lambda}}$$

(c)



Problem 1.4

Given the following wave function:

$$\Psi(x, 0) = \begin{cases} A \left(\frac{x}{a} \right) & , \quad 0 \leq x \leq a \\ A \left(\frac{b-x}{b-a} \right) & , \quad a \leq x \leq b \\ 0 & , \quad x < 0, x > b \end{cases}$$

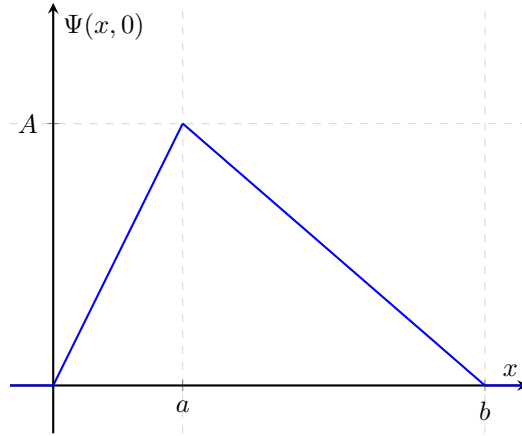
(a) Normalizing the wave function,

$$\int_{-\infty}^{+\infty} |\Psi|^2 dx = \int_0^a A^2 \left(\frac{x}{a} \right)^2 dx + \int_a^b A^2 \left(\frac{b-x}{b-a} \right)^2 dx = 1$$

$$A^2 \left(\frac{x^3}{3a^2} \Big|_0^a - \frac{(b-x)^3}{3(b-a)^2} \Big|_a^b \right) = A^2 \left(\frac{a}{3} - \frac{b-a}{3} \right) = \frac{A^2 b}{3} = 1$$

$$\therefore A = \sqrt{\frac{3}{b}}$$

(b)



(c) From the above curve, we can tell that the maximum probability to find the particle is at $x = a$.

(d) The probability of finding the particle to the left of a would be $P(x < a)$

$$P(x < a) = \int_0^a |\Psi|^2 dx = \int_0^a \frac{3x^2}{a^2 b} dx$$

$$P(x < a) = \frac{x^3}{a^2 b} \Big|_0^a = \frac{a}{b}$$

$$\text{If } b = a \implies P(x < a) = 1 \quad ; \quad b = 2a \implies P(x < a) = \frac{1}{2}$$

(e) The expectation value of the x is given by:

$$\langle x \rangle = \int_{-\infty}^{+\infty} x |\Psi|^2 dx = \int_0^a \frac{3x^3}{a^2 b} dx + \int_a^b \frac{3x(b-x)^2}{b(b-a)^2} dx$$

$$\langle x \rangle = \frac{3x^4}{4a^2 b} \Big|_0^a - \frac{3}{b(b-a)^2} \left(\frac{1}{3}(b-x)^3 x + \frac{1}{12}(b-x)^4 \right) \Big|_a^b$$

$$\langle x \rangle = \frac{3a^2}{4b} + \frac{3}{b(b-a)^2} \frac{(b-a)^3}{3} \left(\frac{3a+b}{4} \right) = \frac{3a^2}{4b} + \frac{b^2 - 3a^2 + 2ab}{4b} = \frac{b+2a}{4}$$

$$\therefore \langle x \rangle = \frac{b+2a}{4}$$

Problem 1.5

The given wave function is

$$\Psi(x, t) = Ae^{-\lambda|x|}e^{-i\omega t}$$

(a) Normalizing the wave function,

$$\int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx = \int_{-\infty}^{+\infty} A^2 e^{-2\lambda|x|} dx = A^2 \left(\int_{-\infty}^0 e^{2\lambda x} dx + \int_0^{+\infty} e^{-2\lambda x} dx \right) = 1$$

$$A^2 \left(\left. \frac{e^{2\lambda x}}{2\lambda} \right|_{-\infty}^0 - \left. \frac{e^{-2\lambda x}}{2\lambda} \right|_0^{+\infty} \right) = A^2 \left(\frac{1}{2\lambda} + \frac{1}{2\lambda} \right) = \frac{A^2}{\lambda} = 1$$

$$\therefore A = \sqrt{\lambda}$$

(b) The expectation value of x and x^2 are,

$$\langle x \rangle = \int_{-\infty}^{+\infty} x |\Psi(x, t)|^2 dx = \lambda \left(\int_{-\infty}^0 x e^{2\lambda x} dx + \int_0^{+\infty} x e^{-2\lambda x} dx \right)$$

$$\langle x \rangle = \lambda \left(\left(x \frac{e^{2\lambda x}}{2\lambda} - \frac{e^{2\lambda x}}{4\lambda^2} \right) \Big|_{-\infty}^0 + \left(-x \frac{e^{-2\lambda x}}{2\lambda} + \frac{e^{-2\lambda x}}{4\lambda^2} \right) \Big|_0^{+\infty} \right) = 0$$

$$\therefore \langle x \rangle = 0$$

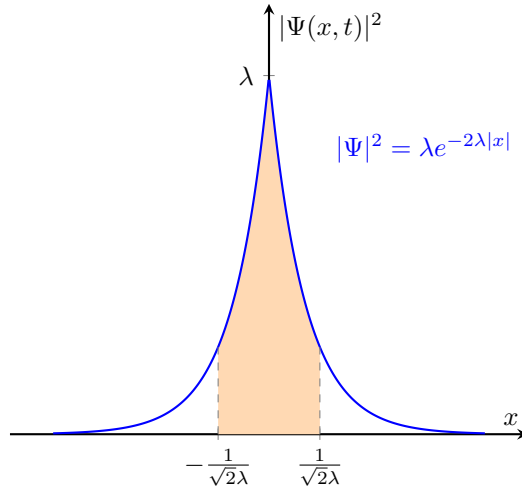
$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} x^2 |\Psi(x, t)|^2 dx = \lambda \left(\int_{-\infty}^0 x^2 e^{2\lambda x} dx + \int_0^{+\infty} x^2 e^{-2\lambda x} dx \right)$$

$$\langle x^2 \rangle = \lambda \left(\left(x^2 \frac{e^{2\lambda x}}{2\lambda} + 2x \frac{e^{2\lambda x}}{4\lambda^2} + 2 \frac{e^{2\lambda x}}{8\lambda^3} \right) \Big|_{-\infty}^0 + \left(-x^2 \frac{e^{-2\lambda x}}{2\lambda} + 2x \frac{e^{-2\lambda x}}{4\lambda^2} - 2 \frac{e^{-2\lambda x}}{8\lambda^3} \right) \Big|_0^{+\infty} \right) = \frac{1}{2\lambda^2}$$

$$\therefore \langle x^2 \rangle = \frac{1}{2\lambda^2}$$

(c) The standard deviation of x is,

$$\sigma = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{1}{2\lambda^2}} = \frac{1}{\sqrt{2}\lambda}$$



The probability of finding the particle outside the interval $I = [\langle x \rangle - \sigma, \langle x \rangle + \sigma]$ is

$$P(x \notin I) = \int_{-\frac{1}{\sqrt{2\lambda}}}^{+\frac{1}{2\lambda}} |\Psi(x, t)|^2 dx = \int_{-\frac{1}{\sqrt{2\lambda}}}^0 \lambda e^{2\lambda x} dx + \int_0^{+\frac{1}{\sqrt{2\lambda}}} \lambda e^{-2\lambda x} dx$$

$$P(x \notin I) = \frac{e^{2\lambda x}}{2} \Big|_{-\frac{1}{\sqrt{2\lambda}}}^0 - \frac{e^{-2\lambda x}}{2} \Big|_0^{+\frac{1}{\sqrt{2\lambda}}}$$

$$P(x \notin I) = e^{-\sqrt{2}} \approx 0.243$$

Problem 1.6

Consider the equation from the question.

$$\frac{d\langle x \rangle}{dt} = \int_{-\infty}^{+\infty} x \frac{\partial}{\partial t} |\Psi|^2 dx$$

The integral can't be done by parts because the derivative and integral are taken with respect to different variables, t and x . The Schrödinger Equation and its complex-conjugated equation are used to eliminate the time derivatives in favor of x so that integration-by-parts can be applied.

Problem 1.7

$$\frac{d\langle p \rangle}{dt} = \frac{d}{dt} \int_{-\infty}^{+\infty} \Psi^* [-i\hbar(\partial/\partial x)] \Psi dx = -i\hbar \int_{-\infty}^{+\infty} \frac{\partial}{\partial t} \left(\Psi^* \frac{\partial \Psi}{\partial x} \right) dx$$

$$\frac{d\langle p \rangle}{dt} = -i\hbar \int_{-\infty}^{+\infty} \frac{\partial \Psi^*}{\partial t} \frac{\partial \Psi}{\partial x} + \Psi^* \frac{\partial}{\partial t} \frac{\partial \Psi}{\partial x} dx = -i\hbar \int_{-\infty}^{+\infty} \frac{\partial \Psi^*}{\partial t} \frac{\partial \Psi}{\partial x} + \Psi^* \frac{\partial}{\partial x} \frac{\partial \Psi}{\partial t} dx$$

Substituting $\frac{\partial \Psi}{\partial t}$ from Schrödinger Equation, we obtain

$$\frac{d\langle p \rangle}{dt} = -i\hbar \int_{-\infty}^{+\infty} \left(\frac{-i\hbar}{2m} \frac{\partial^2 \Psi^*}{\partial x^2} + \frac{i}{\hbar} V \Psi^* \right) \frac{\partial \Psi}{\partial x} + \Psi^* \frac{\partial}{\partial x} \left(\frac{i\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} - \frac{i}{\hbar} V \Psi \right) dx$$

$$\frac{d\langle p \rangle}{dt} = \frac{\hbar^2}{4m^2} \int_{-\infty}^{+\infty} \left(\frac{\partial^2 \Psi^*}{\partial x} \frac{\partial \Psi}{\partial x} - \frac{2m}{\hbar^2} V \Psi^* \frac{\partial \Psi}{\partial x} \right) + \left(-\Psi^* \frac{\partial^3 \Psi}{\partial x^3} + \Psi^* \frac{2m}{\hbar^2} \frac{\partial V}{\partial x} \Psi + \Psi^* \frac{2m}{\hbar^2} V \frac{\partial \Psi}{\partial x} \right) dx$$

$$\frac{d\langle p \rangle}{dt} = \frac{\hbar^2}{4m^2} \int_{-\infty}^{+\infty} \frac{\partial^2 \Psi^*}{\partial x} \frac{\partial \Psi}{\partial x} - \Psi^* \frac{\partial^3 \Psi}{\partial x^3} + \Psi^* \frac{2m}{\hbar^2} \frac{\partial V}{\partial x} \Psi dx$$

$$\frac{d\langle p \rangle}{dt} = \frac{\hbar^2}{4m^2} \int_{-\infty}^{+\infty} 0 + 0 - \Psi^* \frac{2m}{\hbar^2} \frac{\partial V}{\partial x} \Psi dx = \int_{-\infty}^{+\infty} \Psi^* \frac{\partial V}{\partial x} \Psi dx$$

$$\therefore \frac{d\langle p \rangle}{dt} = \left\langle -\frac{\partial V}{\partial x} \right\rangle$$

Problem 1.9

The given wave function is,

$$\Psi(x, t) = Ae^{-a(\frac{m}{\hbar}x^2 + it)} = Ae^{-\frac{ma}{\hbar}x^2}e^{-iat}$$

(a) Normalizing the wave function,

$$\int_{-\infty}^{+\infty} \Psi^* \Psi dx = \int_{-\infty}^{+\infty} A^2 e^{-\frac{2ma}{\hbar}x^2} dx = 1$$

$$A^2 \sqrt{\frac{\pi \hbar}{2ma}} = 1 \implies A = \left(\frac{2ma}{\pi \hbar} \right)^{\frac{1}{4}}$$

(b) To find the potential $V(x)$ for which $\Psi(x, t)$ is a solution, input the wave function into the Schrödinger Equation

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi \implies V = \frac{\hbar}{\Psi} \left(i \frac{\partial \Psi}{\partial t} + \frac{\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} \right)$$

$$V = \frac{\hbar}{Ae^{-\frac{ma}{\hbar}x^2}e^{-iat}} \left(i \left(-iaAe^{-\frac{ma}{\hbar}x^2}e^{-iat} \right) + \frac{\hbar}{2m} \left(\frac{4m^2a^2}{\hbar^2}x^2 Ae^{-\frac{ma}{\hbar}x^2} - \frac{2ma}{\hbar}Ae^{-\frac{ma}{\hbar}x^2} \right) e^{-iat} \right)$$

$$V = \hbar \left(\frac{a}{\hbar} + \frac{2ma^2}{\hbar}x^2 - \frac{a}{\hbar} \right)$$

$$\boxed{\therefore V(x) = 2ma^2x^2}$$

(c)

$$\langle x \rangle = \int_{-\infty}^{+\infty} \Psi^*[x]\Psi dx = \int_{-\infty}^{+\infty} \sqrt{\frac{2ma}{\pi \hbar}} x e^{-\frac{2ma}{\hbar}x^2} dx$$

$$\boxed{\therefore \langle x \rangle = 0 \quad (\because \text{Odd Integrand})}$$

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} \Psi^*[x^2]\Psi dx = \sqrt{\frac{2ma}{\pi \hbar}} \int_{-\infty}^{+\infty} x^2 e^{-\frac{2ma}{\hbar}x^2} dx$$

$$\langle x^2 \rangle = \sqrt{\frac{2ma}{\pi \hbar}} \left(\frac{\hbar}{2ma} \frac{1}{2} \sqrt{\frac{\pi \hbar}{2ma}} \right)$$

$$\boxed{\therefore \langle x^2 \rangle = \frac{\hbar}{4ma}}$$

$$\langle p \rangle = \int_{-\infty}^{+\infty} \Psi^*[-i\hbar \partial / \partial x]\Psi dx = -i\hbar \int_{-\infty}^{+\infty} \Psi^* \frac{\partial \Psi}{\partial x} dx$$

$$\langle p \rangle = -i\hbar \sqrt{\frac{2ma}{\pi \hbar}} \int_{-\infty}^{+\infty} e^{-\frac{ma}{\hbar}x^2} e^{iat} \left(-\frac{2ma}{\hbar} x e^{-\frac{ma}{\hbar}x^2} e^{-iat} \right) dx$$

$$\langle p \rangle = 2ima \sqrt{\frac{2ma}{\pi \hbar}} \int_{-\infty}^{+\infty} x e^{-\frac{2ma}{\hbar}x^2} dx$$

$$\boxed{\therefore \langle p \rangle = 0 \quad (\because \text{Odd Integrand})}$$

$$\langle p^2 \rangle = \int_{-\infty}^{+\infty} \Psi^*[-\hbar^2 \partial^2 / \partial x^2]\Psi dx = -\hbar^2 \int_{-\infty}^{+\infty} \Psi^* \frac{\partial^2 \Psi}{\partial x^2} dx$$

$$\langle p^2 \rangle = -\hbar^2 \sqrt{\frac{2ma}{\pi \hbar}} \int_{-\infty}^{+\infty} e^{-\frac{ma}{\hbar}x^2} \frac{\partial}{\partial x} \left(-\frac{2ma}{\hbar} x e^{-\frac{ma}{\hbar}x^2} \right) dx$$

$$\langle p^2 \rangle = 2ma\hbar \sqrt{\frac{2ma}{\pi\hbar}} \int_{-\infty}^{+\infty} -\frac{2ma}{\hbar} x^2 e^{-\frac{2ma}{\hbar} x^2} + e^{-\frac{2ma}{\hbar} x^2} dx$$

$$\langle p^2 \rangle = 2ma\hbar \sqrt{\frac{2ma}{\pi\hbar}} \left(-\frac{1}{2} \frac{2ma}{\hbar} \frac{\hbar}{2ma} \sqrt{\frac{\pi\hbar}{2ma}} + \sqrt{\frac{\pi\hbar}{2ma}} \right)$$

$$\boxed{\therefore \langle p^2 \rangle = ma\hbar}$$

(d)

$$\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \frac{1}{2} \sqrt{\frac{\hbar}{ma}}$$

$$\sigma_p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{ma\hbar}$$

$$\sigma_x \sigma_p = \frac{\hbar}{2} \implies \text{Uncertainty principle verified}$$

Problem 1.11

The particle given has a mass of m and energy E , in a potential well $V(x)$, sliding between a and b . If it takes time T to go from $x = a$ to $x = b$, then the probability that the particle is found in an interval dx is

$$\rho(x) dx = \frac{dt}{T} \implies \rho(x) = \frac{1}{\left(\frac{dx}{dt}\right) T} = \frac{1}{v(x)T}$$

$$T = \int_0^T dt = \int_a^b \frac{1}{v(x)} dx \quad \left(\because v(x) = \frac{dx}{dt} \implies dt = \frac{1}{v(x)} dx \right)$$

(a) Using the conservation of mechanical energy,

$$E = T + V = \frac{1}{2} m(v(x))^2 + V(x)$$

$$\implies v(x) = \sqrt{\frac{2}{m}(E - V(x))}$$

(b) In the case of simple harmonic oscillator, $V(x) = \frac{1}{2} kx^2$ and $E = \frac{1}{2} kA^2$, where A is the amplitude of oscillation.

$$v(x) = \sqrt{\frac{2}{m} \left(\frac{1}{2} kA^2 - \frac{1}{2} kx^2 \right)} \implies v(x) = \sqrt{\frac{k}{m} (A^2 - x^2)}$$

$$T = \int_{-A}^{+A} \frac{1}{v(x)} dx = \int_{-A}^{+A} \sqrt{\frac{m}{k}} \frac{1}{\sqrt{A^2 - x^2}} dx$$

Applying the following substitution,

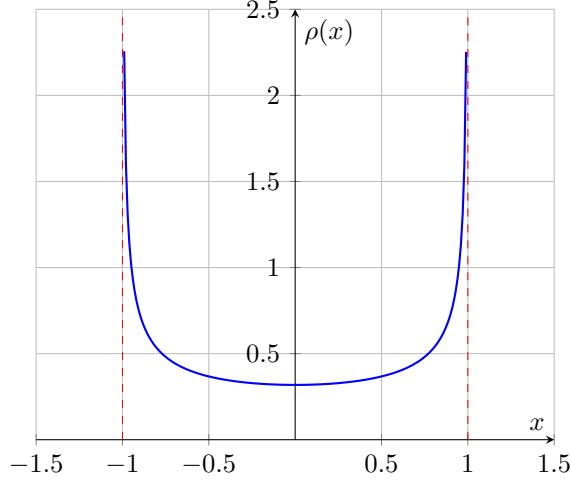
$$x \rightarrow \frac{x}{A} \implies dx \rightarrow \frac{dx}{A}, \quad -A \rightarrow -1 \text{ and } A \rightarrow 1$$

$$T = \int_{-1}^{+1} \sqrt{\frac{m}{k}} \frac{1}{\sqrt{1 - x^2}} dx = \sqrt{\frac{m}{k}} \sin^{-1}(x) \Big|_{-1}^{+1} = \pi \sqrt{\frac{m}{k}}$$

$$\rho(x) = \frac{1}{v(x)T} = \frac{1}{\sqrt{\frac{k}{m} (A^2 - x^2)} \pi \sqrt{\frac{m}{k}}}$$

$$\boxed{\rho(x) = \frac{1}{\pi \sqrt{A^2 - x^2}}}$$

Plot of $\rho(x) = \frac{1}{\pi\sqrt{A^2-x^2}}$ with $A = 1$



(c)

$$\langle x \rangle = \int_{-A}^{+A} x \rho(x) dx = \int_{-A}^{+A} \frac{x}{\pi\sqrt{A^2-x^2}} dx = \int_{-A}^{+A} \frac{1}{\pi A} \frac{x}{\sqrt{1-\left(\frac{x}{A}\right)^2}} dx$$

Since the integrand is an odd function, it evaluates to zero

$$\boxed{\therefore \langle x \rangle = 0}$$

$$\langle x^2 \rangle = \int_{-A}^{+A} x^2 \rho(x) dx = \int_{-A}^{+A} \frac{1}{\pi A} \frac{x^2}{\sqrt{1-\left(\frac{x}{A}\right)^2}} dx$$

Applying the following substitution,

$$\frac{x}{A} \rightarrow x \implies dx \rightarrow A dx, \quad -A \rightarrow -1 \text{ and } +A \rightarrow +1$$

$$\langle x^2 \rangle = \int_{-1}^{+1} \frac{A^2}{\pi} \frac{x^2}{\sqrt{1-x^2}} dx$$

Applying the trigonometric substitution $x = \sin \theta$,

$$dx = \cos \theta d\theta, \quad -1 \rightarrow -\frac{\pi}{2} \text{ and } +1 \rightarrow \frac{\pi}{2}$$

$$\langle x^2 \rangle = \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} \frac{A^2}{\pi} \frac{\sin^2 \theta}{\sqrt{1-\sin^2 \theta}} \cos \theta d\theta = \frac{A^2}{\pi} \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} \sin^2 \theta d\theta$$

We know that $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$, therefore

$$\langle x^2 \rangle = \frac{A^2}{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 - \cos 2\theta d\theta = \frac{A^2}{2\pi} \left(\theta - \frac{1}{2} \sin 2\theta \right) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$\boxed{\therefore \langle x^2 \rangle = \frac{A^2}{2}}$$

$$\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

$$\boxed{\therefore \sigma_x = \frac{A}{\sqrt{2}}}$$

Problem 1.12

In the case of the classical harmonic oscillator,

$$V(x) = \frac{1}{2}kx^2, \quad E = \frac{1}{2}kA^2$$

(a) By the conservation of mechanical energy,

$$\frac{1}{2}kA^2 = \frac{1}{2}kx^2 + \frac{p^2}{2m}$$

$$p = \pm \sqrt{mk(A^2 - x^2)} \iff x = \pm \sqrt{A^2 - \frac{p^2}{mk}}$$

From Problem 1.11, we know that $\rho(x) = \frac{1}{\pi\sqrt{A^2 - x^2}}$

$$\rho(p) = \frac{1}{\pi\sqrt{A^2 - A^2 + \frac{p^2}{mk}}} = \frac{\sqrt{mk}}{\pi p}$$

$$\boxed{\therefore \rho(p) = \frac{\sqrt{mk}}{\pi p}}$$

(b)

$$\langle p \rangle = \int_{-A}^A p \rho(x) dx = \int_{-\sqrt{2mE}}^{\sqrt{2mE}} p \left(\frac{\sqrt{mk}}{\pi p} \right) d \left(\pm \sqrt{A^2 - \frac{p^2}{mk}} \right)$$

$$\langle p \rangle = \int_{-\sqrt{2mE}}^{\sqrt{2mE}} \frac{\sqrt{mk}}{\pi} \frac{\mp 2p}{2\sqrt{A^2 - \frac{p^2}{mk}}} dp$$

$$\boxed{\therefore \langle p \rangle = 0}$$

$$\langle p^2 \rangle = \int_{-A}^A p^2 \rho(x) dx = \int_{-\sqrt{2mE}}^{\sqrt{2mE}} \frac{\sqrt{mk}}{\pi} \frac{\frac{p^2}{mk}}{\sqrt{A^2 - \frac{p^2}{mk}}} dp$$

Substituting $p = A\sqrt{mk} \sin \theta \implies dp = A\sqrt{mk} \cos \theta d\theta$

$$-\sqrt{2mE} = -A\sqrt{mk} \rightarrow -\frac{\pi}{2} \text{ and } \sqrt{2mE} = A\sqrt{mk} \rightarrow \frac{\pi}{2}$$

$$\langle p^2 \rangle = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sqrt{mk}}{\pi} \frac{\frac{A^2 mk \sin^2 \theta}{mk}}{A \cos \theta} A\sqrt{mk} \cos \theta d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{A^2 mk}{\pi} \sin^2 \theta d\theta$$

$$\langle p^2 \rangle = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{A^2 mk}{2\pi} (1 - \cos 2\theta) d\theta = \frac{A^2 mk}{2\pi} \left(\theta - \frac{1}{2} \sin 2\theta \right) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$\boxed{\therefore \langle p^2 \rangle = \frac{1}{2}mkA^2 = mE}$$

(c)

$$\sigma_p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2}$$

$$\boxed{\therefore \sigma_p = \sqrt{mE}}$$

From Problem 1.12, $\sigma_x = \frac{A}{\sqrt{2}}$,

$$\sigma_x \sigma_p = \sqrt{\frac{mk}{2}} A \frac{A}{\sqrt{2}} = \frac{A\sqrt{mk}}{2} = A\sqrt{\frac{mE}{2}}$$

For simple harmonic oscillator, $E \geq \frac{\hbar\omega}{2}$,

$$\boxed{\therefore \sigma_x \sigma_p \geq \frac{A}{2} \sqrt{m\hbar\omega}}$$

Problem 1.14

(a) Let P_{ab} be the probability of finding a particle between $x = a$ and $x = b$. Consider $\frac{dP_{ab}}{dt}$.

$$\frac{dP_{ab}}{dt} = \frac{d}{dt} \int_a^b \Psi^* \Psi dx = \int_a^b \frac{\partial}{\partial t} (\Psi^* \Psi) dx$$

$$\frac{dP_{ab}}{dt} = \int_a^b \Psi^* \frac{\partial \Psi}{\partial t} + \Psi \frac{\partial \Psi^*}{\partial t} dx$$

We know from the Schrödinger Equation that,

$$\frac{\partial \Psi}{\partial t} = \frac{i\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} - \frac{i}{\hbar} V \Psi$$

$$\frac{dP_{ab}}{dt} = \int_a^b \Psi^* \left(\frac{i\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} - \frac{i}{\hbar} V \Psi \right) + \Psi \left(-\frac{i\hbar}{2m} \frac{\partial^2 \Psi^*}{\partial x^2} + \frac{i}{\hbar} V \Psi^* \right) dx$$

$$\frac{dP_{ab}}{dt} = \frac{i\hbar}{2m} \int_a^b \Psi^* \frac{\partial^2 \Psi}{\partial x^2} - \Psi \frac{\partial^2 \Psi^*}{\partial x^2} dx = \frac{i\hbar}{2m} \int_a^b \left(\frac{\partial \Psi^*}{\partial x} \frac{\partial \Psi}{\partial x} + \Psi^* \frac{\partial^2 \Psi}{\partial x^2} \right) - \left(\frac{\partial \Psi}{\partial x} \frac{\partial \Psi^*}{\partial x} + \Psi \frac{\partial^2 \Psi^*}{\partial x^2} \right) dx$$

$$\frac{dP_{ab}}{dt} = \frac{i\hbar}{2m} \int_a^b \frac{\partial}{\partial x} \left(\Psi^* \frac{\partial \Psi}{\partial x} \right) - \frac{\partial}{\partial x} \left(\Psi \frac{\partial \Psi^*}{\partial x} \right) dx$$

$$\frac{dP_{ab}}{dt} = \frac{i\hbar}{2m} \int_a^b \frac{\partial}{\partial x} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) dx = \frac{i\hbar}{2m} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) \Big|_a^b$$

Let $J(x, t) = \frac{i\hbar}{2m} \left(\Psi^*(x, t) \frac{\partial \Psi}{\partial x}(x, t) - \Psi(x, t) \frac{\partial \Psi^*}{\partial x}(x, t) \right)$, then

$$\boxed{\therefore \frac{dP_{ab}}{dt} = J(b, t) - J(a, t)}$$

(b) Given wave function from Problem 1.9 is

$$\Psi(x, t) = A e^{-\alpha((mx^2/\hbar)+it)}$$

The probability current for Ψ here between $x = a$ and $x = b$ is,

$$J(x, t) = \frac{i\hbar}{2m} \left(A e^{-\alpha((mx^2/\hbar)+it)} \left(-\frac{2m\alpha}{\hbar} x A e^{-\alpha((mx^2/\hbar)+it)} \right) - A e^{-\alpha((mx^2/\hbar)+it)} \left(-\frac{2m\alpha}{\hbar} x A e^{-\alpha((mx^2/\hbar)+it)} \right) \right)$$

$$J(x, t) = A^2 \left(-\frac{2m\alpha}{\hbar} \right) \frac{i\hbar}{2m} (x e^{-\alpha(2mx^2/\hbar)} - x e^{-\alpha(2mx^2/\hbar)})$$

$$J(x, t) = 0 \implies \frac{dP_{ab}}{dt} = 0$$

Problem 1.15

Let Ψ_1 and Ψ_2 be two normalizable solutions to the Schrödinger Equation. Then consider the following

$$\frac{d}{dt} \int_{-\infty}^{+\infty} \Psi_1^* \Psi_2 dx = \int_{-\infty}^{+\infty} \frac{\partial}{\partial t} (\Psi_1^* \Psi_2) dx = \int_{-\infty}^{+\infty} \Psi_1^* \frac{\partial \Psi_2}{\partial t} + \Psi_2 \frac{\partial \Psi_1^*}{\partial t} dx$$

From the Schrödinger Equation, we know that,

$$\begin{aligned} \frac{\partial \Psi}{\partial t} &= \frac{i\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} - \frac{i}{\hbar} V \Psi \\ \Rightarrow \int_{-\infty}^{+\infty} \Psi_1^* \left(\frac{i\hbar}{2m} \frac{\partial^2 \Psi_2}{\partial x^2} - \frac{i}{\hbar} V \Psi_2 \right) + \Psi_2 \left(-\frac{i\hbar}{2m} \frac{\partial^2 \Psi_1^*}{\partial x^2} + \frac{i}{\hbar} V \Psi_1^* \right) dx \\ &\Rightarrow \int_{-\infty}^{+\infty} \frac{i\hbar}{2m} \left(\Psi_1^* \frac{\partial^2 \Psi_2}{\partial x^2} - \Psi_2 \frac{\partial^2 \Psi_1^*}{\partial x^2} \right) dx \\ &\Rightarrow \frac{i\hbar}{2m} \int_{-\infty}^{+\infty} \left(\frac{\partial \Psi_1^*}{\partial x} \frac{\partial \Psi_2}{\partial x} + \Psi_1^* \frac{\partial^2 \Psi_2}{\partial x^2} \right) - \left(\frac{\partial \Psi_1^*}{\partial x} \frac{\partial \Psi_2}{\partial x} + \Psi_2 \frac{\partial^2 \Psi_1^*}{\partial x^2} \right) dx \\ &\Rightarrow \frac{i\hbar}{2m} \int_{-\infty}^{+\infty} \frac{\partial}{\partial x} \left(\Psi_1^* \frac{\partial \Psi_2}{\partial x} - \Psi_2 \frac{\partial \Psi_1^*}{\partial x} \right) dx = \frac{i\hbar}{2m} \left(\Psi_1^* \frac{\partial \Psi_2}{\partial x} - \Psi_2 \frac{\partial \Psi_1^*}{\partial x} \right) \Big|_{-\infty}^{+\infty} \end{aligned}$$

We know that for $\int_{-\infty}^{+\infty} |\Psi|^2 dx = 1$ to be true, $\Psi \rightarrow 0$ as $x \rightarrow \infty$ or $-\infty$

$$\boxed{\therefore \frac{d}{dt} \int_{-\infty}^{+\infty} \Psi_1^* \Psi_2 dx = 0}$$

Problem 1.16

A particle (at $t = 0$) is represented by the following wave function

$$\Psi(x, 0) = \begin{cases} A(a^2 - x^2), & |x| < a \\ 0, & |x| \geq a \end{cases}$$

(a) Normalizing the wave function,

$$\begin{aligned} \int_{-\infty}^{+\infty} |\Psi|^2 dx &= 1 \\ A^2 \int_{-a}^a (a^2 - x^2)^2 dx &= A^2 \int_{-a}^a a^4 + x^4 - 2a^2 x^2 dx \\ A^2 \left(a^4 x + \frac{x^5}{5} - \frac{2}{3} a^2 x^3 \right) \Big|_{-a}^a &= 1 \\ A^2 \left(\left(a^5 + \frac{a^5}{5} - \frac{2}{3} a^5 \right) - \left(-a^5 - \frac{a^5}{5} + \frac{2}{3} a^5 \right) \right) &= 1 \Rightarrow A^2 \left(\frac{16}{15} a^5 \right) = 1 \end{aligned}$$

$$\boxed{\therefore A = \frac{1}{4} \sqrt{\frac{15}{a^5}}}$$

(b) The expectation value of the position of the particle is,

$$\begin{aligned}\langle x \rangle &= \int_{-\infty}^{+\infty} \Psi^*[x] \Psi dx = A^2 \int_{-a}^a x(a^2 - x^2)^2 dx \\ \langle x \rangle &= A^2 \int_{-a}^a a^4 x + x^5 - 2a^2 x^3 dx = A^2 \left(\frac{a^4}{2} x^2 + \frac{x^6}{6} - \frac{1}{2} a^2 x^4 \right) \Big|_{-a}^a \\ &\quad \boxed{\therefore \langle x \rangle = 0}\end{aligned}$$

(c) The expectation value of the momentum of the particle is,

$$\begin{aligned}\langle p \rangle &= \int_{-\infty}^{+\infty} \Psi^*[-i\hbar \partial/\partial x] \Psi dx = -i\hbar \int_{-\infty}^{+\infty} \Psi^* \frac{\partial \Psi}{\partial x} dx \\ \langle p \rangle &= -i\hbar \int_{-a}^a (A(a^2 - x^2))(A(-2x)) dx = -i\hbar A^2 \int_{-a}^a 2x^3 - 2a^2 x dx \\ \langle p \rangle &= -i\hbar A^2 \left(\frac{1}{2} x^4 - a^2 x^2 \right) \Big|_{-a}^a \\ &\quad \boxed{\therefore \langle p \rangle = 0}\end{aligned}$$

We cannot calculate $\langle p \rangle$ by $m \frac{d\langle x \rangle}{dt}$ since the given wave function is $\Psi(x, 0)$, the time is constant. Since the time is constant, we will not get an accurate value of $\langle p \rangle$, since it will always be 0.

(d) Expectation value of x^2 is,

$$\begin{aligned}\langle x^2 \rangle &= \int_{-\infty}^{+\infty} \Psi^*[x^2] \Psi dx = \int_{-a}^a A^2 x^2 (a^2 - x^2)^2 dx \\ \langle x^2 \rangle &= A^2 \int_{-a}^a a^4 x^2 + x^6 - 2a^2 x^4 dx = A^2 \left(\frac{a^4 x^3}{3} + \frac{x^7}{7} - \frac{2}{5} a^2 x^5 \right) \Big|_{-a}^a \\ \langle x^2 \rangle &= \frac{15}{16a^5} \left(2 \left(\frac{a^7}{3} + \frac{a^7}{7} - \frac{2}{5} a^7 \right) \right) \\ &\quad \boxed{\therefore \langle x^2 \rangle = \frac{a^2}{7}}\end{aligned}$$

(e) Expectation value of p^2 is,

$$\begin{aligned}\langle p^2 \rangle &= \int_{-\infty}^{+\infty} \Psi^*[-\hbar^2 \partial^2/\partial x^2] \Psi dx = -\hbar^2 \int_{-a}^a A(a^2 - x^2)(-2Aa) dx \\ \langle p^2 \rangle &= -2a\hbar^2 A^2 \int_{-a}^a a^2 - x^2 dx = -2a\hbar^2 A^2 \left(a^2 x - \frac{x^3}{3} \right) \Big|_{-a}^a \\ &\quad \boxed{\therefore \langle p^2 \rangle = \frac{5\hbar^2}{2a^2}}\end{aligned}$$

(f) The uncertainty in position is,

$$\begin{aligned}\sigma_x &= \sqrt{\langle x^2 \rangle - \langle x \rangle^2} \\ &\quad \boxed{\therefore \sigma_x = \frac{a}{\sqrt{7}}}\end{aligned}$$

(g) The uncertainty in momentum is,

$$\sigma_p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2}$$

$$\boxed{\therefore \sigma_p = \sqrt{\frac{5}{2}} \frac{\hbar}{a}}$$

(h) Consider the product of uncertainties,

$$\sigma_x \sigma_p = \sqrt{\frac{5}{14}} \hbar > \frac{\hbar}{2}$$

The uncertainties are consistent with the uncertainty principle.

Problem 1.17

Given that the unstable particle has a lifetime of τ , the equation of probability is

$$P(t) = \int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx = e^{-t/\tau}$$

Let us consider that $\Psi(x, t)$ is a solution to the Schrödinger Equation for a potential V , given below

$$V = V_0 - i\Gamma$$

Therefore, it follows that

$$\frac{\partial \Psi}{\partial t} = \frac{i\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} + \frac{1}{\hbar} (-iV_0 + \Gamma) \Psi$$

(a) Consider the given probability function

$$P(t) = \int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx = e^{-t/\tau}$$

Now, consider its derivative with time,

$$\frac{dP(t)}{dt} = \frac{d}{dt} \int_{-\infty}^{+\infty} \Psi^* \Psi dx = \int_{-\infty}^{+\infty} \frac{\partial}{\partial t} (\Psi^* \Psi) dx$$

$$\frac{dP}{dt} = \int_{-\infty}^{+\infty} \Psi^* \frac{\partial \Psi}{\partial t} + \Psi \frac{\partial \Psi^*}{\partial x} dx$$

From the above equations, substitute in $\frac{\partial \Psi}{\partial t}$ and $\frac{\partial \Psi^*}{\partial x}$

$$\frac{dP}{dt} = \int_{-\infty}^{+\infty} \Psi^* \left(\frac{i\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} + \frac{1}{\hbar} (-iV_0 + \Gamma) \Psi \right) + \Psi \left(-\frac{i\hbar}{2m} \frac{\partial^2 \Psi^*}{\partial x^2} + \frac{1}{\hbar} (iV_0 + \Gamma) \Psi^* \right) dx$$

$$\frac{dP}{dt} = \int_{-\infty}^{+\infty} \frac{i\hbar}{2m} \left(\Psi^* \frac{\partial^2 \Psi}{\partial x^2} - \Psi \frac{\partial^2 \Psi^*}{\partial x^2} \right) + \frac{\Psi^* \Psi}{\hbar} (2\Gamma) dx$$

$$\frac{dP}{dt} = \int_{-\infty}^{+\infty} \frac{i\hbar}{2m} \frac{\partial}{\partial x} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) + \frac{2\Gamma}{\hbar} |\Psi|^2 dx$$

$$\frac{dP}{dt} = \frac{i\hbar}{2m} \frac{\partial}{\partial x} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) \Big|_{-\infty}^{+\infty} + \frac{2\Gamma}{\hbar} \underbrace{\int_{-\infty}^{+\infty} |\Psi|^2 dx}_{P(t)}$$

$$\boxed{\therefore \frac{dP}{dt} = \frac{2\Gamma}{\hbar} P}$$

(b) We know that $P(t) = e^{-t/\tau}$, putting it in our result

$$\begin{aligned}\frac{d}{dt}(e^{-t/\tau}) &= \frac{2\Gamma}{\hbar}e^{-t/\tau} \\ -\frac{1}{\tau}e^{-t/\tau} &= \frac{2\Gamma}{\hbar}e^{-t/\tau}\end{aligned}$$

$$\boxed{\therefore \tau = -\frac{\hbar}{2\Gamma}, P(t) = e^{2\Gamma t/\hbar}}$$

Problem 1.18

The de Broglie wavelength of a particle at temperature T is given by,

$$\lambda = \frac{\hbar}{\sqrt{3mk_B T}}$$

A system is said to be quantum mechanical if its De Broglie wavelength (λ) is greater than the characteristic size of the system (d). In the given problem, the characteristic size of the system would be $d = 0.3$ nm. For the system to be quantum mechanical,

$$\begin{aligned}\lambda > d &\implies \frac{\hbar}{\sqrt{3mk_B T}} > d \\ &\implies T < \frac{\hbar^2}{3mk_B d^2}\end{aligned}$$

(a) For our given system (electron), $\hbar = 1.054 \times 10^{-34}$ Js, $m = 9.1 \times 10^{-31}$ kg, $k_B = 1.38 \times 10^{-23}$ J K⁻¹, $d = 3 \times 10^{-10}$ m

$$\boxed{\therefore T_{e^-} < 1.29 \times 10^5 \text{ K}}$$

In case of silicon $\hbar = 1.054 \times 10^{-34}$ Js, $m = 46.48 \times 10^{-27}$ kg, $k_B = 1.38 \times 10^{-23}$ J K⁻¹, $d = 3 \times 10^{-10}$ m

$$\boxed{\therefore T_{Si} < 2.51 \text{ K}}$$

(b) Consider the ideal gas equation,

$$PV = Nk_B T \implies \frac{V}{N} = \frac{k_B T}{P}$$

Here, $\frac{V}{N}$ is the volume of a singular molecular cube of gas.

$$\frac{V}{N} = d^3 \implies d = \left(\frac{k_B T}{P}\right)^{1/3}$$

Now, putting the above interatomic distance into the condition for a quantum mechanical system,

$$\begin{aligned}T < \frac{\hbar^2}{3mk_B d^2} &\implies T < \frac{\hbar}{3mk_B \left(\frac{k_B T}{P}\right)^{2/3}} \\ T &< \frac{\hbar^{3/5} P^{1/5}}{3^{3/5} m^{3/5} k_B}\end{aligned}$$

For He at atmospheric pressure,

$$\boxed{\therefore T_{He} < 2.92 \text{ K}}$$

For H with $d = 1$ cm,

$$\boxed{\therefore T_H < 6.29 \times 10^{-14}}$$

Therefore, H can be treated as quantum mechanical at $T = 4$ K.

2 Time Independent Schrödinger Equation

Problem 2.1

(a) **Statement:** For normalizable solutions, the separation constant E must be real.

Proof: Let E be a complex number $E = E_0 + i\Gamma$. Then the wave function would become $\Psi(x, t) = \psi(x)e^{(-iE_0 + \Gamma)t/\hbar}$. Given that the solution is normalizable,

$$\int_{-\infty}^{+\infty} |\Psi|^2 dx = 1 \implies \int_{-\infty}^{+\infty} \psi^* e^{(iE_0 + \Gamma)t/\hbar} \psi e^{(-iE_0 + \Gamma)t/\hbar} dx = 1$$

$$\int_{-\infty}^{+\infty} |\psi|^2 e^{2\Gamma t/\hbar} dx = 1 \implies e^{2\Gamma t/\hbar} \int_{-\infty}^{+\infty} |\psi|^2 dx = 1$$

For the above equation to hold true for all of time, $\Gamma = 0$
Therefore, $E = E_0$, and the separation constant is real. QED

□

(b) **Statement:** The time-independent wave function $\psi(x)$ can always be taken to be real.

Proof: Given that $\psi(x)$ solves the Time Independent Schrödinger Equation for some $V(x)$ and E , then $\psi + \psi^*$ and $i(\psi - \psi^*)$ should also solve the equation (by linearity). Since the above two are real solutions to the equation, Let them equal to ψ_R and ψ_I . From this, every wave function with identical energy can be written as a linear combination of real wave functions ψ_R and ψ_I . So, taking the wave function be real always does not hinder us in any way, since we can always write them as linear combinations. QED

□

(c) **Statement:** If $V(x)$ is an even function, then every $\psi(x)$ associated with it is either even or odd.

Proof: From the Time Independent Schrödinger Equation,

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + V(x)\psi(x) = E\psi(x)$$

Given that our potential function is even, $V(-x) = V(x)$, lets perform a transformation $x \rightarrow -x$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(-x)}{\partial x^2} + V(x)\psi(-x) = E\psi(-x)$$

Now, let's subtract the two equations,

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} (\psi(x) - \psi(-x)) + V(x)(\psi(x) - \psi(-x)) = E(\psi(x) - \psi(-x))$$

For the left and right hand sides to be always equal, $\psi(-x) = -\psi(x)$ or $\psi(-x) = \psi(x)$. Our implicit assumption that $\psi(-x)$ is a solution to the Time Independent Schrödinger Equation would only be true if these conditions were followed. Therefore, if the potential $V(x)$ is an even function, the time independent wave function $\psi(x)$ would have to be either even or odd. QED.

□

Problem 2.2

Consider the Time Independent Schrödinger Equation,

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = E$$

For $E > V_{min}$, $E - V_{min} > 0$ for all ψ .

$$E - V = -\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{\partial^2 \psi}{\partial x^2} > 0$$

This means that ψ and $\frac{\partial^2 \psi}{\partial x^2}$ must always have the opposite sign. Consider the inverse of this statement, that ψ and $\frac{\partial^2 \psi}{\partial x^2}$ have the same sign. Then $E - V < 0$. Let us try to normalize such a wave function.

$$\int_{-\infty}^{+\infty} \psi^* \psi dx = 1$$

If $\psi > 0$, then $\frac{\partial^2 \psi}{\partial x^2} < 0$, this means that ψ is concave and always positive, so it shoots up to ∞ . If $\psi < 0$, then $\frac{\partial^2 \psi}{\partial x^2} > 0$. This means ψ is convex and always negative, so it plummets to $-\infty$. Either way, $\int_{-\infty}^{+\infty} |\psi|^2 dx$ would diverge. Hence it is not normalizable. Therefore, ψ and $\frac{\partial^2 \psi}{\partial x^2}$ must have the same sign to be normalizable and as a consequence, $E - V > 0$. QED.

□

Problem 2.3

Let the energy of the system E be zero. The Time Independent Schrödinger Equation then becomes,

$$\hat{H}\psi = 0 \implies \frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = V\psi$$

Consider the case of an infinite square well.

$$V(x) = \begin{cases} 0, & 0 < x < a \\ \infty, & \text{otherwise} \end{cases}$$
$$\implies \frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = 0$$

The solution to the above equation would be $\psi(x) = Ax + B$ Applying the boundary conditions $\psi(0) = \psi(a) = 0$

$$\begin{aligned} \psi(0) = B = 0 \quad \psi(a) = Aa + B = 0 \\ \implies \psi(x) = 0 \end{aligned}$$

This is nonsense, since if $\psi(x) = 0$, then there is no particle. So, a state where $E = 0$ is impossible.

Problem 2.4

The wave function for the infinite square well is,

$$\psi(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$$

Expectation value of position is:

$$\langle x \rangle = \int_{-\infty}^{+\infty} \psi^* x \psi dx = \int_0^a \frac{2}{a} x \sin^2 \left(\frac{n\pi}{a} x \right) dx$$

Performing the substitution $\theta = \frac{n\pi}{a} x$

$$dx = \frac{a}{n\pi} d\theta, \quad 0 \rightarrow 0, \quad a \rightarrow n\pi$$

$$\langle x \rangle = \int_0^{n\pi} \frac{2a}{n^2\pi^2} \theta \sin^2 \theta d\theta$$

$$\langle x \rangle = \frac{a}{n^2\pi^2} \left(\frac{\theta^2}{2} - \frac{\theta}{2} \cos 2\theta + \frac{1}{4} \sin 2\theta \right) \Big|_0^{n\pi}$$

$$\boxed{\therefore \langle x \rangle = \frac{a}{2} \left(1 - \frac{1}{n\pi} \right)}$$

Expectation value of square of position is:

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} \psi^* x^2 \psi dx = \int_0^a \frac{2}{a} x^2 \sin^2 \left(\frac{n\pi}{a} x \right) dx$$

Performing the substitution $\theta = \frac{n\pi}{a} x$

$$dx = \frac{a}{n\pi} d\theta, \quad 0 \rightarrow 0, \quad a \rightarrow n\pi$$

$$\langle x^2 \rangle = \int_0^{n\pi} \frac{2a^2}{n^3\pi^3} \theta^2 \sin^2 \theta d\theta$$

$$\langle x^2 \rangle = \frac{a^2}{n^3\pi^3} \left(\frac{\theta^3}{3} + \left(\frac{1}{4} - \frac{\theta^2}{2} \right) \cos 2\theta + \frac{\theta}{2} \sin 2\theta \right) \Big|_0^{n\pi}$$

$$\langle x^2 \rangle = \frac{a^2}{n^3\pi^3} \left(\frac{n^3\pi^3}{3} + \left(\frac{1}{4} - \frac{n^2\pi^2}{2} \right) \cos 2n\pi + \frac{n\pi}{2} \sin 2n\pi - \frac{1}{4} \right)$$

$$\boxed{\therefore \langle x^2 \rangle = a^2 \left(\frac{1}{3} - \frac{1}{2n\pi} \right)}$$

Expectation value of momentum is:

$$\langle p \rangle = \int_{-\infty}^{+\infty} \psi^* \left[-i\hbar \frac{\partial}{\partial x} \right] \psi dx = \int_0^a \frac{-2in\pi\hbar^2}{a} \sin \left(\frac{n\pi}{a} x \right) \cos \left(\frac{n\pi}{a} x \right) dx$$

$$\langle p \rangle = -\frac{in\pi\hbar^2}{a} \int_0^a \sin \left(\frac{2n\pi}{a} x \right) dx = -\frac{in\pi\hbar^2}{a} \frac{a}{2n\pi} \cos \left(\frac{2n\pi}{a} x \right) \Big|_0^a$$

$$\langle p \rangle = 0$$

Expectation value of momentum squared is:

$$\langle p^2 \rangle = \int_{-\infty}^{+\infty} \psi^* \left[-\hbar^2 \frac{\partial^2}{\partial x^2} \right] \psi dx = \frac{2n^2\hbar^2\pi^2}{a^3} \int_0^a \sin^2 \left(\frac{n\pi}{a} x \right) dx$$

$$\langle p^2 \rangle = \frac{n\pi\hbar^2}{a^2} \left(\frac{n\pi}{a} x - \frac{1}{2} \cos \left(\frac{2n\pi}{a} x \right) \right) \Big|_0^a = \frac{n\pi\hbar^2}{a^2} (n\pi)$$

$$\langle p^2 \rangle = \frac{n^2 \pi^2 \hbar^2}{a^2}$$

Uncertainties in position and momentum are

$$\begin{aligned}\sigma_x &= \sqrt{\langle x^2 \rangle - \langle x \rangle^2} & \sigma_p &= \sqrt{\langle p^2 \rangle - \langle p \rangle^2} \\ \sigma_x &= a \sqrt{\frac{1}{12} - \frac{1}{4n^2 \pi^2}} & \sigma_p &= \frac{n \pi \hbar}{a}\end{aligned}$$

From the uncertainty principle,

$$\begin{aligned}n \pi \hbar \sqrt{\frac{1}{12} - \frac{1}{4n^2 \pi^2}} &= \hbar \sqrt{\frac{n^2 \pi^2}{12} - \frac{1}{4}} \geq \frac{\hbar}{2} \\ n &\geq \frac{\sqrt{6}}{\pi}\end{aligned}$$

$n = 1$ state comes closest to the uncertainty principle limit.

Problem 2.5

Given that the particle in an infinite square well has the following initial wave function:

$$\Psi(x, 0) = A(\Psi_1(x) + \Psi_2(x))$$

(a) We know that $\sum_{n=1}^{\infty} |c_n|^2 = 1$, from that it follows:

$$\begin{aligned}A^2 + A^2 &= 1 \implies A = \pm \frac{1}{\sqrt{2}} \\ \therefore \Psi(x, 0) &= \frac{1}{\sqrt{2}}(\Psi_1 + \Psi_2)\end{aligned}$$

(b) We know that $\Psi_n(x, t) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right) e^{-iE_n t/\hbar}$

$$\Psi(x, t) = \frac{1}{\sqrt{a}} \left(\sin\left(\frac{\pi}{a}x\right) e^{-iE_1 t/\hbar} + \sin\left(\frac{2\pi}{a}x\right) e^{-4iE_1 t/\hbar} \right)$$

$$\text{where } E_1 = \frac{\hbar^2 \pi^2}{2ma^2}$$

$$|\Psi(x, t)|^2 = \frac{1}{a} \left(\sin\left(\frac{\pi}{a}x\right) e^{-iE_1 t/\hbar} + \sin\left(\frac{2\pi}{a}x\right) e^{-4iE_1 t/\hbar} \right) \left(\sin\left(\frac{\pi}{a}x\right) e^{iE_1 t/\hbar} + \sin\left(\frac{2\pi}{a}x\right) e^{4iE_1 t/\hbar} \right)$$

$$|\Psi|^2 = \frac{1}{a} \left(\sin^2\left(\frac{\pi}{a}x\right) + \sin^2\left(\frac{2\pi}{a}x\right) + \sin\left(\frac{\pi}{a}x\right) \sin\left(\frac{2\pi}{a}x\right) (e^{3iE_1 t/\hbar} + e^{-3iE_1 t/\hbar}) \right)$$

$$|\Psi|^2 = \frac{1}{a} \left(\sin^2\left(\frac{\pi}{a}x\right) + \sin^2\left(\frac{2\pi}{a}x\right) + 2 \sin\left(\frac{\pi}{a}x\right) \sin\left(\frac{2\pi}{a}x\right) \cos\left(\frac{3\hbar \pi^2 t}{2ma^2}\right) \right)$$

Let $\omega \equiv \frac{\hbar \pi^2}{2ma^2}$, then

$$\therefore |\Psi|^2 = \frac{1}{2} \psi_1^2 + \frac{1}{2} \psi_2^2 + \psi_1 \psi_2 \cos(3\omega t)$$

(c)

$$\langle x \rangle = \int_{-\infty}^{+\infty} \Psi^*[x] \Psi dx = \int_0^a \left(\frac{1}{2} \psi_1^2 + \frac{1}{2} \psi_2^2 + \psi_1 \psi_2 \cos(3\omega t) \right) x dx$$

$$\langle x \rangle = \frac{1}{2} \langle x \rangle_1 + \frac{1}{2} \langle x \rangle_2 + \frac{1}{a} \cos(3\omega t) \int_0^a x \sin\left(\frac{\pi}{a}x\right) \sin\left(\frac{2\pi}{a}x\right) dx$$

From the previous problem, we got that $\langle x \rangle_n = a \left(\frac{1}{3} - \frac{1}{2n\pi} \right)$

$$\langle x \rangle = a \left(\frac{1}{3} - \frac{1}{2\pi} \right) + a \left(\frac{1}{3} - \frac{1}{4\pi} \right) - \frac{4a}{9\pi} \cos(3\omega t)$$

$$\therefore \langle x \rangle = a \left(\frac{2}{3} - \frac{3}{4\pi} - \frac{4}{9\pi} \cos(3\omega t) \right)$$

We see that $\langle x \rangle$ is oscillatory with an amplitude of $a \left(\frac{2}{3} - \frac{3}{4\pi} \right)$ and angular frequency of 3ω .

(d)

$$\langle p \rangle = m \frac{d\langle x \rangle}{dt}$$

$$\therefore \langle p \rangle = \frac{4m\omega}{3\pi} \sin(3\omega t)$$

(e) If the energy of the particle is measured, you will either get E_1 or $E_2 \equiv 4E_1$. The expectation value of the Hamiltonian would be given by:

$$\langle \hat{H} \rangle = \sum_{n=1}^{\infty} |c_n|^2 E_n$$

$$\therefore \langle \hat{H} \rangle = \frac{5}{2} E_1 = \frac{5\hbar\pi^2}{4ma}$$

Problem 2.6

Problem 2.7

Given initial wave function of the particle is:

$$\Psi(x, 0) = \begin{cases} Ax, & 0 \leq x \leq a/2 \\ A(a-x), & a/2 \leq x \leq a \end{cases}$$

(a) Normalizing the wave function,

$$\int_{-\infty}^{+\infty} |\Psi(x, 0)|^2 dx = \int_0^{\frac{a}{2}} A^2 x^2 dx + \int_{\frac{a}{2}}^a A^2 (a-x)^2 dx = 1$$

$$\frac{A^2 x^3}{3} \Big|_0^{\frac{a}{2}} - \frac{A^2 (a-x)^3}{3} \Big|_{\frac{a}{2}}^a = 1$$

$$\therefore A = \frac{2\sqrt{3}}{a^{\frac{3}{2}}}$$

(b) We know that:

$$c_n = \int_{-\infty}^{+\infty} \psi_n^* \Psi(x, 0) dx$$

$$c_n = \int_0^{\frac{a}{2}} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right) Ax dx + \int_{\frac{a}{2}}^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right) A(a-x) dx$$

$$c_n = A\sqrt{\frac{2}{a}} \left[\left(\frac{a^2}{n^2\pi^2} \sin\left(\frac{n\pi}{a}x\right) - \frac{ax}{n\pi} \cos\left(\frac{n\pi}{a}x\right) \right) \Big|_0^{\frac{a}{2}} + \left(\frac{a^2}{n^2\pi^2} \sin\left(\frac{n\pi}{a}x\right) - \frac{(a-x)a}{n\pi} \cos\left(\frac{n\pi}{a}x\right) \right) \Big|_{\frac{a}{2}}^a \right]$$

$$c_n = \frac{2\sqrt{2}Aa^{\frac{3}{2}}}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) = \frac{4\sqrt{6}}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right)$$

$$\therefore \Psi(x, t) = \sum_{n=1}^{\infty} \frac{4\sqrt{6}}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$$

(c) Probability of measuring the state E_1 is $P(E_1) = |c_1|^2 = \frac{96}{\pi^4} = 98.5\%$

(d)

$$\begin{aligned} \langle \hat{H} \rangle &= \sum_{n=1}^{\infty} |c_n|^2 E_n = \sum_{n=1}^{\infty} \frac{96}{n^4\pi^4} \frac{n^2\hbar^2\pi^2}{2ma} \sin^2\left(\frac{n\pi}{2}\right) \\ \langle \hat{H} \rangle &= \frac{48\hbar^2}{ma\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin^2\left(\frac{n\pi}{2}\right) = \frac{48\hbar^2}{ma\pi^2} \frac{\pi^2}{8} \\ \therefore \langle \hat{H} \rangle &= \frac{6\hbar^2}{ma} \end{aligned}$$

Problem 2.8

Given initial wave function of the particle is:

$$\Psi(x, 0) = \begin{cases} A, & 0 \leq x \leq a/2 \\ 0, & a/2 \leq x \leq a \end{cases}$$

Normalizing, we get:

$$\begin{aligned} \int_{-\infty}^{+\infty} |\Psi(x, 0)|^2 dx &= \int_0^a A^2 dx = 1 \implies A^2 x \Big|_0^{\frac{a}{2}} = 1 \\ \therefore A &= \sqrt{\frac{2}{a}} \end{aligned}$$

We know that:

$$\begin{aligned} c_n &= \int_{-\infty}^{+\infty} \psi_n^* \Psi(x, 0) dx = \int_0^{\frac{a}{2}} \frac{2}{a} \sin\left(\frac{n\pi}{a}x\right) dx + \int_{\frac{a}{2}}^a 0 dx \\ c_n &= -\frac{2}{n\pi} \cos\left(\frac{n\pi}{a}x\right) \Big|_0^{\frac{a}{2}} \\ \therefore c_n &= \frac{2}{n\pi} \end{aligned}$$

Therefore,

$$\therefore \Psi(x, t) = \sum_{n=1}^{\infty} \frac{2\sqrt{2}}{n\pi a} \sin\left(\frac{n\pi}{a}x\right)$$

Probability of state with $n = 1$ is:

$$P(E_1) = |c_1|^2 = \frac{4}{\pi^2} = 40.53\%$$

Problem 2.9

Given initial state of wave function:

$$\Psi(x, 0) = Ax(a - x), \quad (0 < x < a)$$

The expectation value of the Hamiltonian is:

$$\langle H \rangle = \frac{\int_{-\infty}^{+\infty} \Psi^*(x, 0) \hat{H} \Psi(x, 0) dx}{\int_{-\infty}^{+\infty} \Psi^* \Psi dx}$$

$$\begin{aligned}
\langle H \rangle &= \frac{-\frac{\hbar^2}{2m} \int_0^a -2A^2 x(a-x) dx}{\int_0^a A^2 (a^2 x^2 + x^4 - 2ax^3) dx} \\
\langle H \rangle &= \frac{\hbar^2}{m} \frac{ax - \frac{1}{2}x^2 \Big|_0^a}{\frac{1}{3}a^2 x^3 + \frac{1}{5}x^5 - \frac{1}{2}ax^4 \Big|_0^a} \\
\therefore \langle H \rangle &= \frac{5\hbar^2}{ma^3}
\end{aligned}$$

Problem 2.10

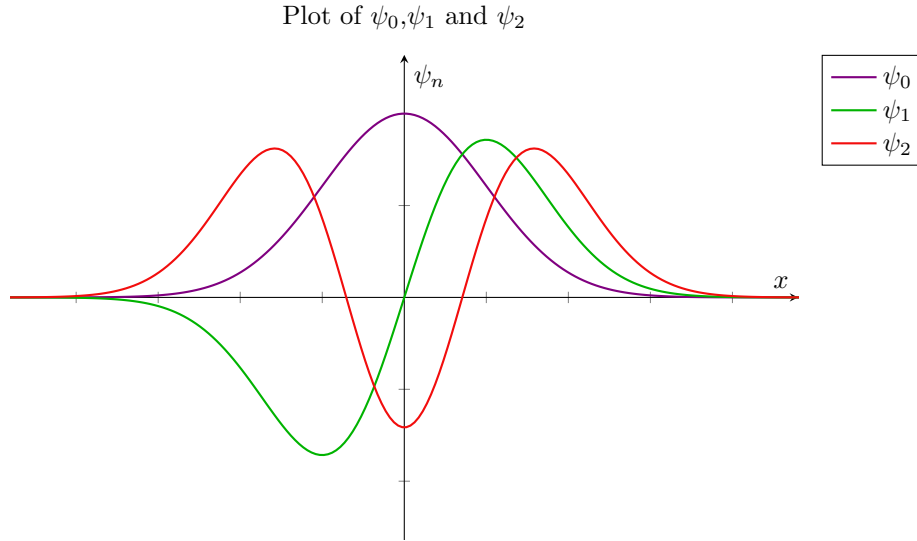
We know that

$$\psi_n = \frac{1}{\sqrt{n!}} (\hat{a}_+)^n \psi_0 = \frac{\alpha}{\sqrt{n!}} (\hat{a}_+)^n e^{-\frac{1}{2}\xi^2}$$

Therefore,

$$\begin{aligned}
\psi_2 &= \frac{\alpha}{\sqrt{2}} \left(\frac{1}{2} \left(\frac{\partial}{\partial \xi} + \xi \right)^2 \right) e^{-\frac{1}{2}\xi^2} \\
\psi_2 &= \frac{\alpha}{2\sqrt{2}} \left(\frac{\partial^2}{\partial \xi^2} + \xi^2 + 2\xi \frac{\partial}{\partial \xi} + 1 \right) e^{-\frac{1}{2}\xi^2} \\
\psi_2 &= \frac{\alpha}{2\sqrt{2}} \left(4\xi^2 e^{-\frac{1}{2}\xi^2} - 2e^{-\frac{1}{2}\xi^2} \right) \\
\therefore \psi_2(\xi) &= \frac{\alpha}{\sqrt{2}} (2\xi^2 - 1) e^{-\frac{1}{2}\xi^2}
\end{aligned}$$

(b) $\psi_1 = \sqrt{2}\alpha\xi e^{-\frac{1}{2}\xi^2}$ and $\psi_0 = \alpha e^{-\frac{1}{2}\xi^2}$. The graphs of ψ_2 , ψ_1 , and ψ_0 are given below:



(c)

$$\begin{aligned}
\int_{-\infty}^{+\infty} \psi_0^* \psi_1 dx &= \int_{-\infty}^{+\infty} \alpha^2 \xi' \sqrt{2} \xi e^{-\xi^2} d\xi = 0 \quad (\because \text{Odd Integrand}) \\
\int_{-\infty}^{+\infty} \psi_1^* \psi_2 dx &= \int_{-\infty}^{+\infty} \alpha^2 \xi' \sqrt{2} (2\xi^3 - \xi) e^{-\xi^2} d\xi = 0 \quad (\because \text{Odd Integrand}) \\
\int_{-\infty}^{+\infty} \psi_2^* \psi_0 dx &= \int_{-\infty}^{+\infty} \frac{\alpha^2 \xi'}{\sqrt{2}} (2\xi^2 - 1) e^{-\xi^2} d\xi = \int_{-\infty}^{+\infty} \frac{\alpha^2}{\sqrt{2}} \xi' \left(\underbrace{2\xi^2 e^{-\xi^2}}_{\sqrt{\pi}} - \underbrace{e^{-\xi^2}}_{\sqrt{\pi}} \right) d\xi = 0
\end{aligned}$$

Problem 2.11

$$\begin{aligned}\psi_0(\xi) &= \alpha e^{-\frac{1}{2}\xi^2} \\ \psi_1(\xi) &= \sqrt{2}\alpha\xi e^{-\frac{1}{2}\xi^2}\end{aligned}$$

For ψ_0

(a)

$$\langle x \rangle = \int_{-\infty}^{+\infty} \psi_0^* x \psi_0 dx = \int_{-\infty}^{+\infty} \alpha^2 \xi e^{-\xi^2} d\xi$$

$$\boxed{\therefore \langle x \rangle = 0}$$

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} \psi_0^* x^2 \psi_0 dx = \int_{-\infty}^{+\infty} \xi'^2 \alpha^2 \xi^2 e^{-\xi^2} d\xi = \frac{\alpha^2 \xi' \sqrt{\pi}}{2}$$

$$\text{However, } \xi' = \sqrt{\frac{\hbar}{m\omega}} \text{ and } \alpha^2 = \sqrt{\frac{m\omega}{\hbar\pi}}$$

$$\boxed{\therefore \langle x^2 \rangle = \frac{\hbar}{2m\omega}}$$

$$\langle p \rangle = \int_{-\infty}^{+\infty} -i\hbar\psi_0^* \frac{\partial\psi_0}{\partial x} dx = \int_{-\infty}^{+\infty} -i\hbar\alpha^2 \left(e^{-\frac{1}{2}\xi^2}\right) \frac{\partial \left(e^{-\frac{1}{2}\xi^2}\right)}{\partial \xi} d\xi = \langle p \rangle = \int_{-\infty}^{+\infty} i\hbar\xi e^{-\xi^2} d\xi$$

$$\boxed{\therefore \langle p \rangle = 0}$$

$$\langle p^2 \rangle = \int_{-\infty}^{+\infty} \psi_0^* \left(-\hbar^2 \frac{\partial^2 \psi_0}{\partial x^2}\right) dx = \int_{-\infty}^{+\infty} -\hbar^2 \alpha^2 e^{-\frac{1}{2}\xi^2} \frac{\partial^2 e^{-\frac{1}{2}\xi^2}}{\partial \xi^2} \xi' d\xi$$

$$\langle p^2 \rangle = \int_{-\infty}^{+\infty} -\hbar^2 \alpha^2 \xi' (\xi^2 e^{-\xi^2} - e^{-\xi^2}) dx = -\hbar^2 \alpha^2 \xi' \left(\frac{\sqrt{\pi}}{2} - \sqrt{\pi}\right) = \frac{1}{2} \sqrt{\pi} \hbar^2 \alpha^2 \xi'$$

$$\text{However, } \alpha^2 = \sqrt{\frac{m\omega}{\hbar\pi}} \text{ and } \xi' = \sqrt{\frac{m\omega}{\hbar}}$$

$$\boxed{\therefore \langle p^2 \rangle = \frac{m\omega\hbar}{2}}$$

$$\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} \quad \sigma_p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2}$$

(b)

$$\sigma_x = \sqrt{\frac{\hbar}{2m\omega}} \quad \sigma_p = \sqrt{\frac{\hbar\omega m}{2}}$$

$$\sigma_x \sigma_p = \frac{\hbar}{2} \geq \frac{\hbar}{2}$$

(c) By Ehrenfest's Principle,

$$\langle T \rangle = \frac{\langle p^2 \rangle}{2m} = \frac{1}{4} \hbar\omega$$

$$\langle V \rangle = \frac{1}{2} m\omega^2 \langle x^2 \rangle = \frac{1}{4} \hbar\omega$$

$$\langle T \rangle + \langle V \rangle = \frac{1}{2} \hbar\omega = E_0$$

For ψ_1

(a)

$$\langle x \rangle = \int_{-\infty}^{+\infty} \psi_1^* x \psi_1 dx = \int_{-\infty}^{+\infty} \frac{2\alpha^2}{\xi'} \xi^3 e^{-\xi^2} d\xi$$

$$\boxed{\therefore \langle x \rangle = 0}$$

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} \psi_1^* x^2 \psi_1 dx = \int_{-\infty}^{+\infty} 2\alpha^2 \xi'^3 \xi^4 e^{-\xi^2} d\xi = \frac{3}{2} \alpha^2 \xi'^3 \sqrt{\pi}$$

$$\text{However, } \alpha^2 = \sqrt{\frac{m\omega}{\hbar\pi}} \text{ and } \xi' = \sqrt{\frac{m\omega}{\hbar}}$$

$$\boxed{\therefore \langle x^2 \rangle = \frac{3\hbar}{2m\omega}}$$

$$\langle p \rangle = \int_{-\infty}^{+\infty} -i\hbar\psi_1^* \left(\frac{\partial\psi_1}{\partial x} \right) dx = \int_{-\infty}^{+\infty} -2i\hbar\alpha^2 \xi' \xi e^{-\frac{1}{2}\xi^2} \left(\frac{\partial \left(\xi e^{-\frac{1}{2}\xi^2} \right)}{\partial \xi} \right) d\xi$$

$$\langle p \rangle = -2i\hbar\alpha^2 \xi' \int_{-\infty}^{+\infty} \xi e^{-\xi^2} - \xi^3 e^{-\xi^2} d\xi$$

$$\boxed{\therefore \langle p \rangle = 0}$$

$$\langle p^2 \rangle = \int_{-\infty}^{+\infty} \psi_1^* \left(-\hbar^2 \frac{\partial^2 \psi_1}{\partial x^2} \right) dx = -2\alpha^2 \hbar^2 \int_{-\infty}^{+\infty} \xi e^{-\frac{1}{2}\xi^2} \frac{\partial^2 \left(\xi e^{-\frac{1}{2}\xi^2} \right)}{\partial \xi^2} d\xi$$

$$\langle p^2 \rangle = -2\alpha^2 \hbar^2 \int_{-\infty}^{+\infty} -3\xi^2 e^{-\xi^2} + \xi^4 e^{-\xi^2} d\xi = -\frac{3}{2} \alpha^2 \hbar^2 \sqrt{\pi}$$

$$\text{However, } \alpha^2 = \sqrt{\frac{m\omega}{\hbar\pi}} \text{ and } \xi' = \sqrt{\frac{m\omega}{\hbar}}$$

$$\boxed{\therefore \langle p^2 \rangle = \frac{3\hbar m\omega}{2}}$$

(b)

$$\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} \quad \sigma_p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2}$$

$$\sigma_x = \sqrt{\frac{3\hbar}{2m\omega}} \quad \sigma_p = \sqrt{\frac{3\hbar m\omega}{2}}$$

$$\sigma_x \sigma_p = \frac{3\hbar}{2} \geq \frac{\hbar}{2}$$

(c) By Ehrenfest's Principle,

$$\langle T \rangle = \frac{\langle p^2 \rangle}{2m} = \frac{3}{4} \hbar\omega$$

$$\langle V \rangle = \frac{1}{2} m\omega^2 \langle x^2 \rangle = \frac{3}{4} \hbar\omega$$

$$\langle T \rangle + \langle V \rangle = \frac{3}{2} \hbar\omega = E_1$$

Problem 2.12

We know that,

$$\begin{aligned}\hat{a}_+ &= \frac{1}{\sqrt{2\hbar m\omega}}(-i\hat{p} + m\omega\hat{x}) & \hat{a}_- &= \frac{1}{\sqrt{2\hbar m\omega}}(i\hat{p} + m\omega\hat{x}) \\ \Rightarrow \hat{x} &= \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}_+ + \hat{a}_-) & \hat{p} &= \sqrt{\frac{\hbar m\omega}{2}}i(\hat{a}_+ - \hat{a}_-) \\ \langle x \rangle_n &= \int_{-\infty}^{+\infty} \psi_n^* \hat{x} \psi_n dx = \sqrt{\frac{\hbar}{2m\omega}} \int_{-\infty}^{+\infty} \psi_n^* (\hat{a}_+ + \hat{a}_-) \psi_n dx \\ \langle x \rangle_n &= \sqrt{\frac{\hbar}{2m\omega}} \int_{-\infty}^{+\infty} \psi_n^* (\sqrt{n+1}\psi_{n+1} + \sqrt{n}\psi_{n-1}) dx = \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n+1}\delta_{n,n+1} + \sqrt{n}\delta_{n,n-1})\end{aligned}$$

$$\boxed{\therefore \langle x \rangle_n = 0}$$

$$\begin{aligned}\langle x^2 \rangle_n &= \int_{-\infty}^{+\infty} \psi_n^* \hat{x}^2 \psi_n dx = \int_{-\infty}^{+\infty} \frac{\hbar}{2m\omega} (\psi_n^* (\hat{a}_+ + \hat{a}_-)^2 \psi_n) dx \\ \langle x^2 \rangle_n &= \frac{\hbar}{2m\omega} \int_{-\infty}^{+\infty} \psi_n^* (\hat{a}_+^2 + \hat{a}_-^2 + 2\hat{a}_+ \hat{a}_- + 1) \psi_n dx \\ \langle x^2 \rangle_n &= \frac{\hbar}{2m\omega} \int_{-\infty}^{+\infty} \sqrt{(n+2)(n+1)} \psi_n^* \psi_{n+2} + \sqrt{n(n-1)} \psi_n^* \psi_{n-2} + (2n+1) \psi_n^* \psi_n dx \\ \langle x^2 \rangle_n &= \frac{\hbar}{2m\omega} \left(\sqrt{(n+2)(n+1)} \delta_{n,n+2} + \sqrt{n(n-1)} \delta_{n,n-2} + (2n+1) \delta_{n,n} \right)\end{aligned}$$

$$\boxed{\therefore \langle x^2 \rangle_n = \frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right)}$$

By Ehrenfest's Principle,

$$\langle p \rangle_n = m \frac{d\langle x \rangle}{dt}$$

$$\boxed{\therefore \langle p \rangle_n = 0}$$

$$\begin{aligned}\langle p^2 \rangle_n &= \int_{-\infty}^{+\infty} \psi_n^* \hat{p}^2 \psi_n dx = -\frac{\hbar m\omega}{2} \int_{-\infty}^{+\infty} \psi_n^* (\hat{a}_+ - \hat{a}_-)^2 \psi_n dx \\ \langle p^2 \rangle_n &= -\frac{\hbar m\omega}{2} \int_{-\infty}^{+\infty} \psi_n^* (\hat{a}_+^2 + \hat{a}_-^2 - 2\hat{a}_+ \hat{a}_- - 1) \psi_n dx \\ \langle p^2 \rangle_n &= -\frac{\hbar m\omega}{2} \int_{-\infty}^{+\infty} \sqrt{(n+2)(n+1)} \psi_n^* \psi_{n+2} + \sqrt{n(n-1)} \psi_n^* \psi_{n-2} - (2n+1) \psi_n^* \psi_n dx \\ \langle p^2 \rangle_n &= -\frac{\hbar m\omega}{2} \left(\sqrt{(n+2)(n+1)} \delta_{n,n+2} + \sqrt{n(n-1)} \delta_{n,n-2} - (2n+1) \delta_{n,n} \right)\end{aligned}$$

$$\boxed{\therefore \langle p^2 \rangle_n = \hbar m\omega \left(n + \frac{1}{2} \right)}$$

By Ehrenfest's Principle,

$$\begin{aligned}\langle T \rangle_n &= \frac{\langle p^2 \rangle}{2m} = \frac{1}{2} \hbar \omega \left(n + \frac{1}{2} \right) \\ \sigma_{x_n} &= \sqrt{\langle x^2 \rangle - \langle x \rangle^2} & \sigma_{p_n} &= \sqrt{\langle p^2 \rangle - \langle p \rangle^2} \\ \sigma_{x_n} &= \sqrt{\frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right)} & \sigma_{p_n} &= \sqrt{\hbar m\omega \left(n + \frac{1}{2} \right)} \\ \sigma_{x_n} \sigma_{p_n} &= \hbar \left(n + \frac{1}{2} \right) \geq \frac{\hbar}{2}\end{aligned}$$

Problem 2.13

The given initial wave function is:

$$\Psi(x, 0) = A[3\psi_0 + 4\psi_1]$$

(a) Normalizing the wave function,

$$\begin{aligned} \int_{-\infty}^{+\infty} \Psi^*(x, 0) \Psi(x, 0) dx &= 1 \implies \int_{-\infty}^{+\infty} A^2 [3\psi_0^* + 4\psi_1^*] [3\psi_0 + 4\psi_1] dx = 1 \\ \int_{-\infty}^{+\infty} A^2 (9\psi_0^* \psi_0 + 16\psi_1^* \psi_1 + 12\psi_0^* \psi_1 + 12\psi_1^* \psi_0) dx &= 1 \\ A^2 (9\delta_{0,0} + 16\delta_{1,1} + 24\delta_{1,0}) &= 1 \implies 25A^2 = 1 \\ \therefore A &= \frac{1}{5} \end{aligned}$$

(b)

$$\begin{aligned} \Psi(x, t) &= \sum_{n=1}^{\infty} c_n \psi_n e^{-iE_n t / \hbar} \\ c_n &= \int_{-\infty}^{+\infty} \psi_n^* \Psi(x, 0) dx = \frac{1}{5} \int_{-\infty}^{+\infty} 3\psi_n^* \psi_0 + 4\psi_n^* \psi_1 dx \\ c_n &= \frac{1}{5} (3\delta_{n,0} + 4\delta_{n,1}) \\ \therefore c_0 &= \frac{3}{5}, \quad c_1 = \frac{4}{5}, \quad c_{n \geq 2} = 0 \\ \therefore \Psi(x, t) &= \frac{\alpha}{5} \left(3e^{-\frac{1}{2}\xi^2} e^{-i\omega t/2} + 4\sqrt{2}\xi e^{-\frac{1}{2}\xi^2} e^{-3i\omega t/2} \right) \\ |\Psi(x, t)|^2 &= \frac{\alpha^2}{25} e^{-\xi^2} (3e^{i\omega t/2} + 4\sqrt{2}\xi e^{3i\omega t/2})(3e^{-i\omega t/2} + 4\sqrt{2}\xi e^{-3i\omega t/2}) \\ |\Psi(x, t)|^2 &= \frac{\alpha^2}{25} e^{-\xi^2} (9 + 32\xi^2 + 12\sqrt{2}\xi \underbrace{(e^{i\omega t} + e^{-i\omega t})}_{2 \cos \omega t}) \\ \therefore |\Psi(x, t)|^2 &= \frac{\alpha^2}{25} e^{-\xi^2} (32\xi^2 + 24\sqrt{2} \cos \omega t + 9) \implies \text{Oscillation frequency} = \omega \end{aligned}$$

If in place of ψ_1 , ψ_2 was mentioned,

$$\begin{aligned} |\Psi(x, t)|^2 &= \frac{\alpha^2}{25} e^{-\xi^2} (3e^{-i\omega t/2} + 2\sqrt{2}(2\xi^2 - 1)e^{-5i\omega t/2})(3e^{i\omega t/2} + 2\sqrt{2}(2\xi^2 - 1)e^{5i\omega t/2}) \\ |\Psi(x, t)|^2 &= \frac{\alpha^2}{25} e^{-\xi^2} (9 + (2\xi^2 - 1)^2 + 6\sqrt{2}(2\xi^2 - 1) \underbrace{(e^{-2i\omega t} + e^{2i\omega t})}_{2 \cos 2\omega t}) \\ \therefore |\Psi(x, t)|^2 &= \frac{\alpha^2}{25} e^{-\xi^2} (9 + (2\xi^2 - 1)^2 + 12\sqrt{2} \cos 2\omega t) \implies \text{Oscillation Frequency} = 2\omega \end{aligned}$$

(c)

$$\begin{aligned} \langle x \rangle &= \frac{\alpha^2}{25} \int_{-\infty}^{+\infty} (3\psi_0^* e^{i\omega t/2} + 4\psi_1^* e^{3i\omega t/2}) \hat{x} (3\psi_0 e^{-i\omega t/2} + 4\psi_1 e^{-3i\omega t/2}) dx \\ \langle x \rangle &= \frac{\alpha^2}{25} \sqrt{\frac{\hbar}{2m\omega}} \int_{-\infty}^{+\infty} (3\psi_0^* e^{i\omega t/2} + 4\psi_1^* e^{3i\omega t/2}) (\hat{a}_+ + \hat{a}_-) (3\psi_0 e^{-i\omega t/2} + 4\psi_1 e^{-3i\omega t/2}) dx \\ \langle x \rangle &= \frac{\alpha^2}{25} \sqrt{\frac{\hbar}{2m\omega}} \int_{-\infty}^{+\infty} (3\psi_0^* e^{i\omega t/2} + 4\psi_1^* e^{-i\omega t/2}) (3\psi_1 e^{-i\omega t/2} + 4\sqrt{2}\psi_2 e^{-3i\omega t/2} + 4\psi_0 e^{-i\omega t/2}) dx \end{aligned}$$

$$\langle x \rangle = \frac{\alpha^2}{25} \sqrt{\frac{\hbar}{2m\omega}} \int_{-\infty}^{+\infty} 9\psi_0^*\psi_1 + 12(\sqrt{2}\psi_0^*\psi_2 + \psi_0^*\psi_0)e^{-i\omega t} + 12\psi_1^*\psi_1e^{i\omega t} + 12(\sqrt{2}\psi_1^*\psi_2 + \psi_1^*\psi_0) dx$$

$$\langle x \rangle = \frac{\alpha^2}{25} \sqrt{\frac{\hbar}{2m\omega}} (9\delta_{0,1} + 12(\sqrt{2}\delta_{0,2} + \delta_{0,0}) + 12\delta_{1,1} + 12(\sqrt{2}\delta_{1,2} + \delta_{1,0}))$$

$$\boxed{\therefore \langle x \rangle = \frac{24\alpha^2}{25} \sqrt{\frac{\hbar}{2m\omega}} \cos \omega t}$$

$$\langle p \rangle = \frac{\alpha^2}{25} \int_{-\infty}^{+\infty} (3\psi_0^*e^{i\omega t/2} + 4\psi_1^*e^{3i\omega t/2})\hat{p}(3\psi_0e^{-i\omega t/2} + 4\psi_1e^{-3i\omega t/2}) dx$$

$$\langle p \rangle = \frac{-i\alpha^2}{25} \sqrt{\frac{\hbar m\omega}{2}} \int_{-\infty}^{+\infty} (3\psi_0^*e^{i\omega t/2} + 4\psi_1^*e^{3i\omega t/2})(\hat{a}_+ - \hat{a}_-)(3\psi_0e^{-i\omega t/2} + 4\psi_1e^{-3i\omega t/2}) dx$$

$$\langle p \rangle = -\frac{i\alpha^2}{25} \sqrt{\frac{\hbar m\omega}{2}} \int_{-\infty}^{+\infty} (3\psi_0^*e^{i\omega t/2} + 4\psi_1^*e^{-i\omega t/2})(3\psi_1e^{-i\omega t/2} + 4\sqrt{2}\psi_2e^{-3i\omega t/2} - 4\psi_0e^{-i\omega t/2}) dx$$

$$\langle p \rangle = -\frac{i\alpha^2}{25} \sqrt{\frac{\hbar m\omega}{2}} \int_{-\infty}^{+\infty} 9\psi_0^*\psi_1 + 12(\sqrt{2}\psi_0^*\psi_2 - \psi_0^*\psi_0)e^{-i\omega t} + 12\psi_1^*\psi_1e^{i\omega t} + 12(\sqrt{2}\psi_1^*\psi_2 - \psi_1^*\psi_0) dx$$

$$\langle p \rangle = -\frac{i\alpha^2}{25} \sqrt{\frac{\hbar m\omega}{2}} (9\delta_{0,1} + 12(\sqrt{2}\delta_{0,2} - \delta_{0,0}) + 12\delta_{1,1} + 12(\sqrt{2}\delta_{1,2} - \delta_{1,0}))$$

$$\boxed{\therefore \langle p \rangle = -\frac{24\alpha^2}{25} \sqrt{\frac{\hbar m\omega}{2}} \sin \omega t}$$

Notice that $\langle p \rangle = m \frac{d\langle x \rangle}{dt}$. Therefore, Ehrenfest's principle is verified.

(d) The wave function $\Psi(x, t)$ is a superposition of $n = 0$ and $n = 1$ states, so only E_0 and E_1 are the energies you will measure. The probability of each state is the amplitude squared of the corresponding coefficients of the linear combination of ψ .

$$P(E_n) = |c_n|^2 \implies P(E_0) = |c_0|^2 = \frac{9}{25} = 36\%, \quad P(E_1) = |c_1|^2 = \frac{16}{25} = 64\%$$

Problem 2.14

Let the classically allowed region be $x = -a$ to $x = a$. We must find the probability of the particle to be outside this region. Let this region be A .

$$P(A) = \int_{-\infty}^{-a} \psi^* \psi dx + \int_a^{+\infty} \psi^* \psi dx = \int_{-\infty}^{-\xi(a)} \frac{\alpha^2}{\xi'} e^{-\xi^2} d\xi + \int_{\xi(a)}^{+\infty} \frac{\alpha^2}{\xi'} e^{-\xi^2} d\xi$$

$$P(A) = \frac{\alpha^2 \sqrt{\pi}}{2\xi'} (erf(\xi)|_{-\infty}^{-\xi(a)} + erf(\xi)|_{\xi(a)}^{+\infty}) = \frac{\alpha^2 \sqrt{\pi}}{2\xi'} (erf(\xi)|_{-\infty}^{+\infty} - erf(\xi)|_{-\xi(a)}^{\xi(a)})$$

$$\text{But, } \xi' = \sqrt{\frac{\hbar}{m\omega}}, \quad \alpha^2 = \sqrt{\frac{m\omega}{\hbar\pi}}$$

$$P(A) = 1 - erf(1) = 0.157 = 15.7\%$$

Problem 2.15

The recursion formula for a Hermite polynomial is,

$$a_{j+2} = -\frac{2(n-j)}{(j+2)(j+1)} a_j$$

$$n = 5$$

$$\begin{aligned} a_3 &= -\frac{2(5-3)}{2 \cdot 3} a_1 = -\frac{4}{3} a_1 \\ a_5 &= -\frac{2(5-3)}{4 \cdot 5} a_3 = \frac{4}{15} a_1 = 2^5 = 32 \implies a_1 = 120 \\ \therefore H_5(\xi) &= 32\xi^5 - 160\xi^3 + 120\xi \end{aligned}$$

$$n = 6$$

$$\begin{aligned} a_2 &= -\frac{2(6-0)}{1 \cdot 2} = -6a_0 \\ a_4 &= -\frac{2(6-2)}{3 \cdot 4} a_2 = 4a_0 \\ a_6 &= -\frac{2(6-4)}{5 \cdot 6} a_4 = -\frac{8}{15} a_0 = 2^6 = 64 \implies a_0 = -120 \\ \therefore H_6(\xi) &= 64\xi^6 - 480\xi^4 + 640\xi^2 - 120 \end{aligned}$$

Problem 2.16

(a) By the Rodrigues Formula,

$$\begin{aligned} H_n(\xi) &= (-1)^n e^{\xi^2} \frac{d^n}{d\xi^n} (e^{-\xi^2}) \\ H_3(\xi) &= -e^{\xi^2} \frac{d^3}{d\xi^3} (e^{-\xi^2}) = -e^{\xi^2} \frac{d^2}{d\xi^2} (-2\xi e^{-\xi^2}) \\ H_3(\xi) &= -e^{\xi^2} \frac{d}{d\xi} (-2e^{-\xi^2} + 4\xi^2 e^{-\xi^2}) = -e^{\xi^2} (4\xi e^{-\xi^2} + 8\xi e^{-\xi^2} - 8\xi^3 e^{-\xi^2}) \\ \therefore H_3(\xi) &= 8\xi^3 - 12\xi \\ H_4 &= e^{\xi^2} \frac{d^4}{d\xi^4} (e^{-\xi^2}) = -e^{\xi^2} \frac{d}{d\xi} (e^{-\xi^2} H_3) = -e^{\xi^2} \frac{d}{d\xi} (-12\xi e^{-\xi^2} + 8\xi^3 e^{-\xi^2}) \\ \therefore H_4 &= 16\xi^4 - 48\xi^2 + 12 \end{aligned}$$

(b)

$$\begin{aligned} H_{n+1}(\xi) &= 2\xi H_n(\xi) - 2n H_{n-1}(\xi) \\ H_5(\xi) &= 2\xi H_4(\xi) - 8H_3(\xi) \\ \therefore H_5(\xi) &= 32\xi^5 - 160\xi^3 + 120 \\ H_6(\xi) &= 2\xi H_5(\xi) - 12H_4(\xi) \\ \therefore H_6(\xi) &= 64\xi^6 - 480\xi^4 + 720\xi^2 - 120 \end{aligned}$$

(c)

$$\begin{aligned} \frac{dH_n(\xi)}{d\xi} &= 2n H_{n-1}(\xi) \\ \frac{dH_6(\xi)}{d\xi} &= 384\xi^5 - 1920\xi^3 + 1440\xi = 2 \cdot 6(32\xi^5 - 160\xi^3 + 120\xi) \\ \therefore \frac{dH_6(\xi)}{d\xi} &= 2 \cdot 6H_5(\xi) \end{aligned}$$

(d)

$$H_n(\xi) = \left. \frac{\partial^n}{\partial z^n} (e^{-z^2+2z\xi}) \right|_{z=0}$$

$$\begin{aligned}
H_1(\xi) &= \frac{\partial}{\partial z} (e^{-z^2+2z\xi}) \Big|_{z=0} = (-2z + 2\xi) e^{-z^2+2z\xi} \Big|_{z=0} = 2\xi \\
H_2(\xi) &= \frac{\partial}{\partial z} ((-2z + 2\xi) e^{-z^2+2z\xi}) \Big|_{z=0} = 2\xi^2 - 1 \\
H_3(\xi) &= \frac{\partial^2}{\partial z^2} ((-2z + 2\xi) e^{-z^2+2z\xi}) \Big|_{z=0} = 8\xi^3 - 12\xi
\end{aligned}$$

Problem 2.17

Given function is,

$$f(x) = Ae^{ikx} + Be^{-ikx}$$

From Euler's Formula, $e^{i\theta} = \cos \theta + i \sin \theta$

$$f(x) = A \cos kx + iA \sin kx + B \cos kx - iB \sin kx$$

$$f(x) = (A + B) \cos kx + i(A - B) \sin kx$$

Let $A + B = C$ and $i(A - B) = D$, then

$$\therefore f(x) = C \cos kx + D \sin kx$$

Problem 2.18

The probability flux is given by,

$$\begin{aligned}
J(x, t) &= \frac{i\hbar}{2m} \left(\Psi \frac{\partial \Psi^*}{\partial x} - \Psi^* \frac{\partial \Psi}{\partial x} \right) \\
\Psi &= Ae^{i\left(kx - \frac{\hbar k^2}{2m}t\right)} \quad \Psi^* = Ae^{-i\left(kx - \frac{\hbar k^2}{2m}t\right)} \\
\frac{\partial \Psi}{\partial x} &= ikAe^{i\left(kx - \frac{\hbar k^2}{2m}t\right)} \quad \frac{\partial \Psi^*}{\partial x} = -ikAe^{-i\left(kx - \frac{\hbar k^2}{2m}t\right)} \\
J(x, t) &= \frac{i\hbar}{2m} (-ikA^2 - ikA^2) = -i^2 \hbar \frac{A^2}{m} \\
J(x, t) &= \frac{kA^2}{m} \implies \text{Flows in direction of the wave}
\end{aligned}$$

Problem 2.19

(a) By Dirichlet's Theorem, on the interval $[-a, a]$

$$f(x) = \sum_{n=0}^{\infty} a_n \sin\left(\frac{n\pi}{a}x\right) + b_n \cos\left(\frac{n\pi}{a}x\right)$$

We know that $\sin \theta = -i \left(\frac{e^{i\theta} - e^{-i\theta}}{2} \right)$ and $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$

$$\begin{aligned}
f(x) &= \sum_{n=0}^{\infty} -\frac{ia_n}{2} (e^{in\pi x/a} - e^{-in\pi x/a}) + \frac{b_n}{2} (e^{in\pi x/a} + e^{-in\pi x/a}) \\
f(x) &= \sum_{n=0}^{\infty} \left(\frac{b_n - ia_n}{2} \right) e^{in\pi x/a} + \left(\frac{b_n + ia_n}{2} \right) e^{-in\pi x/a} \\
f(x) &= \sum_{n=0}^{\infty} \left(\frac{b_n - ia_n}{2} \right) e^{in\pi x/a} + \sum_{n=-\infty}^0 \left(\frac{b_n + ia_n}{2} \right) e^{in\pi x/a}
\end{aligned}$$

Let $c_n = b_n - \frac{|n|}{n} a_n$,

$$\therefore f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/a}$$

(b) Consider $\int_{-a}^{+a} f(x) e^{-im\pi x/a} dx$

$$\begin{aligned} \int_{-a}^{+a} f(x) e^{-im\pi x/a} dx &= \int_{-a}^{+a} \sum_{n=-\infty}^{\infty} c_n e^{-im\pi x/a} e^{in\pi x/a} dx \\ \Rightarrow \sum_{n=-\infty}^{\infty} c_n \int_{-a}^{+a} e^{i\pi x(n-m)/a} dx &= \sum_{n=-\infty}^{\infty} 2ac_n \delta_{n,m} = 2ac_m \\ \therefore c_n &= \frac{1}{2a} \int_{-a}^{+a} f(x) e^{-in\pi x/a} dx \end{aligned}$$

(c) Let $k \equiv \frac{n\pi}{a}$ and $F(k) = \sqrt{\frac{2}{\pi}} ac_n$

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} \frac{1}{a} \sqrt{\frac{\pi}{2}} F(k) e^{ikx} = \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} F(k) e^{ikx} \underbrace{\Delta k}_{\frac{\pi}{a}} \\ c_n &= \sqrt{\frac{\pi}{2}} \frac{1}{a} F(k) = \frac{1}{2a} \int_{-a}^{+a} f(x) e^{-ikx} dx \\ \therefore F(k) &= \frac{1}{\sqrt{2\pi}} \int_0^1 f(x) e^{-ikx} dx \end{aligned}$$

(d)

$$f(x) = \lim_{a \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} F(k) e^{ikx} \Delta k = \lim_{\Delta k \rightarrow 0} \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} F(k) e^{ikx} \Delta k$$

$$\boxed{\therefore f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} F(k) e^{ikx} dk}$$

$$F(k) = \lim_{a \rightarrow \infty} \int_{-a}^{+a} f(x) e^{-ikx} dx$$

$$\boxed{\therefore F(k) = \int_{-\infty}^{+\infty} f(x) e^{-ikx} dx}$$

Problem 2.20

Given initial wave function is,

$$\Psi(x, 0) = A e^{-a|x|}$$

(a) Normalizing the wave function,

$$\int_{-\infty}^{+\infty} A^2 e^{-2a|x|} dx = \int_{-\infty}^0 A^2 e^{2ax} dx + \int_0^{\infty} A^2 e^{-2ax} dx = 1$$

$$\frac{A^2}{2a} (e^{2ax} \Big|_{-\infty}^0 - e^{-2ax} \Big|_0^{+\infty}) = 1$$

$$\therefore A = \sqrt{a}$$

(b) Applying the Fourier Transform,

$$\begin{aligned}\phi(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \Psi(x, 0) e^{-ikx} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \sqrt{a} e^{-(ikx + a|x|)} dx \\ \phi(k) &= \sqrt{\frac{a}{2\pi}} \left(\int_{-\infty}^0 e^{(a-ik)x} dx + \int_0^{+\infty} e^{-(ik+a)x} dx \right) \\ \phi(k) &= \sqrt{\frac{a}{2\pi}} \left(\frac{1}{a-ik} e^{(a-ik)x} \Big|_{-\infty}^0 - \frac{1}{a+ik} e^{-(a+ik)x} \Big|_0^{+\infty} \right) \\ \phi(k) &= \sqrt{\frac{a}{2\pi}} \left(\frac{1}{a+ik} + \frac{1}{a-ik} \right) = \sqrt{\frac{2a^3}{\pi}} \frac{1}{a^2 + k^2}\end{aligned}$$

(c) Applying the inverse Fourier Transform,

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{ikx} dk = \frac{\sqrt{a^3}}{\pi} \int_{-\infty}^{+\infty} \frac{e^{ikx}}{a^2 + k^2} dk$$

(d) For $a \rightarrow 0$

$$\Psi(x, t) = \lim_{a \rightarrow 0} \frac{a^{\frac{3}{2}}}{\pi} \int_{-\infty}^{+\infty} \frac{e^{ikx}}{a^2 + k^2} dk = \int_{-\infty}^{+\infty} \frac{e^{ikx}}{\pi} \lim_{a \rightarrow 0} \frac{a^{\frac{3}{2}}}{a^2 + k^2} dk = 0$$

For $a \rightarrow \infty$

$$\Psi(x, t) = \lim_{a \rightarrow \infty} \frac{a^{\frac{3}{2}}}{\pi} \int_{-\infty}^{+\infty} \frac{e^{ikx}}{a^2 + k^2} dk = \int_{-\infty}^{+\infty} \frac{e^{ikx}}{\pi} \lim_{a \rightarrow \infty} \frac{a^{\frac{3}{2}}}{a^2 + k^2} dk = 0$$

Problem 2.21

Given initial wave function is,

$$\Psi(x, 0) = A e^{-ax^2}$$

(a) Normalizing the wave function,

$$\begin{aligned}\int_{-\infty}^{+\infty} A^2 e^{-2ax^2} dx &= 1 \implies A^2 \sqrt{\frac{\pi}{2a}} = 1 \\ \therefore A &= \left(\frac{2a}{\pi} \right)^{\frac{1}{4}}\end{aligned}$$

(b) Applying the Fourier Transform,

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \Psi(x, 0) e^{-ikx} dx = \frac{A}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-(ikx + ax^2)} dx$$

Applying the substitution $y = \sqrt{a}x + \frac{ik}{2\sqrt{a}}$

$$\implies ikx + ax^2 = y^2 + \frac{k^2}{4a}, \quad dx = \frac{1}{\sqrt{a}} dy$$

$$\phi(k) = \frac{A}{\sqrt{2\pi a}} \int_{-\infty}^{+\infty} e^{-y^2} e^{-\frac{k^2}{4a}} dy$$

$$\therefore \phi(k) = \frac{A e^{-\frac{k^2}{4a}}}{\sqrt{2a}}$$

Now, applying inverse Fourier Transform,

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{i(kx - \frac{\hbar k^2}{2m} t)} dk$$

$$\Psi(x, t) = \frac{A}{2\sqrt{a\pi}} \int_{-\infty}^{+\infty} e^{i(kx - \frac{\hbar k^2}{2m} t) - \frac{k^2}{4a}} dk$$

Let $\gamma \equiv \sqrt{1 + \frac{2\hbar i a t}{m}}$,

$$\Psi(x, t) = \frac{A}{2\sqrt{a\pi}} \int_{-\infty}^{+\infty} e^{ikx - \frac{k^2 \gamma^2}{4a}} dk$$

Applying the substitution $y = \frac{k\gamma}{2\sqrt{a}} - \frac{i\sqrt{a}}{\gamma} x$

$$\Rightarrow y^2 + \frac{ax^2}{\gamma^2} = \frac{k^2 \gamma^2}{4a} - ikx \quad dk = 2\sqrt{a} dy$$

$$\Psi(x, t) = \frac{A}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-y^2} e^{-\frac{ax^2}{\gamma^2}} dk$$

$$\boxed{\therefore \Psi(x, t) = \frac{A}{\gamma} e^{-ax^2/\gamma^2}}$$

(c)

$$|\Psi(x, t)|^2 = \frac{A^2}{\gamma \gamma^*} e^{-ax^2/\gamma^2 - ax^2/\gamma^{*2}} = \sqrt{\frac{2a}{\pi}} \frac{1}{\sqrt{1 + (2\hbar a t/m)^2}} e^{-ax^2 \left(\frac{1}{1 + (2\hbar a t/m)} + \frac{1}{1 - (2\hbar a t/m)} \right)}$$

$$|\Psi(x, t)|^2 = \frac{1}{\sqrt{\pi/2}} \sqrt{\frac{a}{1 + (2\hbar a t/m)^2}} e^{-2x^2 \left(\frac{a}{1 + (2\hbar a t/m)^2} \right)}$$

$$\therefore |\Psi(x, t)|^2 = \sqrt{\frac{2}{\pi}} w e^{-2x^2 w^2}$$

$$|\Psi(x, 0)|^2 = \sqrt{\frac{2a}{\pi}} e^{-2ax^2}$$

For a large t, $|\Psi(x, t)|^2 \rightarrow 0$

(d)

$$\langle x \rangle = \int_{-\infty}^{+\infty} \sqrt{\frac{2}{\pi}} w e^{-2x^2 w^2} x dx$$

$$\boxed{\therefore \langle x \rangle = 0}$$

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} \sqrt{\frac{2}{\pi}} w e^{-2x^2 w^2} x^2 dx$$

Applying the substitution $y \equiv \sqrt{2}wx$

$$dx = \frac{1}{\sqrt{2}w} dy$$

$$\langle x^2 \rangle = \frac{1}{2w^2 \sqrt{\pi}} \int_{-\infty}^{+\infty} y^2 e^{-y^2} dy$$

$$\boxed{\therefore \langle x^2 \rangle = \frac{1}{4w^2} = \frac{1 + (2\hbar a t/m)^2}{4a}}$$

By Ehrenfest's principle,

$$\begin{aligned}\langle p \rangle &= m \frac{d\langle x \rangle}{dt} \\ \boxed{\therefore \langle p \rangle &= 0} \\ \langle p^2 \rangle &= -\hbar^2 \int_{-\infty}^{+\infty} \Psi^* \frac{\partial^2 \Psi}{\partial x^2} dx = -\frac{2a\hbar^2 A^2}{\gamma^3} \int_{-\infty}^{+\infty} \left(-\frac{x}{\gamma\gamma^*} + \frac{2ax^2}{\gamma^2\gamma^*} \right) e^{-ax^2(\frac{1}{\gamma} + \frac{1}{\gamma^*})} dx \\ \langle p^2 \rangle &= -\frac{4a^2\hbar^2 A^2}{\gamma^5\gamma^*} \int_{-\infty}^{+\infty} x^2 e^{-2x^2w^2} dx\end{aligned}$$

Applying the substitution $y = \sqrt{2}xw$

$$\begin{aligned}dx &= \frac{1}{\sqrt{2}w} dy \\ \langle p^2 \rangle &= -\frac{4a^2\hbar^2 A^2}{\gamma^5\gamma^*} \frac{1}{2\sqrt{2}w^3} \int_{-\infty}^{+\infty} y^2 e^{-y^2} dy \\ \boxed{\therefore \langle p^2 \rangle &= a\hbar^2}\end{aligned}$$

$$\begin{aligned}\sigma_x &= \frac{1}{2w} & \sigma_p &= \sqrt{a}\hbar \\ \sigma_x \sigma_p &= \frac{\hbar \sqrt{a}}{2w} \geq \frac{\hbar}{2}\end{aligned}$$

Closest to uncertainty limit at $t = 0$.

Problem 2.22

(a)

$$\begin{aligned}&\int_{-3}^1 (x^3 - 3x^2 + 2x - 1)\delta(x+2) dx \\ &= ((-2)^3 - 3(-2)^2 + 2(-2) - 1) \int_{-3}^1 \delta(x+2) dx \\ &= -25 \int_{-3}^1 \delta(x+2) dx = -25(1) = -25\end{aligned}$$

(b)

$$\begin{aligned}&\int_0^\infty [\cos(3x) + 2]\delta(x - \pi) dx \\ &= [\cos(3\pi) + 2] \int_0^\infty \delta(x - \pi) dx \\ &= 2 \int_0^\infty \delta(x - \pi) dx = 2(1) = 2\end{aligned}$$

(c)

$$\begin{aligned}&\int_{-1}^1 \exp(|x| + 3)\delta(x - 2) dx \\ &= \int_{-1}^0 \exp(3 - x)\delta(x - 2) dx + \int_0^1 \exp(3 + x)\delta(x - 2) dx \\ &= \exp(1) \int_0^1 \delta(x - 2) dx + \exp(5) \int_0^1 \delta(x - 2) dx \\ &= 0 + 0 = 0\end{aligned}$$

Problem 2.23

Given that $D_1(x) = D_2(x)$ if

$$\int_{-\infty}^{+\infty} f(x) D_1(x) dx = \int_{-\infty}^{+\infty} f(x) D_2(x) dx$$

(a)

$$\begin{aligned} D_1(x) &= \delta(cx), \quad D_2(x) = \frac{1}{|c|} \delta(x) \\ \int_{-\infty}^{+\infty} f(x) D_1(x) dx &= \int_{-\infty}^{+\infty} f(x) \delta(cx) dx \\ y &\equiv cx, \quad dx = \frac{dy}{c} \\ \text{If } c > 0, \int_{-\infty}^{+\infty} f(x) D_1(x) dx &= \int_{-\infty}^{+\infty} \frac{1}{c} f\left(\frac{y}{c}\right) \delta(y) dy = \frac{1}{c} f(0) \\ \text{If } c < 0, \int_{-\infty}^{+\infty} f(x) D_1(x) dx &= \int_{+\infty}^{-\infty} \frac{1}{c} f\left(\frac{y}{c}\right) \delta(y) dy = -\frac{1}{c} f(0) \\ \therefore \int_{-\infty}^{+\infty} f(x) D_1(x) dx &= \frac{1}{|c|} f(0) \\ \int_{-\infty}^{+\infty} f(x) D_2(x) dx &= \int_{-\infty}^{+\infty} \frac{1}{|c|} f(x) \delta(x) dx = \frac{1}{|c|} f(0) \\ \therefore \int_{-\infty}^{+\infty} f(x) D_1(x) dx &= \int_{-\infty}^{+\infty} f(x) D_2(x) dx \implies \boxed{\therefore \delta(cx) = \frac{1}{|c|} \delta(x)} \end{aligned}$$

(b)

$$\begin{aligned} \theta(x) &\equiv \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases} \\ D_1(x) &= \frac{d\theta}{dx}(x), \quad D_2(x) = \delta(x) \\ \int_{-\infty}^{+\infty} f(x) D_1(x) dx &= \int_{-\infty}^{+\infty} f(x) \frac{d\theta}{dx}(x) dx \\ &= \theta(x) f(x) \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \theta(x) f'(x) dx = \lim_{x \rightarrow \infty} f(x) - \int_0^{+\infty} f'(x) dx \\ &= \lim_{x \rightarrow \infty} f(x) - [f(x)] \Big|_0^{+\infty} = \lim_{x \rightarrow \infty} \cancel{f(x)} - \lim_{x \rightarrow \infty} \cancel{f(x)} + f(0) = f(0) \\ \therefore \int_{-\infty}^{+\infty} f(x) D_1(x) dx &= f(0) \\ \int_{-\infty}^{+\infty} f(x) D_2(x) dx &= \int_{-\infty}^{+\infty} f(x) \delta(x) dx = f(0) \\ \therefore \int_{-\infty}^{+\infty} f(x) D_1(x) dx &= \int_{-\infty}^{+\infty} f(x) D_2(x) dx \implies \boxed{\therefore \delta(x) = \frac{d\theta}{dx}(x)} \end{aligned}$$

Problem 2.24

$$\begin{aligned}
 \psi(x) &= \frac{\sqrt{m\alpha}}{\hbar} e^{-m\alpha|x|/\hbar^2} \\
 \langle p^2 \rangle &= \int_{-\infty}^{+\infty} \psi^* (-\hbar^2 \partial_x^2) \psi dx = -m\alpha \int_{-\infty}^{+\infty} e^{-m\alpha|x|/\hbar^2} \partial_x^2 (e^{-m\alpha|x|/\hbar^2}) dx \\
 \langle p^2 \rangle &= -m\alpha \int_{-\infty}^{+\infty} e^{-m\alpha|x|/\hbar^2} \partial_x \left(-\frac{m\alpha}{\hbar^2} \frac{|x|}{x} e^{-m\alpha|x|/\hbar^2} \right) dx \\
 \langle p^2 \rangle &= \frac{m^2 \alpha^2}{\hbar^2} \int_{-\infty}^{+\infty} e^{-2m\alpha|x|/\hbar^2} \left(-\frac{m\alpha}{\hbar^2} \frac{|x|^2}{x^2} + \partial_x \left(\frac{|x|}{x} \right) \right) dx \\
 \langle p^2 \rangle &= \frac{m^2 \alpha^2}{\hbar^2} \int_{-\infty}^{+\infty} \underbrace{-\frac{m\alpha}{\hbar^2} \frac{|x|^2}{x^2} e^{-2m\alpha|x|/\hbar^2}}_{I_1} + \underbrace{\partial_x \left(\frac{|x|}{x} \right) e^{-2m\alpha|x|/\hbar^2}}_{I_2} dx \\
 \langle p^2 \rangle &= -\frac{m^3 \alpha^3}{\hbar^4} I_1 + \frac{m^2 \alpha^2}{\hbar^2} I_2 \\
 I_1 &= \int_{-\infty}^{+\infty} \frac{|x|^2}{x^2} e^{-2m\alpha|x|/\hbar^2} dx = \int_{-\infty}^0 e^{2m\alpha x/\hbar^2} dx + \int_0^{+\infty} e^{-2m\alpha x/\hbar^2} dx \\
 I_1 &= \frac{\hbar^2}{2m\alpha} (e^{2m\alpha x/\hbar^2} \Big|_{-\infty}^0 - e^{-2m\alpha x/\hbar^2} \Big|_0^{+\infty}) = 0 \\
 I_2 &= \int_{-\infty}^{+\infty} \partial_x \left(\frac{|x|}{x} \right) e^{-2m\alpha|x|/\hbar^2} dx = \frac{|x|}{x} e^{-2m\alpha|x|/\hbar^2} \Big|_{-\infty}^{+\infty} + \frac{2m\alpha}{\hbar^2} \int_{-\infty}^{+\infty} \frac{|x|}{x^2} e^{-2m\alpha|x|/\hbar^2} dx \\
 I_2 &= 1 + 0 = 1 \\
 \boxed{\therefore \langle p^2 \rangle} &= \frac{m^2 \alpha^2}{\hbar^2}
 \end{aligned}$$

Problem 2.25

$$\begin{aligned}
 \psi_B(x) &= \frac{\sqrt{m\alpha}}{\hbar} e^{-m\alpha|x|/\hbar^2}, \quad \psi_S(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0 \\ Fe^{ikx} + Ge^{-ikx}, & x > 0 \end{cases} \\
 \langle \psi_B | \psi_S \rangle &= \int_{-\infty}^{+\infty} \psi_B^* \psi_S dx \\
 \langle \psi_B | \psi_S \rangle &= \frac{\sqrt{m\alpha}}{\hbar} \left(\int_0^{\infty} Ae^{(ik+m\alpha/\hbar^2)x} + Be^{(-ik+m\alpha/\hbar^2)x} dx + \int_{-\infty}^0 Fe^{-(ik+m\alpha/\hbar^2)x} + Ge^{-(ik+m\alpha/\hbar^2)x} dx \right) \\
 \langle \psi_B | \psi_S \rangle &= \frac{\sqrt{m\alpha}}{\hbar} \left(\frac{A}{ik+m\alpha/\hbar^2} e^{(ik+m\alpha/\hbar^2)x} + \frac{B}{-ik+m\alpha/\hbar^2} e^{(-ik+m\alpha/\hbar^2)x} \right) \Big|_{-\infty}^0 \\
 &\quad + \frac{\sqrt{m\alpha}}{\hbar^2} \left(-\frac{F}{-ik+m\alpha/\hbar^2} e^{-(ik+m\alpha/\hbar^2)x} - \frac{G}{ik+m\alpha/\hbar^2} e^{-(ik+m\alpha/\hbar^2)x} \right) \Big|_0^{\infty} \\
 \langle \psi_B | \psi_S \rangle &= \frac{\sqrt{m\alpha}}{\hbar} \left(\frac{A}{ik+m\alpha/\hbar^2} + \frac{B}{-ik+m\alpha/\hbar^2} + \frac{F}{-ik+m\alpha/\hbar^2} + \frac{G}{ik+m\alpha/\hbar^2} \right) \\
 \langle \psi_B | \psi_S \rangle &= \frac{\sqrt{m\alpha}}{\hbar} \left(\frac{A+G}{ik+m\alpha/\hbar^2} + \frac{B+F}{-ik+m\alpha/\hbar^2} \right) \\
 \langle \psi_B | \psi_S \rangle &= \frac{\sqrt{m\alpha}}{\hbar} \left(\frac{(m\alpha/\hbar^2)(A+B+F+G) + ik(B+F-A-G)}{k^2 + (m\alpha/\hbar^2)^2} \right)
 \end{aligned}$$

Substituting the boundary conditions,

$$\begin{aligned}
 A + B &= F + G \\
 B + F - A - G &= \frac{2mi\alpha}{k\hbar^2}(A + B) \\
 \langle \psi_B | \psi_S \rangle &= \frac{\sqrt{m\alpha}}{\hbar} \left(\frac{(2m\alpha/\hbar^2)(A + B) - (2m\alpha/\hbar^2)(A + B)}{k^2 + (m\alpha/\hbar^2)^2} \right) \\
 \therefore \langle \psi_B | \psi_S \rangle &= 0 \implies \text{Bound and scattering states are orthogonal}
 \end{aligned}$$

Problem 2.26

The Fourier transform of $\delta(x)$ is

$$\begin{aligned}
 \phi(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-ikx} \delta(x) dx = \frac{1}{\sqrt{2\pi}} e^{-ik0} \int_{-\infty}^{+\infty} \delta(x) dx \\
 \phi(k) &= \frac{1}{\sqrt{2\pi}}
 \end{aligned}$$

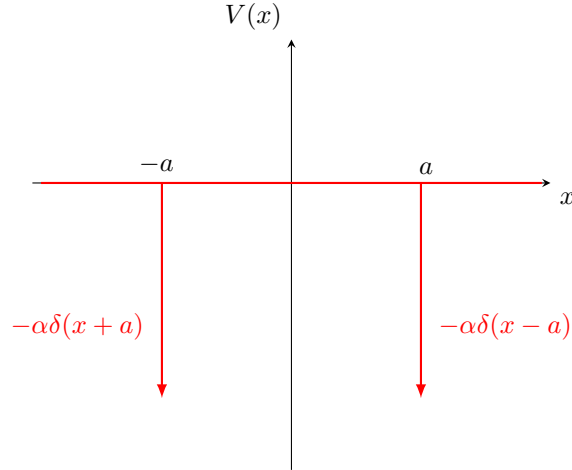
By Plancherel's Theorem, the inverse Fourier transform is,

$$\begin{aligned}
 \delta(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{ikx} dk \\
 \therefore \delta(x) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ikx} dk
 \end{aligned}$$

Problem 2.27

$$V(x) = -\alpha[\delta(x + a) + \delta(x - a)] \quad (\alpha, a > 0)$$

(a)



(b) For $x < -a$,

$$\begin{aligned}
 \partial_x^2 \psi &= -\frac{2mE}{\hbar^2} \psi = \kappa^2 \psi \\
 \text{where } \kappa^2 &= -\frac{2mE}{\hbar^2} \\
 \psi(x) &= Ae^{\kappa x} + Be^{-\kappa x} \\
 \psi(x) &= Ae^{\kappa x} \quad (x < -a) \quad (\because \text{for normalizability})
 \end{aligned}$$

For $|x| < a$,

$$\begin{aligned}\partial_x^2 \psi &= \kappa^2 \psi \\ \psi(x) &= Ce^{\kappa x} + De^{-\kappa x} \quad (|x| < a)\end{aligned}$$

For $x > a$,

$$\begin{aligned}\partial_x^2 \psi &= \kappa^2 \psi \\ \psi(x) &= Fe^{-\kappa x} + Ge^{\kappa x} \\ \psi(x) &= Fe^{-\kappa x} \quad (x > a) \quad (\because \text{for normalizability}) \\ \psi(x) &= \begin{cases} Ae^{\kappa x}, & x < -a \\ Ce^{\kappa x} + De^{-\kappa x}, & |x| < a \\ Fe^{-\kappa x}, & x > a \end{cases} \\ \psi'(x) &= \begin{cases} \kappa Ae^{\kappa x}, & x < -a \\ \kappa[Ce^{\kappa x} - De^{-\kappa x}], & |x| < a \\ -\kappa Fe^{-\kappa x}, & x > a \end{cases}\end{aligned}$$

Applying the continuity requirement at $\pm a$,

$$\begin{aligned}Ae^{-\kappa a} &= Ce^{-\kappa a} + De^{\kappa a} \implies De^{2\kappa a} = A - C \\ Fe^{-\kappa a} &= Ce^{\kappa a} + De^{-\kappa a} \implies Ce^{2\kappa a} = F - D \\ \Delta\psi'|_{\pm a} &= -\frac{2m\alpha}{\hbar^2}\psi(\pm a) \\ De^{2\kappa a} + A - C &= -\frac{2m\alpha}{\kappa\hbar^2}A \implies De^{2\kappa a} = -\frac{m\alpha}{\kappa\hbar^2}A \\ Ce^{2\kappa a} + F - D &= -\frac{2m\alpha}{\kappa\hbar^2}F \implies Ce^{2\kappa a} = -\frac{m\alpha}{\kappa\hbar^2}F\end{aligned}$$

From all the above relations, we obtain the following equation:

$$\left(\frac{m\alpha}{\kappa\hbar^2} - 1\right) Ae^{2\kappa a} = \frac{m\alpha}{\kappa\hbar^2} F$$

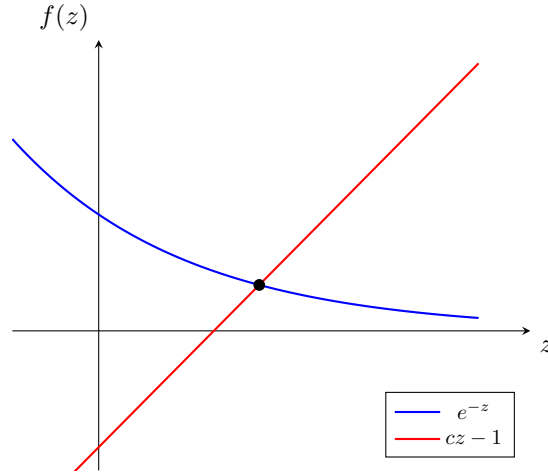
Since the potential is even, the wave function is either odd or even.

Even Solution

If the wave function is even, then

$$\begin{aligned}A &= F \\ e^{-2\kappa a} &= \frac{\kappa\hbar^2}{m\alpha} - 1 \\ z \equiv 2\kappa a, \quad c \equiv \frac{\hbar^2}{2m\alpha a} \\ \boxed{\therefore e^{-z} = cz - 1}\end{aligned}$$

This is a transcendental equation, the graphical solution is given below.



There will always be one even bound state for any valid value of c .

Odd Solution

If the wave function is odd, then

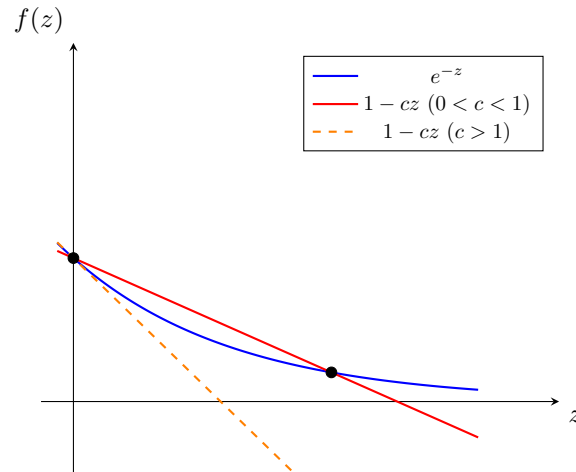
$$A = -F$$

$$e^{-2\kappa a} = 1 - \frac{\kappa \hbar^2}{m\alpha}$$

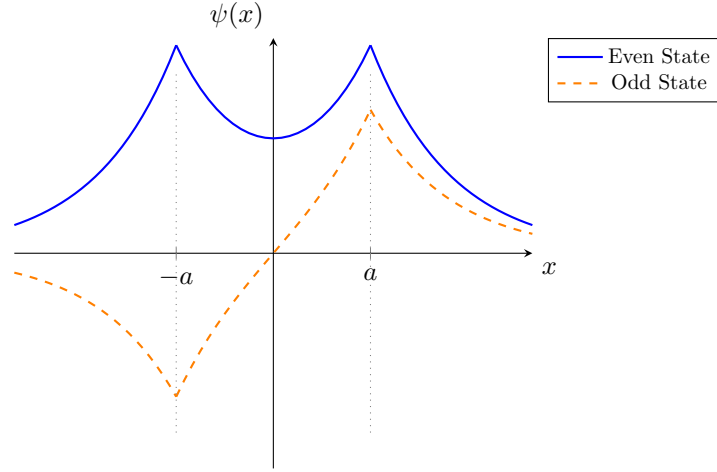
$$z \equiv 2\kappa a, \quad c \equiv \frac{\hbar^2}{2m\alpha a}$$

$\therefore e^{-z} = 1 - cz$

The graphical solution is given below.



For $0 < c < 1$, we will have a odd bound state, but no odd bound state $c > 1$. $z = 0$ is a trivial solution for both, but that does not represent any particle, since it leads to $E = 0$.



$$\text{For } \alpha = \frac{\hbar^2}{ma}, \quad c = \frac{\hbar^2}{2m\alpha a} = \frac{\hbar^2}{2ma(\hbar^2/ma)} = \frac{1}{2}$$

$$\text{Even state} \implies e^{-z} = \frac{z}{2} - 1$$

$$z \approx 2.218 = \frac{2a\sqrt{-2mE}}{\hbar}$$

$$E = -\frac{0.615\hbar^2}{ma^2}$$

$$\text{Odd State} \implies e^{-z} = 1 - \frac{z}{2}$$

$$z \approx 1.594 = \frac{2a\sqrt{-2mE}}{\hbar}$$

$$E = -\frac{0.318\hbar^2}{ma^2}$$

$$\text{For } \alpha = \frac{\hbar^2}{4ma}, \quad c = \frac{\hbar^2}{2m\alpha a} = \frac{\hbar^2}{2ma(\hbar^2/4ma)} = 2$$

$$\text{Even State} \implies e^{-z} = 2z - 1$$

$$z \approx 0.739 = \frac{2a\sqrt{-2mE}}{\hbar}$$

$$E = -\frac{0.068\hbar^2}{ma^2}$$

Since $c > 1$, no odd bound state exists.

(c) (i) In the limiting case $a \rightarrow 0$,

$$\lim_{a \rightarrow 0} e^{-2\kappa a} = \pm \left(1 - \frac{\kappa \hbar^2}{m\alpha}\right)$$

$$\implies 1 = \pm \left(1 - \frac{\kappa \hbar^2}{m\alpha}\right)$$

$$\text{Odd state} \implies \frac{\kappa \hbar^2}{m\alpha} = 0 \implies \kappa = 0 \implies E = 0$$

This is not a physically realisable state

$$\text{Even state} \implies \frac{\kappa \hbar^2}{m\alpha} = 2 \implies \kappa = \frac{2m\alpha}{\hbar^2} \implies E = -\frac{2m\alpha^2}{\hbar^2}$$

The potential for the condition becomes $V(x) = -2\alpha\delta(x)$

$\therefore$ This is the same energy as the single delta function well of strength 2α .

(ii) In the limiting case $a \rightarrow \infty$,

$$\lim_{a \rightarrow \infty} e^{-2\kappa a} = \pm \left(1 - \frac{\kappa \hbar^2}{m\alpha}\right)$$

$$0 = 1 - \frac{\kappa \hbar^2}{m\alpha} \implies \kappa = \frac{m\alpha}{\hbar^2}$$

$$E = -\frac{m\alpha^2}{2\hbar^2}$$

$\therefore$ This is the same energy as the single delta function well.

Problem 2.28

$$V(x) = -\alpha[\delta(x+a) + \delta(x-a)]$$

For the scattering state, $E > 0$

For $x < -a$,

$$\frac{d^2\phi}{dx^2} = -\frac{2mE}{\hbar^2}\phi = -k^2\phi$$

$$\text{where } k^2 \equiv \frac{2mE}{\hbar^2}$$

$$\phi(x) = Ae^{ikx} + Be^{-ikx}, (x < -a)$$

For $|x| < a$,

$$\frac{d^2\phi}{dx^2} = -k^2\phi$$

$$\psi(x) = Ce^{ikx} + De^{-ikx}, (|x| < a)$$

For $x > a$,

$$\frac{d^2\phi}{dx^2} = -k^2\phi$$

$$\phi(x) = Fe^{ikx}, (x > a) \text{ (Assuming no leftward wave)}$$

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < -a \\ Ce^{ikx} + De^{-ikx}, & |x| < a \\ Fe^{ikx}, & x > a \end{cases}$$

$$\phi'(x) = \begin{cases} ik[Ae^{ikx} - Be^{-ikx}], & x < -a \\ ik[Ce^{ikx} - De^{-ikx}], & |x| < a \\ ikFe^{ikx}, & x > a \end{cases}$$

Applying the continuity conditions at $\pm a$,

$$\begin{aligned} Ae^{-ika} + Be^{ika} &= Ce^{-ika} + De^{ika} \\ Ce^{ika} + De^{-ika} &= Fe^{ika} \\ \Delta\phi'|_{\pm a} &= -\frac{2m\alpha}{\hbar^2}\phi(\pm a) \\ ik[Ce^{-ika} - De^{ika} - (Ae^{-ika} - Be^{ika})] &= -\frac{2m\alpha}{\hbar^2}(Ce^{-ika} + De^{ika}) \\ \Rightarrow ik[A - C]e^{-ika} &= \frac{m\alpha}{\hbar^2}(Ce^{-ika} + De^{ika}) \\ \Rightarrow A &= C(1 - i\frac{m\alpha}{\hbar^2 k}) - i\frac{m\alpha}{\hbar^2 k}De^{2ika} \\ ik[Fe^{ika} - (Ce^{ika} - De^{-ika})] &= -\frac{2m\alpha}{\hbar^2}Fe^{ika} \\ ik[F - C + De^{-2ika}] &= -\frac{2m\alpha}{\hbar^2}F \rightarrow C = (1 + i\frac{m\alpha}{\hbar^2 k})F \\ D = (F - C)e^{2ika} &= -i\frac{m\alpha}{\hbar^2 k}Fe^{2ika}A = F - (\frac{i m\alpha}{\hbar^2 k}e^{2ika})^2 F = F(1 + \frac{m^2\alpha^2 e^{4ika}}{\hbar^4 k^2}) \\ \frac{F}{A} &= \frac{\hbar^4 k^2}{\hbar^4 k^2 + m^2\alpha^2 e^{4ika}} \\ T = |\frac{F}{A}|^2 &= \frac{\hbar^4 k^2}{\hbar^4 k^2 + m^2\alpha^2 e^{4ika}} \cdot \frac{\hbar^4 k^2}{\hbar^4 k^2 + m^2\alpha^2 e^{-4ika}} \\ T &= \frac{\hbar^8 k^4}{\hbar^8 k^4 + m^2\alpha^2 \hbar^4 k^2 (e^{4ika} + e^{-4ika}) + m^4\alpha^4} \\ T &= \frac{\hbar^8 k^4}{\hbar^8 k^4 + m^4\alpha^4 + 2m^2\alpha^2 \hbar^4 k^2 \cos(4ka)} \\ T &= \frac{1}{1 + (\frac{m\alpha}{\hbar^2 k})^4 + 2(\frac{m\alpha}{\hbar^2 k})^2 \cos(4ka)} \\ \text{Let } \beta &\equiv \frac{m\alpha}{\hbar^2 k}, \gamma \equiv 4ka \\ \therefore T &= \frac{1}{1 + \beta^4 + 2\beta^2 \cos \gamma} \end{aligned}$$

Problem 2.29

For the odd solutions, we have

$$\psi(x) = \begin{cases} -Fe^{\kappa x}, & x < -a \\ C \sin(lx), & |x| < a \\ Fe^{-\kappa x}, & x > a \end{cases}$$

$$\psi'(x) = \begin{cases} -F\kappa e^{\kappa x}, & x < -a \\ Cl \cos(lx), & |x| < a \\ -F\kappa e^{-\kappa x}, & x > a \end{cases}$$

Imposing the continuity of ψ and ψ' at a ,

$$Fe^{-\kappa a} = C \sin(la)$$

$$-\kappa Fe^{-\kappa a} = Cl \cos(la)$$

Dividing the two equations, we get

$$\kappa = -l \cot(la)$$

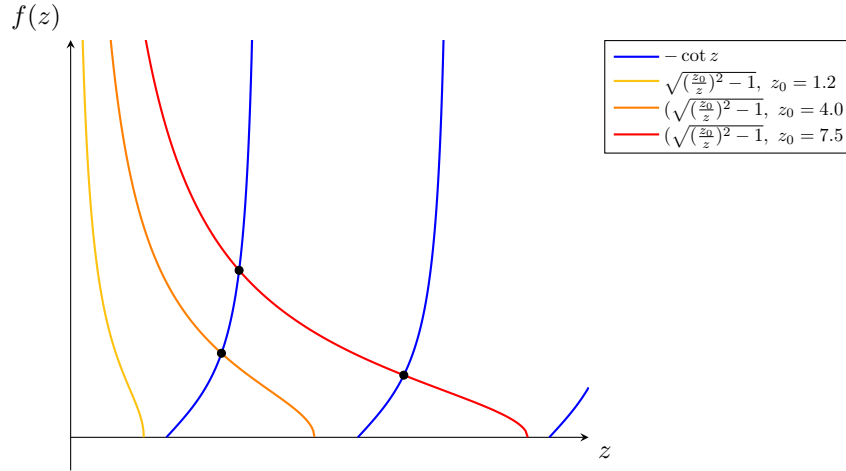
And from the relation between κ and l ,

$$\kappa^2 + l^2 = \frac{2mV_0}{\hbar^2}$$

We get the following by taking $z \equiv la$ and $z_0 \equiv \frac{a}{\hbar} \sqrt{2mV_0}$,

$$-\cot z = \sqrt{\left(\frac{z_0}{z}\right)^2 - 1}$$

This is a transcendental equation for z , and can be solved numerically/graphically. The plot of the two functions is given below.



For values of $z_0 < \frac{\pi}{2}$, there is no odd bound state. So for a shallow and narrow well, there is no bound state. For the case of a wide and deep well, the intersections will happen at the asymptotes, i.e. at $z \approx n\pi$, where $n = 2, 4, 6, \dots$. Since $z = la$, then

$$la \approx n\pi \implies E_n + V_0 = \frac{n^2 \pi^2 \hbar^2}{2m(2a)^2}, \quad n = 2, 4, 6, \dots$$

$E + V_0$ has the same energy as a infinite square well of width $2a$, and this is the other half the energies that we obtained from the even solutions.

Problem 2.30

$$\psi(x) = \begin{cases} Fe^{\kappa x}, & x < -a \\ D \cos(lx), & |x| < a \\ Fe^{-\kappa x}, & x > a \end{cases}$$

Normalizing the wave functions,

$$\int_{-\infty}^{+\infty} \psi^* \psi dx = 1$$

$$\begin{aligned}
2 \left(\int_0^a |D|^2 \cos^2(lx) dx + \int_a^{+\infty} |F|^2 e^{-2\kappa x} dx \right) &= 1 \\
2 \left(\frac{|D|^2}{2} \int_0^a \cos(2lx) + 1 dx + |F|^2 \int_a^{+\infty} e^{-2\kappa x} dx \right) &= 1 \\
|D|^2 \left(\frac{\sin(2la)}{2l} + a \right) \Big|_0^a - \frac{|F|^2 e^{-2\kappa x}}{\kappa} \Big|_a^{+\infty} &= 1 \\
|D|^2 \left(\frac{\sin(2la)}{2l} + a \right) + |F|^2 \frac{e^{-2\kappa a}}{\kappa} &= 1
\end{aligned}$$

From imposing continuity, we got

$$\begin{aligned}
F &= D e^{\kappa a} \cos(la), \quad \kappa = l \tan(la) \\
|D|^2 \left(\frac{1}{l} \cos(la) \sin(la) + a \right) + \frac{|D|^2 \cos^2(la)}{\kappa} &= 1 \\
|D|^2 \left(\frac{\sin^2(la)}{\kappa} + a + \frac{\cos^2(la)}{\kappa} \right) = 1 &\implies |D|^2 \left(\frac{1}{\kappa} + a \right) = 1 \\
\therefore |D| &= \frac{1}{\sqrt{a + \kappa^{-1}}} \\
|F| &= \frac{e^{\kappa a} \cos(la)}{\sqrt{a + \kappa^{-1}}}
\end{aligned}$$

Problem 2.31

Finite well potential is,

$$V(x) = \begin{cases} 0 & , |x| > a \\ -V_0 & , |x| < a \end{cases}$$

The limiting case of $V_0 \rightarrow \infty$ and $a \rightarrow 0$ while $2aV_0 = \alpha$ becomes the delta function well $V(x) = -\alpha\delta(x)$. The even solution for the energy in the finite well is,

$$\tan z = \sqrt{\left(\frac{z_0}{z}\right)^2 - 1}$$

$$\text{where } z_0 \equiv \frac{a}{\hbar} \sqrt{2mV_0}, \quad z \equiv la = \frac{a}{\hbar} \sqrt{2m(E + V_0)}$$

For this limiting case, $z_0, z \rightarrow 0$

$$z \tan z = \sqrt{z_0^2 - z^2}$$

For a small z , $\tan z \approx z$

$$\begin{aligned}
z^2 &= \sqrt{z_0^2 - z^2} \\
z^4 &= z_0^2 - z^2 = -\frac{2mEa^2}{\hbar^2} \\
\frac{a^4 4m^2 V_0^2}{\hbar^4} &= -\frac{2mEa^2}{\hbar^2} \\
E &= -\frac{2ma^2 V_0^2}{\hbar^2} = -\frac{m(2aV_0)^2}{2\hbar^2} \\
\therefore E &= -\frac{m\alpha^2}{2\hbar^2}
\end{aligned}$$

Therefore, taking the delta function well as a limiting case of a finite potential well is valid. For the scattering state transmission coefficient in the finite well case,

$$T^{-1} = 1 + \frac{V_0^2}{4E(E + V_0)} \sin^2 \left(\frac{2a}{\hbar} \sqrt{2m(E + V_0)} \right)$$

For a small x , $\sin x \approx x$, since $a \rightarrow 0$

$$\begin{aligned} T^{-1} &= 1 + \frac{V_0^2}{4E(E + V_0)} \left(\frac{2a}{\hbar} \sqrt{2m(E + V_0)} \right)^2 \\ T^{-1} &= 1 + \frac{V_0^2}{4E(E + V_0)} \frac{4a^2}{\hbar^2} (2m(E + V_0)) \\ T^{-1} &= 1 + \frac{m(2aV_0)^2}{2\hbar^2 E} = 1 + \frac{m\alpha^2}{2\hbar^2 E} \\ \therefore T &= \frac{1}{1 + (m\alpha^2/2\hbar^2 E)} \end{aligned}$$

This is consistent with the transmission coefficient in the delta function well.

Problem 2.32

From imposing continuity on ϕ and ϕ' ,

$$Ae^{-ika} + Be^{ika} = C \sin(la) + D \cos(la) \quad (1)$$

$$ik[Ae^{-ika} - Be^{ika}] = l[C \cos(la) - D \sin(la)] \quad (2)$$

$$C \sin(la) + D \cos(la) = Fe^{ika} \quad (3)$$

$$C \cos(la) - D \sin(la) = \frac{ik}{l} Fe^{ika} \quad (4)$$

From equations (3) and (4),

$$\begin{aligned} C &= Fe^{ika} \left(\sin(la) + \frac{ik}{l} \cos(la) \right) \\ D &= Fe^{ika} \left(\cos(la) - \frac{ik}{l} \sin(la) \right) \end{aligned}$$

From equations (1) and (2),

$$\begin{aligned} 2Ae^{-ika} &= C \left(\sin(la) - \frac{il}{k} \cos(la) \right) + D \left(\cos(la) - \frac{ik}{l} \sin(la) \right) \\ A &= \frac{Fe^{2ika}}{2} \left[2(\cos^2(la) + \sin^2(la)) + i \sin(2la) \left(\frac{k}{l} + \frac{l}{k} \right) \right] \\ \frac{A}{F} &= e^{2ika} \left[1 + \frac{1}{2} i \sin(2la) \left(\frac{l^2 + k^2}{lk} \right) \right] \end{aligned}$$

$$T^{-1} = \left| \frac{A}{F} \right|^2 = 1 + \frac{1}{4} \left(\frac{l^2 + k^2}{lk} \right)^2 \sin^2(2la)$$

Since $k^2 = \frac{2mE}{\hbar^2}$ and $l^2 = \frac{2m(E+V_0)}{\hbar^2}$,

$$\begin{aligned} T^{-1} &= 1 + \frac{(2m(E + V_0)/\hbar^2 - 2mE/\hbar^2)^2}{4(2m/\hbar^2)^2 E(E + V_0)} \sin^2 \left(\frac{2a}{\hbar} \sqrt{2m(E + V_0)} \right) \\ \therefore T^{-1} &= 1 + \frac{V_0^2}{4E(E + V_0)} \sin^2 \left(\frac{2a}{\hbar} \sqrt{2m(E + V_0)} \right) \end{aligned}$$

Problem 2.33

$$V(x) = \begin{cases} 0, & |x| \geq a \\ V_0, & |x| \leq a \end{cases} \quad a, V_0 > 0$$

For $x \leq -a$,

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} &= E\psi \implies \frac{\partial^2 \psi}{\partial x^2} = -\alpha^2 \psi \\ \text{where } \alpha^2 &\equiv \frac{2mE}{\hbar^2} \\ \psi(x) &= Ae^{i\alpha x} + Be^{-i\alpha x} \quad (x \leq -a) \end{aligned}$$

For $|x| < a$,

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V_0 \psi &= E\psi \implies \frac{\partial^2 \psi}{\partial x^2} = \frac{2m}{\hbar^2} (V_0 - E) \psi \\ \beta^2 &= \frac{2m}{\hbar^2} |V_0 - E| \\ \frac{\partial^2 \psi}{\partial x^2} &= \begin{cases} \beta^2 \psi, & E < V_0 \\ 0, & E = V_0 \\ -\beta^2 \psi, & E > V_0 \end{cases} \implies \psi(x) = \begin{cases} Ce^{\beta x} + De^{-\beta x}, & E < V_0 \\ Cx + D, & E = V_0 \\ Ce^{i\beta x} + De^{i\beta x}, & E > V_0 \end{cases} \quad (|x| < a) \end{aligned}$$

For $x \geq a$, and assuming there no wave from the left,

$$\frac{\partial^2 \psi}{\partial x^2} = -\alpha^2 \psi \implies \psi(x) = Fe^{i\alpha x} \quad (x \geq a) \quad (5)$$

For $E > V_0$,

$$\begin{aligned} \psi(x) &= \begin{cases} Ae^{i\alpha x} + Be^{-i\alpha x}, & x \leq -a \\ Ce^{i\beta x} + De^{-i\beta x}, & |x| < a \\ Fe^{i\alpha x}, & x \geq a \end{cases} \\ \frac{d\psi}{dx}(x) &= \begin{cases} i\alpha[Ae^{i\alpha x} - Be^{-i\alpha x}], & x \leq -a \\ i\beta[Ce^{i\beta x} - De^{-i\beta x}], & |x| < a \\ i\alpha Fe^{i\alpha x}, & x \geq a \end{cases} \end{aligned}$$

Applying the continuity of ψ and $\frac{d\psi}{dx}$ at $\pm a$,

$$\begin{aligned} Ae^{-i\alpha a} + Be^{i\alpha a} &= Ce^{-i\alpha a} + De^{i\alpha a} \\ i\alpha[Ae^{-i\alpha a} - Be^{i\alpha a}] &= i\beta[Ce^{-i\alpha a} - De^{i\alpha a}] \\ Ce^{i\beta a} + De^{-i\beta a} &= Fe^{i\alpha a} \\ i\beta[Ce^{i\beta a} - De^{-i\beta a}] &= i\alpha Fe^{i\alpha a} \end{aligned}$$

From the above equations, we get that

$$\begin{aligned}
C &= \left(1 + \frac{\alpha}{\beta}\right) \frac{1}{2} F e^{i(\alpha-\beta)a}, \quad D = \left(1 - \frac{\alpha}{\beta}\right) \frac{1}{2} F e^{i(\alpha+\beta)a} \\
A &= \frac{F}{4\alpha\beta} [(\alpha + \beta)^2 e^{i(\alpha-\beta)a} - (\alpha - \beta)^2 e^{i(3\alpha+\beta)a}] \\
T = \left| \frac{F}{A} \right|^2 &\Rightarrow \frac{1}{T} = \frac{1}{16\alpha^2\beta^2} [(\alpha + \beta)^4 + (\alpha - \beta)^4 - 2(\alpha^2 - \beta^2)^2 \cos(2a(\alpha + \beta))] \\
&\Rightarrow T^{-1} = 1 + \frac{V_0^2}{8E(E - V_0)} (1 - \cos(2a(\alpha + \beta))) \\
&\boxed{\therefore T^{-1} = 1 + \frac{V_0^2}{4E(E - V_0)} \sin^2((\alpha + \beta)a)}
\end{aligned}$$

For $E = V_0$,

$$\begin{aligned}
\psi(x) &= \begin{cases} A e^{i\alpha x} + B e^{-i\alpha x}, & x \leq -a \\ Cx + D, & |x| < a \\ F e^{i\alpha x}, & x \geq a \end{cases} \\
\frac{d\psi}{dx}(x) &= \begin{cases} i\alpha[A e^{i\alpha x} - B e^{-i\alpha x}], & x \leq -a \\ C, & |x| < a \\ i\alpha F e^{i\alpha x}, & x \geq a \end{cases}
\end{aligned}$$

Applying continuity of ψ and $\frac{d\psi}{dx}$ at $\pm a$,

$$\begin{aligned}
A e^{-i\alpha a} + B e^{i\alpha a} &= D - aC \\
F e^{i\alpha a} &= D + aC \\
i\alpha[A e^{-i\alpha a} - B e^{i\alpha a}] &= C \\
i\alpha F e^{i\alpha a} &= C
\end{aligned}$$

From the obtained relations, we get

$$\begin{aligned}
C &= i\alpha F e^{i\alpha a}, \quad D = (1 - i\alpha a) F e^{i\alpha a} \\
A &= (1 - i\alpha a) F e^{-2i\alpha a} \\
T^{-1} &= |A/F|^2 \\
&\boxed{\therefore T^{-1} = 1 + \alpha^2 a^2}
\end{aligned}$$

For $E < V_0$,

$$\begin{aligned}
\psi(x) &= \begin{cases} A e^{i\alpha x} + B e^{-i\alpha x}, & x \leq -a \\ C e^{\beta x} + D e^{-\beta x}, & |x| < a \\ F e^{i\alpha x}, & x \geq a \end{cases} \\
\frac{d\psi}{dx}(x) &= \begin{cases} i\alpha[A e^{i\alpha x} - B e^{-i\alpha x}], & x \leq -a \\ \beta[C e^{\beta x} - D e^{-\beta x}], & |x| < a \\ i\alpha F e^{i\alpha x}, & x \geq a \end{cases}
\end{aligned}$$

Applying continuity of ψ and $\frac{d\psi}{dx}$ at $\pm a$,

$$\begin{aligned} Ae^{-i\alpha a} + Be^{i\alpha a} &= Ce^{-\beta a} + De^{\beta a} \\ Ce^{\beta a} + De^{-\beta a} &= Fe^{i\alpha a} \\ i\alpha[Ae^{-i\alpha a} - Be^{i\alpha a}] &= \beta[Ce^{-\beta a} - De^{\beta a}] \\ \beta[Ce^{\beta a} - De^{-\beta a}] &= i\alpha Fe^{i\alpha a} \end{aligned}$$

From the derived relations, we get

$$\begin{aligned} C &= \left(1 + \frac{i\alpha}{\beta}\right) \frac{F}{2} e^{i\alpha a} e^{-\beta a}, \quad D = \left(1 - \frac{i\alpha}{\beta}\right) \frac{F}{2} e^{i\alpha a} e^{\beta a} \\ A &= \frac{1}{2} F e^{2i\alpha a} \left[\left(1 + \frac{i\alpha}{\beta}\right) e^{-2\beta a} + \left(1 - \frac{i\alpha}{\beta}\right) e^{2\beta a} \right] \\ T^{-1} &= |A/F|^2 = \frac{1}{2} \left(1 - \frac{\alpha^2}{\beta^2} + \left(1 + \frac{\alpha^2}{\beta^2}\right) \cosh(4\beta a)\right) \\ \therefore T^{-1} &= 1 + \frac{V_0}{4E(V_0 - E)} \sinh^2 \left(\frac{2a}{\hbar} \sqrt{2m(V_0 - E)} \right) \end{aligned}$$

Problem 2.34

$$V(x) = \begin{cases} 0, & x \leq 0 \\ V_0, & x > 0 \end{cases} \quad V_0 > 0$$

For $x \leq 0$,

$$\begin{aligned} -\frac{\hbar^2}{2m} \partial_x^2 \psi &= E\psi \Rightarrow \partial_x^2 \psi = -\frac{2mE}{\hbar^2} \psi \\ \partial_x^2 \psi &= -\alpha^2 \psi \text{ where } \alpha^2 \equiv \frac{2mE}{\hbar^2} \\ \psi(x) &= Ae^{i\alpha x} + Be^{-i\alpha x} \quad (x \leq 0) \end{aligned}$$

For $x > 0$

$$-\frac{\hbar^2}{2m} \partial_x^2 \psi + V_0 \psi = E\psi \Rightarrow \partial_x^2 \psi = \frac{2m}{\hbar^2} (V_0 - E) \psi$$

(a)

Case I - $E < V_0$

$$\begin{aligned} \partial_x^2 \psi &= \beta^2 \psi, \text{ where} \\ \beta^2 &= \frac{2m}{\hbar^2} (V_0 - E) \\ \psi(x) &= Ce^{-\beta x} \quad (x > 0) \quad (\because \psi \rightarrow 0 \text{ as } x \rightarrow \infty) \end{aligned}$$

$$\begin{aligned} \psi(x) &= \begin{cases} Ae^{i\alpha x} + Be^{-i\alpha x}, & x \leq 0 \\ Ce^{-\beta x}, & x > 0 \end{cases} \\ \psi'(x) &= \begin{cases} i\alpha [Ae^{i\alpha x} - Be^{-i\alpha x}], & x \leq 0 \\ -\beta Ce^{-\beta x}, & x > 0 \end{cases} \end{aligned}$$

Applying continuity of ψ and ψ' at $x = 0$,

$$\begin{aligned}
A + B &= C \\
i\alpha(A - B) &= -\beta C \implies A - B = \frac{i\beta}{\alpha}C \\
A &= \left(1 + \frac{i\beta}{\alpha}\right) \frac{C}{2}; \quad B = \left(1 - \frac{i\beta}{\alpha}\right) \frac{C}{2} \\
R &= \left|\frac{B}{A}\right|^2 = \left|\frac{1 - \frac{i\beta}{\alpha}}{1 + \frac{i\beta}{\alpha}}\right|^2 = \left|\frac{\alpha - i\beta}{\alpha + i\beta}\right|^2 \\
R &= \left(\frac{\alpha - i\beta}{\alpha + i\beta}\right) \left(\frac{\alpha - i\beta}{\alpha + i\beta}\right)^* = \frac{\alpha - i\beta}{\alpha + i\beta} \frac{\alpha + i\beta}{\alpha - i\beta} = 1
\end{aligned}$$

We obtain a reflection coefficient of 1 since $E < V_0$, and the wave $Ae^{i\alpha x}$ did not have enough energy E to cross over the step potential barrier V_0 , hence it reflects back completely.

(b)

Case II - $E > V_0$

$$\begin{aligned}
\partial_x^2 \psi &= -\beta^2 \psi, \text{ where } \beta^2 = \frac{2m(E - V_0)}{\hbar^2} \\
\psi(x) &= Fe^{i\beta x} \quad (x > 0), \text{ assuming no leftward wave} \\
\psi(x) &= \begin{cases} Ae^{i\alpha x} + Be^{-i\alpha x}, & x \leq 0 \\ Fe^{i\beta x}, & x > 0 \end{cases} \\
\psi'(x) &= \begin{cases} i\alpha[Ae^{i\alpha x} - Be^{-i\alpha x}], & x \leq 0 \\ i\beta Fe^{i\beta x}, & x > 0 \end{cases}
\end{aligned}$$

Applying the continuity of ψ and ψ' at $x = 0$,

$$\begin{aligned}
A + B &= F; \quad A - B = \frac{\beta}{\alpha}F \\
A &= \left(1 + \frac{\beta}{\alpha}\right) \frac{F}{2}; \quad B = \left(1 - \frac{\beta}{\alpha}\right) \frac{F}{2} \\
\frac{B}{A} &= \frac{1 - \frac{\alpha}{\beta}}{1 + \frac{\alpha}{\beta}} = \frac{\beta - \alpha}{\beta + \alpha} \\
\therefore R &= \left|\frac{B}{A}\right|^2 = \left(\frac{\beta - \alpha}{\beta + \alpha}\right)^2
\end{aligned}$$

(c)

$$\begin{aligned}
T &= 1 - R = \frac{J_t}{J_i} \\
J(x) &= \frac{\hbar}{2mi} (\psi^* \partial_x \psi - \psi \partial_x \psi^*) \\
J_t &= \frac{\hbar}{2mi} (F^* e^{-i\beta x} \cdot i\beta F e^{i\beta x} + F e^{i\beta x} \cdot i\beta F^* e^{-i\beta x}) \\
J_t &= \frac{\hbar}{2mi} (2i\beta |F|^2) = \frac{\hbar}{m} \beta |F|^2 \\
J_i &= \frac{\hbar}{2mi} (A^* e^{-i\alpha x} \cdot i\alpha A e^{i\alpha x} + A e^{i\alpha x} \cdot i\alpha A^* e^{-i\alpha x}) \\
J_i &= \frac{\hbar}{2mi} (2i\alpha |A|^2) = \frac{\hbar}{m} \alpha |A|^2 \\
T &= \frac{\beta}{\alpha} \left| \frac{F}{A} \right|^2 = \sqrt{\frac{E - V_0}{E}} \left| \frac{F}{A} \right|^2
\end{aligned}$$

For $E < V_0$,

$$\begin{aligned}
T &= 1 - R = 1 - 1 \\
\therefore T &= 0
\end{aligned}$$

(d) For $E > V_0$,

$$T = 1 - R = \frac{4\alpha\beta}{(\alpha + \beta)^2}$$

Problem 2.35

The potential drop is described by the potential

$$V(x) = \begin{cases} 0, & x < 0 \\ -V_0, & x > 0 \end{cases}$$

For the region $x < 0$,

$$\begin{aligned}
-\frac{\hbar^2}{2m} \partial_x^2 \psi &= E\psi \Rightarrow \partial_x^2 \psi = -k^2 \psi \\
\text{where } k^2 &\equiv \frac{2mE}{\hbar^2} \\
\psi(x) &= Ae^{ikx} + Be^{-ikx} \quad (x < 0)
\end{aligned}$$

For the region $x > 0$,

$$\begin{aligned}
-\frac{\hbar^2}{2m} \partial_x^2 \psi - V_0 \psi &= E\psi \Rightarrow \partial_x^2 \psi = -l^2 \psi \\
\text{where } l^2 &\equiv \frac{2m(E + V_0)}{\hbar^2} \\
\psi(x) &= Fe^{ilx} \quad (x > 0), \text{ assuming no leftward wave.}
\end{aligned}$$

$$\begin{aligned}
\psi(x) &= \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0 \\ Fe^{ilx}, & x > 0 \end{cases} \\
\psi'(x) &= \begin{cases} ik[Ae^{ikx} - Be^{-ikx}], & x < 0 \\ ilFe^{ilx}, & x > 0 \end{cases}
\end{aligned}$$

Imposing the continuity condition on ψ and ψ' at $x = 0$,

$$\begin{aligned}
 A + B &= F ; A - B = \frac{l}{k} F \\
 A &= \left(1 + \frac{l}{k}\right) \frac{F}{2} ; B = \left(1 - \frac{l}{k}\right) \frac{F}{2} \\
 R &= \left|\frac{B}{A}\right|^2 = \left(\frac{1 - l/k}{1 + l/k}\right)^2 = \left(\frac{k - l}{k + l}\right)^2 \\
 R &= \frac{k^2 + l^2 - 2kl}{k^2 + l^2 + 2kl} = \frac{2E + V_0 - 2\sqrt{E(E + V_0)}}{2E + V_0 + 2\sqrt{E(E + V_0)}} \\
 R &= \frac{2 + (V_0/E) - 2\sqrt{1 + (V_0/E)}}{2 + (V_0/E) + 2\sqrt{1 + (V_0/E)}}
 \end{aligned}$$

Let $\beta \equiv \frac{V_0}{E}$

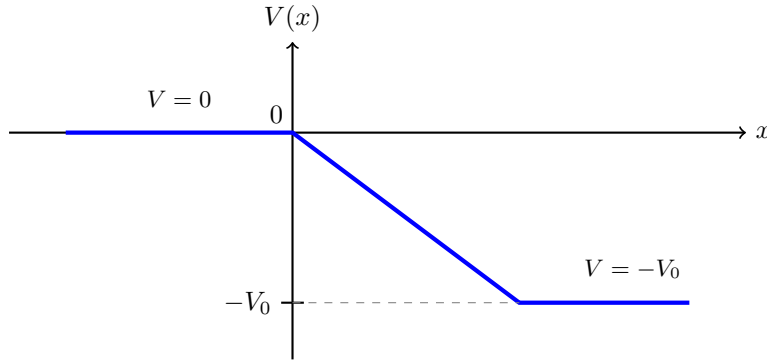
$$R = \frac{2 + \beta - 2\sqrt{1 + \beta}}{2 + \beta + 2\sqrt{1 + \beta}}$$

(a) If $E = V_0/3$, then $\beta = 3$

$$R = \frac{2 + 3 - 2\sqrt{1 + 3}}{2 + 3 + 2\sqrt{1 + 3}} = \frac{5 - 4}{5 + 4} = \frac{1}{9} \approx 11.11\%$$

It has a 11.11% chance of being reflected.

(b) This does not represent a cliff because of the discontinuous potential, it goes immediately from 0 to $-V_0$. However, in a true cliff, the potential must be a continuous decrease.



(c) Given that the potential drops from 0 to -120MeV when the neutron enters the nucleus, and a neutron of energy $E = 3\text{MeV}$ strikes such a nucleus, then

$$\begin{aligned}
 V_0 &= 120\text{MeV}, E = 3\text{MeV} \\
 \beta &= \frac{V_0}{E} = \frac{120\text{MeV}}{3\text{MeV}} = 40 \\
 R &= \frac{2 + 40 - 2\sqrt{1 + 40}}{2 + 40 + 2\sqrt{1 + 40}} = \frac{42 - 2\sqrt{41}}{42 + 2\sqrt{41}} \approx 52.05\% \\
 T &= 1 - R = 47.95\%
 \end{aligned}$$

The neutron has a 47.95% chance of being absorbed and initiating another fission.

3 Formalism

Problem 3.1

(a) The set $\mathcal{H}$ consists of all functions f on (a, b) such that:

$$\int_a^b |f(x)|^2 dx < \infty$$

For any scalar $c \in \mathbb{C}$ and any $f \in \mathcal{H}$:

$$\int_a^b |cf(x)|^2 dx = |c|^2 \int_a^b |f(x)|^2 dx$$

Since the integral of $|f|^2$ is finite, the resulting integral is finite, implying $cf \in \mathcal{H}$. For any $f, g \in \mathcal{H}$, we check if $f + g \in \mathcal{H}$ by expanding the integrand:

$$\int_a^b |f(x) + g(x)|^2 dx = \int_a^b (|f(x)|^2 + f(x)g^*(x) + g(x)f^*(x) + |g(x)|^2) dx$$

The integrals of $|f|^2$ and $|g|^2$ are finite by definition. The finiteness of the remaining terms is established by the Cauchy-Schwarz Inequality:

$$\left| \int_a^b f(x)g^*(x) dx \right| \leq \sqrt{\int_a^b |f(x)|^2 dx \int_a^b |g(x)|^2 dx}$$

Since both terms on the right are finite, the integral of the cross-terms is finite. Consequently:

$$\int_a^b |f(x) + g(x)|^2 dx < \infty$$

Thus, $f + g \in \mathcal{H}$. Because $\mathcal{H}$ is preserved under addition and scalar multiplication and inherits the algebraic properties of the general function space such as the existence of a zero element $0(x) = 0$ and additive inverses, $\mathcal{H}$ is a vector space.

(b) Consider the following operation on $f, g \in \mathcal{H}$ on (a, b) :

$$\langle f|g \rangle = \int_a^b f^* g dx$$

Notice that

$$\begin{aligned} \langle f|g \rangle^* &= \int_a^b (f^*)^* g^* dx = \int_a^b f g^* dx = \langle g|f \rangle \\ \therefore \langle g|f \rangle &= \langle f|g \rangle^* \end{aligned}$$

Now consider $\langle f|f \rangle$,

$$\begin{aligned} \langle f|f \rangle &= \int_a^b |f|^2 dx \geq 0 \quad (\because |f| \geq 0 \quad \forall f \in \mathcal{H}) \\ \therefore \langle f|f \rangle &\geq 0 \end{aligned}$$

Now consider $f, g, h \in \mathcal{H}$ and $\langle f|\alpha g + \beta h \rangle$ where $\alpha, \beta \in \mathbb{C}$

$$\begin{aligned} \langle f|\alpha g + \beta h \rangle &= \int_a^b f^* (\alpha g + \beta h) dx = \alpha \int_a^b f^* g dx + \beta \int_a^b f^* h dx \\ \therefore \langle f|\alpha g + \beta h \rangle &= \alpha \langle f|g \rangle + \beta \langle f|h \rangle \end{aligned}$$

Therefore $\langle f|g \rangle$ where $f, g \in \mathcal{H}$ obeys all the rules of an inner product on $\mathcal{H}$.

Problem 3.2

(a) The given function is $f(x) = x^\nu$. If $f \in \mathcal{H}$, then its square integral on $(0, 1)$ must converge.

$$\int_0^1 x^\nu dx = \left. \frac{x^{\nu+1}}{\nu+1} \right|_0^1 < \infty \quad (\nu \neq -1)$$

The integral diverges for $\nu < -1$. Therefore the $f(x) = x^\nu \in \mathcal{H}$ for $\nu > -1$.

(b) For the case $\nu = \frac{1}{2}$, the function $f(x) = x^\nu$ does exist in the Hilbert space since $\nu \geq -1$. The square integral of $xf(x) = x^{\frac{3}{2}}$ also converges and is in the Hilbert space. $D_x f = \frac{1}{2}x^{-1/2}$ also converges since $\nu > -1$.

Problem 3.3

Given that $\langle h|\hat{Q}h\rangle = \langle \hat{Q}h|h\rangle$, i.e. $\hat{Q} = \hat{Q}^\dagger \forall h \in \mathcal{H}$

Let $h = f + g$

$$\begin{aligned} \langle f + g|\hat{Q}(f + g)\rangle &= \langle \hat{Q}(f + g)|f + g\rangle \\ \implies \langle f|\hat{Q}f\rangle + \langle g|\hat{Q}g\rangle + \langle f|\hat{Q}g\rangle + \langle g|\hat{Q}f\rangle &= \langle \hat{Q}f|f\rangle + \langle \hat{Q}g|g\rangle + \langle \hat{Q}f|g\rangle + \langle \hat{Q}g|f\rangle \\ \implies \langle f|\hat{Q}g\rangle + \langle g|\hat{Q}f\rangle &= \langle \hat{Q}f|g\rangle + \langle \hat{Q}g|f\rangle \rightarrow (1) \end{aligned}$$

Now consider the case $h = f + ig$

$$\begin{aligned} \langle f + ig|\hat{Q}(f + ig)\rangle &= \langle \hat{Q}(f + ig)|f + ig\rangle \\ \implies \langle f|\hat{Q}f\rangle - i\langle g|\hat{Q}f\rangle + i\langle f|\hat{Q}g\rangle + \langle g|\hat{Q}g\rangle &= \langle \hat{Q}f|f\rangle + \langle \hat{Q}g|g\rangle + i\langle \hat{Q}f|g\rangle - i\langle \hat{Q}g|f\rangle \\ \implies \langle f|\hat{Q}g\rangle - \langle g|\hat{Q}f\rangle &= \langle \hat{Q}f|g\rangle - \langle \hat{Q}g|f\rangle \rightarrow (2) \end{aligned}$$

From eqns. (1) and (2), we finally get,

$$\begin{aligned} \langle f|\hat{Q}g\rangle &= \langle \hat{Q}f|g\rangle \\ \langle g|\hat{Q}f\rangle &= \langle \hat{Q}g|f\rangle \end{aligned}$$

Problem 3.4

(a) Let $\hat{A}$ and $\hat{B}$ be two hermitian operators, i.e. $\langle f|\hat{A}f\rangle = \langle \hat{A}f|f\rangle$ and $\langle \hat{B}f|f\rangle = \langle f|\hat{B}f\rangle \forall f$

Consider $\langle (\hat{A} + \hat{B})f|f\rangle = \langle \hat{A}f + \hat{B}f|f\rangle$

$$= \langle \hat{A}f|f\rangle + \langle \hat{B}f|f\rangle$$

Now consider $\langle f|(\hat{A} + \hat{B})f\rangle = \langle f|\hat{A}f\rangle + \langle f|\hat{B}f\rangle$

$$= \langle \hat{A}f|f\rangle + \langle \hat{B}f|f\rangle$$

Notice that,

$$\begin{aligned} \langle (\hat{A} + \hat{B})f|f\rangle &= \langle f|(\hat{A} + \hat{B})f\rangle \\ \implies \hat{A} + \hat{B} &\text{ is hermitian} \end{aligned}$$

(b) Let $\hat{Q}$ be a hermitian operator and $\alpha \in \mathbb{C}$. If $\alpha\hat{Q}$ is hermitian, then

$$\begin{aligned} \langle (\alpha\hat{Q})f|f\rangle &= \langle f|(\alpha\hat{Q})f\rangle \\ \alpha^* \langle \hat{Q}f|f\rangle &= \alpha \langle f|\hat{Q}f\rangle \\ \implies \alpha^* &= \alpha \end{aligned}$$

$\implies \alpha$ is a real number.

(c) Let $\hat{P}$ and $\hat{Q}$ be two hermitian operators. If $\hat{P}\hat{Q}$ is hermitian, then

$$\begin{aligned}\langle \hat{P}\hat{Q}f|f \rangle &= \langle f|\hat{P}\hat{Q}f \rangle \\ \implies \langle \hat{P}(\hat{Q}f)|f \rangle &= \langle f|\hat{P}(\hat{Q}f) \rangle \\ \implies \langle \hat{Q}f|\hat{P}f \rangle &= \langle \hat{P}f|\hat{Q}f \rangle \\ \implies \langle f|\hat{Q}\hat{P}f \rangle &= \langle \hat{Q}\hat{P}f|f \rangle \\ \implies \hat{P}\hat{Q} &= \hat{Q}\hat{P} \implies [\hat{P}, \hat{Q}] = 0\end{aligned}$$

(d)

$$\begin{aligned}\langle \hat{x}\phi|\phi \rangle &= \int_{-\infty}^{\infty} (x\phi)^* \phi dx = \int_{-\infty}^{\infty} x |\phi|^2 dx \\ \langle \phi|\hat{x}\phi \rangle &= \int_{-\infty}^{\infty} \phi^* x \phi dx = \int_{-\infty}^{\infty} x |\phi|^2 dx \\ \implies \hat{x} &\text{ is Hermitian}\end{aligned}$$

Consider the operator $\hat{Q} = \partial_{xx}$

$$\begin{aligned}\langle \hat{Q}\phi|\phi \rangle &= \int_{-\infty}^{\infty} (\partial_x^2 \phi)^* \phi dx = \phi^* \partial_x \phi|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} |\partial_x \phi|^2 dx \\ \langle \phi|\hat{Q}\phi \rangle &= \int_{-\infty}^{\infty} \phi^* \partial_x^2 \phi dx = \phi^* \partial_x \phi|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} |\partial_x \phi|^2 dx \\ \implies \hat{Q} &= \partial_{xx} \text{ is hermitian.}\end{aligned}$$

Consider $\hat{V} = V(x)$,

$$\begin{aligned}\langle \hat{V}\psi|\psi \rangle &= \int_{-\infty}^{\infty} (V(x)\psi)^* \psi dx = \int_{-\infty}^{\infty} V(x) |\psi|^2 dx = \langle \psi|\hat{V}\psi \rangle \\ \implies \hat{V} &\text{ is hermitian}\end{aligned}$$

Since a real scalar multiple of a hermitian operator is also hermitian, and the sum of two hermitian operators is also hermitian, then

$$\hat{H} = -\frac{\hbar^2}{2m}\partial_{xx} + V(x) \text{ is also hermitian.}$$

Problem 3.5

(a) $\hat{X} = x$

$$\begin{aligned}\langle \hat{X}f|f \rangle &= \langle f|\hat{X}f \rangle \\ \implies \int_{-\infty}^{\infty} x |f(x)|^2 dx &= \int_{-\infty}^{\infty} f^* x f dx \\ \implies \hat{X}^\dagger &= \hat{X} = x\end{aligned}$$

$\hat{Q} = i$

$$\begin{aligned}\langle if|f \rangle &= \langle f|i^\dagger f \rangle \implies i^* \langle f|f \rangle = \langle f|i^\dagger f \rangle \\ -i \langle f|f \rangle &= \langle f|i^\dagger f \rangle \\ \implies \hat{i}^\dagger &= -i\end{aligned}$$

$$\hat{D}_x = \frac{d}{dx}$$

$$\begin{aligned} \langle \hat{D}_x f | f \rangle &= \langle f | \hat{D}_x^\dagger f \rangle \\ \int_{-\infty}^{\infty} \left(\frac{df}{dx} \right)^* f dx &= \int_{-\infty}^{\infty} f^* (\hat{D}_x^\dagger f) dx \\ |f|^2 \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f^* \left(\frac{df}{dx} \right) dx &= \int_{-\infty}^{\infty} f^* (\hat{D}_x^\dagger f) dx \\ \int_{-\infty}^{\infty} f^* \left(-\frac{d}{dx} f \right) dx &= \int_{-\infty}^{\infty} f^* (\hat{D}_x^\dagger f) dx \\ \implies \hat{D}_x^\dagger &= -\hat{D}_x = -\frac{d}{dx} \end{aligned}$$

(b) Show that $(\hat{Q}\hat{R})^\dagger = \hat{R}^\dagger\hat{Q}^\dagger$ Proof:

$$\begin{aligned} \langle \hat{Q}\hat{R}f | g \rangle &= \langle g | (\hat{Q}\hat{R})^\dagger f \rangle \\ \langle (\hat{Q}\hat{R})f | g \rangle &= \langle \hat{Q}(\hat{R}f) | g \rangle = \langle \hat{R}f | \hat{Q}^\dagger g \rangle = \langle f | \hat{R}^\dagger \hat{Q}^\dagger g \rangle \\ \implies (\hat{Q}\hat{R})^\dagger &= \hat{R}^\dagger \hat{Q}^\dagger \text{ QED} \end{aligned}$$

Show that $(\hat{Q} + \hat{R})^\dagger = \hat{Q}^\dagger + \hat{R}^\dagger$ Proof:

$$\begin{aligned} \langle (\hat{Q} + \hat{R})^\dagger f | g \rangle &= \langle g | (\hat{Q} + \hat{R}) f \rangle \\ \langle g | (\hat{Q} + \hat{R}) f \rangle &= \langle g | \hat{Q}f \rangle + \langle g | \hat{R}f \rangle \\ &= \langle \hat{Q}^\dagger g | f \rangle + \langle \hat{R}^\dagger g | f \rangle = \langle (\hat{Q}^\dagger + \hat{R}^\dagger) g | f \rangle \\ \implies (\hat{Q} + \hat{R})^\dagger &= \hat{Q}^\dagger + \hat{R}^\dagger \text{ QED} \end{aligned}$$

Show that $(c\hat{Q})^\dagger = c^* \hat{Q}^\dagger$ Proof:

$$\begin{aligned} \langle (c\hat{Q})^\dagger f | g \rangle &= \langle g | c\hat{Q}f \rangle = c \langle g | \hat{Q}f \rangle \\ &= c \langle \hat{Q}^\dagger g | f \rangle = \langle c^* \hat{Q}^\dagger g | f \rangle \\ \implies (c\hat{Q})^\dagger &= c^* \hat{Q}^\dagger \text{ QED} \end{aligned}$$

(c)

$$\begin{aligned} \hat{a}_- &= \frac{1}{\sqrt{2\hbar m\omega}} (-i\hat{p} + m\omega\hat{x}) \\ \langle \hat{a}_- \phi | \psi \rangle &= \langle \frac{1}{\sqrt{2\hbar m\omega}} (-i\hat{p} + m\omega\hat{x}) \phi | \psi \rangle \\ &= \frac{1}{\sqrt{2\hbar m\omega}} (i\langle \hat{p}\phi | \psi \rangle + m\omega\langle \hat{x}\phi | \psi \rangle) \end{aligned}$$

Since $\hat{x}$ & $\hat{p}$ are observables, they must be hermitian,

$$\begin{aligned} &= \frac{1}{\sqrt{2\hbar m\omega}} (i\langle \phi | \hat{p}\psi \rangle + m\omega\langle \phi | \hat{x}\psi \rangle) \\ &= \langle \phi | \frac{1}{\sqrt{2\hbar m\omega}} (i\hat{p} + m\omega\hat{x}) \psi \rangle \\ \implies \hat{a}_-^\dagger &= \hat{a}_+ \end{aligned}$$

Problem 3.6

$$\begin{aligned}
\hat{Q} &= \partial_\phi^2 \\
f(\phi + 2\pi) &= f(\phi) \\
\langle \hat{Q}f | f \rangle &= \int_0^{2\pi} (\partial_\phi^2 f)^* f d\phi = \int_0^{2\pi} (\partial_\phi^2 f^*) f d\phi \\
&= [f \partial_\phi f^* - f^* \partial_\phi f]_0^{2\pi} - \int_0^{2\pi} |\partial_\phi f|^2 d\phi \\
\langle \hat{Q}f | f \rangle &= - \int_0^{2\pi} |\partial_\phi f|^2 d\phi \\
\langle f | \hat{Q}f \rangle &= \int_0^{2\pi} f^* \partial_\phi^2 f d\phi = - \int_0^{2\pi} |\partial_\phi f|^2 d\phi \\
\implies \langle f | \hat{Q}f \rangle &= \langle \hat{Q}f | f \rangle = - \int_0^{2\pi} |\partial_\phi f|^2 d\phi \\
\implies \hat{Q} &= \partial_\phi^2 \text{ is a hermitian operator}
\end{aligned}$$

The eigenvalue eqn is,

$$\begin{aligned}
\hat{Q}f &= qf \implies \partial_\phi^2 f = qf \\
f(\phi) &= \begin{cases} Ae^{i\sqrt{-q}\phi} + Be^{-i\sqrt{-q}\phi}, & q \leq 0 \\ Ae^{\sqrt{q}\phi} + Be^{-\sqrt{q}\phi}, & q > 0 \end{cases} \\
f(\phi + 2\pi) &= f(\phi) \implies q > 0 \text{ has no periodic solution} \\
q \leq 0 &\implies e^{i\sqrt{-q}2\pi} = 1 \implies \sqrt{-q} = n \\
q &= 0, -1, -4, -9, \dots
\end{aligned}$$

Yes, it is degenerate for $n > 0$ since both $e^{in\phi}$ and $e^{-in\phi}$ share the same eigenvalue.

Problem 3.7

(a) Given that $f(x)$ and $g(x)$ are degenerate eigenstates of $\hat{Q}$.

$$\hat{Q}f = qf \text{ and } \hat{Q}g = qg$$

Consider a linear combination of f and g .

$$\begin{aligned}
\hat{Q}(af + bg) &= a\hat{Q}f + b\hat{Q}g = aqf + bqg \\
\hat{Q}(af + bg) &= (af + bg)q
\end{aligned}$$

$\implies af + bg$ is also an eigenstate with eigenvalue q .

(b) $f(x) = e^x$, $g(x) = e^{-x}$ on $[-1, 1]$, $\hat{D} = \partial_x^2$

$$\begin{aligned}
\hat{D}f &= \partial_x^2(e^x) = e^x \implies \hat{D}f = f \implies |e_1\rangle = f \\
\hat{D}g &= \partial_x^2(e^{-x}) = e^{-x} \implies \hat{D}g = g \implies |e_2\rangle = g
\end{aligned}$$

By Gram-Schmidt orthogonalization,

$$\begin{aligned}
|e'_1\rangle &= \frac{|e_1\rangle}{\|e_1\|} \\
\|e_1\|^2 &= \langle e_1|e_1\rangle = \int_{-1}^1 e^{2x} dx = \left. \frac{e^{2x}}{2} \right|_{-1}^1 = \frac{e^2 - e^{-2}}{2} = \sinh(2) \\
\Rightarrow \|e_1\| &= \sqrt{\sinh(2)} \\
|e'_1\rangle &= \frac{e^x}{\sqrt{\sinh(2)}}
\end{aligned}$$

Now finding $|e'_2\rangle$:

$$\begin{aligned}
|e''_2\rangle &= |e_2\rangle - \langle e'_1|e_2\rangle |e'_1\rangle \\
\langle e'_1|e_2\rangle &= \int_{-1}^1 \frac{e^x}{\sqrt{\sinh(2)}} e^{-x} dx = \frac{1}{\sqrt{\sinh(2)}} \int_{-1}^1 1 dx = \frac{2}{\sqrt{\sinh(2)}} \\
|e''_2\rangle &= e^{-x} - \left(\frac{2}{\sqrt{\sinh(2)}} \right) \frac{e^x}{\sqrt{\sinh(2)}} = e^{-x} - \frac{2e^x}{\sinh(2)}
\end{aligned}$$

The normalized second basis vector is obtained by dividing $|e''_2\rangle$ by its norm.

Problem 3.8

(a) $\hat{Q} = i\partial_\phi$, $f(\phi + 2\pi) = f(\phi)$

$$\begin{aligned}
\hat{Q}f &= qf \\
i\partial_\phi f = qf &\Rightarrow i \frac{df}{d\phi} = qf \Rightarrow \int \frac{1}{f} df = \int -iq d\phi \\
f(\phi) &= Ae^{-iq\phi}
\end{aligned}$$

But $f(\phi + 2\pi) = f(\phi)$

$$\begin{aligned}
Ae^{-iq(\phi+2\pi)} &= Ae^{-iq\phi} \Rightarrow e^{-2i\pi q} = 1 \\
\Rightarrow q &= 0, \pm 1, \pm 2, \dots
\end{aligned}$$

Eigenvalues of $\hat{Q}$ are real.

$$\begin{aligned}
f_n(\phi) &= Ae^{-in\phi} \\
\langle f_n|f_m\rangle &= \int_0^{2\pi} |A|^2 e^{i(n-m)\phi} d\phi = \left. \frac{|A|^2 e^{i(n-m)\phi}}{i(n-m)} \right|_0^{2\pi} = 0 \quad (n \neq m) \\
\Rightarrow \langle f_n|f_m\rangle &= 0 \quad (n \neq m)
\end{aligned}$$

(b) $\hat{Q} = \partial_\phi^2$, $f(\phi + 2\pi) = f(\phi)$

$$\begin{aligned}
\partial_\phi^2 f &= qf \\
f_n(\phi) &= Ae^{in\phi} + Be^{-in\phi} \\
n &= 0, 1, 2, 3, \dots \\
\langle f_n|f_m\rangle &= \int_0^{2\pi} (A_n^* e^{-in\phi} + B_n^* e^{in\phi}) (A_m e^{im\phi} + B_m e^{-im\phi}) d\phi = 0 \quad (n \neq m) \\
\Rightarrow \langle f_n|f_m\rangle &= 0 \quad (n \neq m)
\end{aligned}$$

Problem 3.11

$$\begin{aligned}\Psi(x, t) &= \alpha e^{-\frac{m\omega}{2\hbar}x^2} e^{-i\omega t/2} \\ \Phi(p, t) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-\frac{ip}{\hbar}x} \Psi(x, t) dx \\ &= \frac{\alpha e^{-i\omega t/2}}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-\left(\frac{ip}{\hbar}x + \frac{m\omega}{2\hbar}x^2\right)} dx\end{aligned}$$

Completing the square in the exponent:

$$\frac{ip}{\hbar}x + \frac{m\omega}{2\hbar}x^2 = \left(\sqrt{\frac{m\omega}{2\hbar}}x + \frac{ip}{\sqrt{2\hbar m\omega}}\right)^2 + \frac{p^2}{2\hbar m\omega}$$

Let $y \equiv \sqrt{\frac{m\omega}{2\hbar}}x + \frac{ip}{\sqrt{2\hbar m\omega}} \implies dy = \sqrt{\frac{m\omega}{2\hbar}}dx$:

$$\begin{aligned}\Phi(p, t) &= \frac{\alpha e^{-i\omega t/2} \int_{-\infty}^{\infty} e^{-y^2} dy e^{-p^2/2\hbar m\omega}}{\sqrt{2\pi\hbar} \sqrt{\frac{m\omega}{2\hbar}}} \\ \Phi(p, t) &= \frac{\alpha e^{-i\omega t/2} e^{-p^2/2\hbar m\omega}}{\sqrt{m\omega}}\end{aligned}$$

Classical range boundary momentum occurs when:

$$\frac{1}{2}\hbar\omega = \frac{p^2}{2m} \implies p_c = \sqrt{\hbar m\omega}$$

The probability of finding the particle outside the classical region is:

$$P(|p| > p_c) = 2 \int_{p_c}^{\infty} |\Phi(p, t)|^2 dp = \frac{2\alpha^2}{m\omega} \int_{p_c}^{\infty} e^{-\frac{p^2}{\hbar m\omega}} dp$$

Let $z \equiv \frac{p}{\sqrt{\hbar m\omega}} \implies dp = \sqrt{\hbar m\omega} dz$:

$$\begin{aligned}P(|p| > p_c) &= \frac{2}{\sqrt{\pi}} \int_1^{\infty} e^{-z^2} dz \\ P(|p| > p_c) &= 0.16\end{aligned}$$

Problem 3.12

Given the free particle wave function:

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{i\left(kx - \frac{\hbar k^2}{2m}t\right)} dk$$

The momentum space wave function is found using the Fourier transform:

$$\begin{aligned}\Phi(p, t) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \Psi(x, t) e^{-\frac{ipx}{\hbar}} dx \\ \Phi(p, t) &= \frac{1}{2\pi\sqrt{\hbar}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \phi(k) e^{-i\frac{p}{\hbar}x} e^{ikx} e^{-i\frac{\hbar k^2}{2m}t} dk dx \\ \Phi(p, t) &= \frac{1}{2\pi\sqrt{\hbar}} \int_{-\infty}^{\infty} \phi(k) e^{-i\frac{\hbar k^2}{2m}t} \left[\int_{-\infty}^{\infty} e^{ix\left(k - \frac{p}{\hbar}\right)} dx \right] dk\end{aligned}$$

Since $\int_{-\infty}^{\infty} e^{ix(k-\frac{p}{\hbar})} dx = 2\pi\delta(k - \frac{p}{\hbar})$:

$$\begin{aligned}\Phi(p, t) &= \frac{1}{\sqrt{\hbar}} \int_{-\infty}^{\infty} \phi(k) e^{-i\frac{\hbar k^2}{2m}t} \delta\left(k - \frac{p}{\hbar}\right) dk \\ \Phi(p, t) &= \frac{1}{\sqrt{\hbar}} \phi\left(\frac{p}{\hbar}\right) e^{-\frac{ip^2 t}{2m\hbar}}\end{aligned}$$

Taking the absolute square shows it is time independent:

$$|\Phi(p, t)|^2 = \frac{1}{\hbar} \left| \phi\left(\frac{p}{\hbar}\right) \right|^2 \rightarrow \text{Time independent}$$

Problem 3.13

Using Plancherel's theorem form:

$$\begin{aligned}\Psi(x, t) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{\frac{ipx}{\hbar}} \Phi(p, t) dp \\ \langle x \rangle &= \langle \Psi | \hat{x} | \Psi \rangle = \int_{-\infty}^{\infty} \Psi^* x \Psi dx \\ &= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{ip'x}{\hbar}} \Phi^*(p', t) x e^{\frac{ipx}{\hbar}} \Phi(p, t) dp' dp dx\end{aligned}$$

Using the identity $xe^{\frac{ipx}{\hbar}} = -i\hbar\partial_p(e^{\frac{ipx}{\hbar}})$:

$$\langle x \rangle = \frac{-i}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{ip'x}{\hbar}} \Phi^*(p', t) \int_{-\infty}^{\infty} \left[\partial_p \left(e^{\frac{ipx}{\hbar}} \right) \right] \Phi(p, t) dp dp' dx$$

Integrating by parts with respect to p :

$$\begin{aligned}\langle x \rangle &= \frac{i}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Phi^*(p', t) \partial_p \Phi(p, t) e^{ix(\frac{p-p'}{\hbar})} dx dp dp' \\ \langle x \rangle &= i \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Phi^*(p', t) \partial_p \Phi(p, t) \hbar \delta(p - p') dp dp' \\ \langle x \rangle &= i\hbar \int_{-\infty}^{\infty} \partial_p \Phi(p, t) \int_{-\infty}^{\infty} \Phi^*(p', t) \delta(p - p') dp' dp \\ \langle x \rangle &= \int_{-\infty}^{\infty} \Phi^*(p, t) (i\hbar\partial_p) \Phi(p, t) dp\end{aligned}$$

Problem 3.14

(a)

$$\begin{aligned}[\hat{A} + \hat{B}, \hat{C}] &= (\hat{A} + \hat{B})\hat{C} - \hat{C}(\hat{A} + \hat{B}) \\ &= (\hat{A}\hat{C} - \hat{C}\hat{A}) + (\hat{B}\hat{C} - \hat{C}\hat{B}) \\ &= [\hat{A}, \hat{C}] + [\hat{B}, \hat{C}]\end{aligned}$$

(b)

$$\begin{aligned}[\hat{A}\hat{B}, \hat{C}] &= \hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B} \\ &= \hat{A}\hat{B}\hat{C} - \hat{A}\hat{C}\hat{B} + \hat{A}\hat{C}\hat{B} - \hat{C}\hat{A}\hat{B} \\ &= \hat{A}(\hat{B}\hat{C} - \hat{C}\hat{B}) + (\hat{A}\hat{C} - \hat{C}\hat{A})\hat{B} \\ &= \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}\end{aligned}$$

(c)

$$\begin{aligned}
[x^n, \hat{p}] f &= x^n (-i\hbar \partial_x) f + i\hbar \partial_x (x^n f) \\
&= -i\hbar x^n \partial_x f + i\hbar n x^{n-1} f + i\hbar x^n \partial_x f \\
&= i\hbar n x^{n-1} f \\
\implies [x^n, \hat{p}] &= i\hbar n x^{n-1}
\end{aligned}$$

Alternatively, for a general function $f(x)$:

$$\begin{aligned}
[f(x), \hat{p}] g &= f(-i\hbar \partial_x) g + i\hbar \partial_x (fg) \\
&= -i\hbar f \partial_x g + i\hbar f' g + i\hbar f \partial_x g \\
&= i\hbar f'(x) g \\
\implies [f(x), \hat{p}] &= i\hbar f'(x)
\end{aligned}$$

(d)

$$\begin{aligned}
[\hat{H}, \hat{a}_-] f &= \hat{H}(\hat{a}_- f) - \hat{a}_- (\hat{H} f) \\
\hat{H} &= \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right) \\
\hat{H}(\hat{a}_- f) &= \hbar\omega \left(\hat{a}_+ \hat{a}_- \hat{a}_- f + \frac{1}{2} \hat{a}_- f \right) \\
\hat{a}_- (\hat{H} f) &= \hbar\omega \left(\hat{a}_- \hat{a}_+ \hat{a}_- f + \frac{1}{2} \hat{a}_- f \right) \\
\implies [\hat{H}, \hat{a}_-] f &= \hbar\omega (\hat{a}_+ \hat{a}_- - \hat{a}_- \hat{a}_+) \hat{a}_- f
\end{aligned}$$

Since $[\hat{a}_-, \hat{a}_+] = 1$:

$$[\hat{H}, \hat{a}_-] = -\hbar\omega \hat{a}_-$$

Similarly,

$$[\hat{H}, \hat{a}_+] = \hbar\omega \hat{a}_+$$

Problem 3.15

From the generalized uncertainty principle:

$$\sigma_A^2 \sigma_B^2 \geq \left(\frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right)^2$$

Let $\hat{A} = \hat{x} = x$ and $\hat{B} = \hat{H} = -\frac{\hbar^2}{2m} \partial_x^2 + V(x)$:

$$\begin{aligned}
[\hat{x}, \hat{H}] f &= x \left(-\frac{\hbar^2}{2m} \partial_x^2 f + V f \right) - \left(-\frac{\hbar^2}{2m} \partial_x^2 (xf) + V xf \right) \\
&= -\frac{\hbar^2}{2m} x \partial_x^2 f + \frac{\hbar^2}{2m} \partial_x (f + x \partial_x f) \\
&= -\frac{\hbar^2}{2m} x \partial_x^2 f + \frac{\hbar^2}{2m} (2 \partial_x f + x \partial_x^2 f) \\
&= \frac{\hbar^2}{m} \partial_x = \frac{i\hbar}{m} (-i\hbar \partial_x) = \frac{i\hbar}{m} \hat{p}
\end{aligned}$$

Taking the expectation value:

$$\langle [\hat{x}, \hat{H}] \rangle = \frac{i\hbar}{m} \langle p \rangle$$

Substituting this into the uncertainty relation:

$$\begin{aligned} \sigma_x^2 \sigma_E^2 &\geq \left(\frac{1}{2i} \frac{i\hbar}{m} \langle p \rangle \right)^2 \\ \sigma_x \sigma_E &\geq \frac{\hbar |\langle p \rangle|}{2m} \end{aligned}$$

Problem 3.16

Consider two operators $\hat{Q}$ and $\hat{R}$ such that $[\hat{Q}, \hat{R}] \neq 0$. If they admit a complete set of common eigenfunctions, then let f be an eigenfunction:

$$\hat{Q}f = qf \quad \text{and} \quad \hat{R}f = rf$$

Applying both operators sequentially:

$$\begin{aligned} \hat{R}\hat{Q}f &= \hat{R}(qf) = q\hat{R}f = qrf \\ \hat{Q}\hat{R}f &= \hat{Q}(rf) = r\hat{Q}f = rqf \\ \implies \hat{Q}\hat{R}f - \hat{R}\hat{Q}f &= (qr - rq)f = 0 \\ \implies [\hat{Q}, \hat{R}]f &= 0 \end{aligned}$$

Since this must hold for a complete set of eigenfunctions, it implies $[\hat{Q}, \hat{R}] = 0$. Therefore, non-commuting operators cannot share a complete set of common eigenfunctions.

Problem 3.17

The differential equation for the wave function that minimizes the uncertainty relation is:

$$\begin{aligned} (-i\hbar\partial_x - \langle p \rangle) \Psi &= ia(x - \langle x \rangle) \Psi \\ -i\hbar\partial_x \Psi &= [ia(x - \langle x \rangle) + \langle p \rangle] \Psi \\ \partial_x \Psi &= \left[\frac{a}{\hbar} (\langle x \rangle - x) + \frac{i}{\hbar} \langle p \rangle \right] \Psi \\ \frac{d\Psi}{\Psi} &= \left[\frac{a}{\hbar} (\langle x \rangle - x) + \frac{i}{\hbar} \langle p \rangle \right] dx \end{aligned}$$

Integrating both sides:

$$\begin{aligned} \int \frac{d\Psi}{\Psi} &= \int \left[\frac{a}{\hbar} (\langle x \rangle - x) + \frac{i}{\hbar} \langle p \rangle \right] dx \\ \ln \Psi &= \frac{a}{\hbar} \left(x\langle x \rangle - \frac{x^2}{2} \right) + \frac{i}{\hbar} \langle p \rangle x + C' \\ \Psi &= B' e^{\frac{a}{\hbar} \left(x\langle x \rangle - \frac{x^2}{2} \right)} e^{\frac{i}{\hbar} \langle p \rangle x} \end{aligned}$$

Completing the square in the exponent for the real part:

$$x\langle x \rangle - \frac{x^2}{2} = -\frac{1}{2} (x^2 - 2x\langle x \rangle + \langle x \rangle^2) + \frac{\langle x \rangle^2}{2} = -\frac{1}{2} (x - \langle x \rangle)^2 + \frac{\langle x \rangle^2}{2}$$

Absorbing the constant term $e^{\frac{a\langle x \rangle^2}{2\hbar}}$ into a new normalization constant B :

$$\therefore \Psi(x) = B e^{-\frac{a}{2\hbar} (x - \langle x \rangle)^2} e^{\frac{i\langle p \rangle x}{\hbar}}$$

Problem 3.18

The generalized Ehrenfest principle, where Q is an observable, is

$$D_t\langle Q \rangle = \frac{i}{\hbar} \langle [\hat{H}, \hat{Q}] \rangle + \langle \partial_t \hat{Q} \rangle$$

(a) $Q = 1$

$$\begin{aligned} D_t\langle 1 \rangle &= \frac{i}{\hbar} \langle [\hat{H}, 1] \rangle + \langle \partial_t 1 \rangle \\ \therefore D_t \int_{-\infty}^{+\infty} |\Psi|^2 dx &= 0 \end{aligned}$$

(b) $Q = H = \frac{p^2}{2m} + V(x)$

$$\begin{aligned} D_t\langle H \rangle &= \frac{i}{\hbar} \langle [\hat{H}, \hat{H}] \rangle + \langle \partial_t \hat{H} \rangle \\ D_t\langle H \rangle &= \langle \partial_t \hat{H} \rangle \quad (\because [\hat{H}, \hat{H}] = 0) \end{aligned}$$

Consider $\langle \partial_t \hat{H} \rangle$,

$$\begin{aligned} \langle \partial_t \hat{H} \rangle &= \langle \Psi | \partial_t \hat{H} | \Psi \rangle \\ \langle \partial_t \hat{H} \rangle &= \int_{-\infty}^{+\infty} \Psi^* \partial_t \hat{H} \Psi dx \end{aligned}$$

And now consider $D_t\langle H \rangle$,

$$D_t\langle H \rangle = D_t\langle \Psi | \hat{H} | \Psi \rangle = D_t \int_{-\infty}^{+\infty} \Psi^* \hat{H} \Psi dx$$

From the Feynman/Leibniz Rule,

$$D_t\langle H \rangle = \int_{-\infty}^{+\infty} \partial_t (\Psi^* \hat{H} \Psi) dx = \int_{-\infty}^{+\infty} \partial_t \Psi^* \hat{H} \Psi + \Psi^* \partial_t \hat{H} \Psi dx$$

Now, equating the two, we obtain,

$$\int_{-\infty}^{+\infty} \partial_t \Psi^* \hat{H} \Psi dx = 0$$

From the Schrödinger Equation, $\hat{H}\Psi = i\hbar\partial_t\Psi$

$$\int_{-\infty}^{+\infty} |\partial_t \Psi|^2 dx = 0$$

or we can use $\partial_t \Psi^* = -\frac{i}{\hbar} \hat{H} \Psi^*$

$$\int_{-\infty}^{+\infty} |\hat{H} \Psi|^2 dx = 0$$

For a stationary state, $\partial_t \hat{H} \Psi = 0$, then

$$D_t\langle H \rangle = 0$$

It says that the energy is stationary in time.

(c) $Q = x$

$$\begin{aligned} D_t\langle x \rangle &= \frac{i}{\hbar} \langle [\hat{H}, \hat{x}] \rangle + \langle \partial_t \hat{x} \rangle \\ D_t\langle x \rangle &= \frac{i}{\hbar} \langle [\hat{H}, \hat{x}] \rangle \\ [\hat{H}, \hat{x}]f &= \left(-\frac{\hbar^2}{2m} \partial_{xx} + V \right) (xf) - x \left(-\frac{\hbar^2}{2m} \partial_{xx} f + Vf \right) \end{aligned}$$

$$\begin{aligned}
[\hat{H}, \hat{x}]f &= -\frac{\hbar^2}{2m}(\partial_{xx}(xf) - x\partial_{xx}f) \\
[\hat{H}, \hat{x}]f &= -\frac{\hbar^2}{2m}(x\partial_{xx}f + 2\partial_x f - x\partial_{xx}f) \\
\therefore [\hat{H}, \hat{x}] &= -\frac{\hbar^2}{m}\partial_x \\
\frac{i}{\hbar} \left\langle -\frac{\hbar^2}{m}\partial_x \right\rangle &= \frac{1}{m} \langle -i\hbar\partial_x \rangle = \frac{1}{m} \langle p \rangle \\
\therefore \langle p \rangle &= m D_t \langle x \rangle
\end{aligned}$$

(d) $Q = p$

$$\begin{aligned}
D_t \langle p \rangle &= \frac{i}{\hbar} \langle [\hat{H}, \hat{p}] \rangle + \langle \partial_t \hat{p} \rangle \\
[\hat{H}, \hat{p}]f &= \hat{H}\hat{p}f - \hat{p}\hat{H}f \\
[\hat{H}, \hat{p}]f &= \left(-\frac{\hbar^2}{2m}\partial_{xx} + V \right) (-i\hbar\partial_x f) - (-i\hbar\partial_x) \left(-\frac{\hbar^2}{2m}\partial_{xx}f + Vf \right) \\
[\hat{H}, \hat{p}]f &= \frac{i\hbar^3}{2m}\partial_{xx}f - i\hbar V\partial_x f - \frac{i\hbar^3}{2m}\partial_{xxx}f + i\hbar V\partial_x f + i\hbar f\partial_x V = i\hbar f\partial_x V \\
\therefore [\hat{H}, \hat{p}] &= i\hbar\partial_x V
\end{aligned}$$

Since $\partial_t \hat{p} = 0$,

$$\begin{aligned}
D_t \langle p \rangle &= \frac{i}{\hbar} \langle i\hbar\partial_x V \rangle \\
\therefore D_t \langle p \rangle &= \langle -\partial_x V \rangle
\end{aligned}$$

Problem 3.19

(a) The potential of the free particle is $V(x) = 0$. From the previous problem (3.18), we derived that:

$$D_t \langle p \rangle = \langle -\partial_x V \rangle \quad D_t \langle x \rangle = \frac{1}{m} \langle p \rangle$$

In this case, $-\partial_x V = 0$

$$D_t \langle p \rangle = 0 \implies \langle p \rangle = p_0 \quad (p_0 \text{ is a constant})$$

$$D_t \langle x \rangle = \frac{1}{m} \langle p \rangle = \frac{p_0}{m}$$

Therefore, $\langle x \rangle$ moves at a constant velocity $D_t \langle x \rangle = p_0/m$.

(b) The harmonic oscillator potential is $V(x) = \frac{1}{2}m\omega^2 x^2$. Then $-\partial_x V = -m\omega^2 x$.

$$D_t \langle p \rangle = \langle -m\omega^2 x \rangle = -m\omega^2 \langle x \rangle$$

$$D_t \langle x \rangle = \frac{1}{m} \langle p \rangle \implies D_{tt} \langle x \rangle = \frac{1}{m} D_t \langle p \rangle$$

$$D_{tt} \langle x \rangle = -\omega^2 \langle x \rangle$$

$$\langle x \rangle = A \cos(\omega t) + B \sin(\omega t)$$

Therefore, $\langle x \rangle$ oscillates at the classical frequency ω .

Problem 3.20

The initial wave function given is

$$\Psi(x, 0) = A[\psi_1(x) + \psi_2(x)]$$

From Problem 2.5, we got that

$$\Psi(x, t) = \frac{1}{\sqrt{a}} \left[\sin\left(\frac{\pi x}{a}\right) e^{-i\omega t} + \sin\left(\frac{2\pi x}{a}\right) e^{-4i\omega t} \right] \quad \left(\because \omega \equiv \frac{E_1}{\hbar} \right)$$

$$\langle x \rangle = \frac{a}{2} \left[1 - \frac{32}{9\pi^2} \cos(3\omega t) \right]$$

$$\langle H \rangle = \frac{1}{2}(E_1 + E_2) = \frac{5}{2}E_1 \quad \left(\because E_n = n^2 E_1 = \frac{n^2 \hbar^2 \pi^2}{2ma^2} \right)$$

Now, calculating $\langle x^2 \rangle$ and σ_x

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} x^2 |\Psi|^2 dx$$

$$\langle x^2 \rangle = \frac{1}{a} \int_0^a x^2 \sin^2\left(\frac{\pi x}{a}\right) + x^2 \sin^2\left(\frac{2\pi x}{a}\right) + 2x^2 \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{2\pi x}{a}\right) \cos(3\omega t) dx$$

$$I_1 = \int_0^a x^2 \sin^2\left(\frac{\pi x}{a}\right) dx = \frac{1}{2} \int_0^a x^2 - x^2 \cos\left(\frac{2\pi x}{a}\right) dx$$

$$I_1 = \frac{1}{2} \left[\frac{x^3}{3} - \frac{a}{2\pi} x^2 \sin\left(\frac{2\pi x}{a}\right) - \frac{a^2}{2\pi^2} x \cos\left(\frac{2\pi x}{a}\right) + \frac{a^3}{4\pi^3} \sin\left(\frac{2\pi x}{a}\right) \right] \Big|_0^a$$

$$I_1 = \frac{1}{2} \left[\frac{a^3}{3} - \frac{a^3}{2\pi} \sin(2\pi) - \frac{a^3}{2\pi^2} \cos(2\pi) + \frac{a^3}{4\pi^3} \sin(2\pi) \right] = a^3 \left[\frac{1}{6} - \frac{1}{4\pi^2} \right]$$

$$I_2 = \int_0^a x^2 \sin\left(\frac{2\pi x}{a}\right) dx = \frac{1}{2} \int_0^a x^2 - x^2 \cos\left(\frac{4\pi x}{a}\right) dx$$

$$I_2 = \frac{1}{2} \left[\frac{x^3}{3} - \frac{a}{4\pi} x^2 \sin\left(\frac{4\pi x}{a}\right) - \frac{a^2}{8\pi^2} x \cos\left(\frac{4\pi x}{a}\right) + \frac{a^3}{32\pi^3} \sin\left(\frac{4\pi x}{a}\right) \right] \Big|_0^a$$

$$I_2 = \frac{1}{2} \left[\frac{a^3}{3} - \frac{a^3}{4\pi} \sin(4\pi) - \frac{a^3}{8\pi^2} \cos(4\pi) + \frac{a^3}{32\pi^3} \sin(4\pi) \right] = a^3 \left[\frac{1}{6} - \frac{1}{16\pi^2} \right]$$

$$I_3 = 2 \int_0^a x^2 \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{2\pi x}{a}\right) dx = \int_0^a x^2 \cos\left(\frac{\pi x}{a}\right) - x^2 \cos\left(\frac{3\pi x}{a}\right) dx$$

$$I_3 = \frac{a}{\pi} x^2 \sin\left(\frac{\pi x}{a}\right) + \frac{2a^2}{\pi^2} x \cos\left(\frac{\pi x}{a}\right) - \frac{2a^3}{\pi^3} \sin\left(\frac{\pi x}{a}\right) + \frac{a}{3\pi} x^2 \sin\left(\frac{3\pi x}{a}\right) + \frac{2a^2}{9\pi^2} \cos\left(\frac{3\pi x}{a}\right) - \frac{2a^3}{27\pi^3} \sin\left(\frac{3\pi x}{a}\right) \Big|_0^a$$

$$I_3 = \frac{a^3}{\pi} \sin \pi + \frac{2a^3}{\pi^2} \cos \pi - \frac{2a^3}{\pi^3} \sin \pi + \frac{a^3}{3\pi} \sin 3\pi + \frac{2a^3}{9\pi^2} \cos 3\pi - \frac{2a^3}{27\pi^3} \sin 3\pi$$

$$I_3 = \frac{a^3}{\pi} \left(-1 - \frac{1}{3} + \frac{2}{\pi^2} + \frac{2}{27\pi^2} \right) = \frac{a^3}{\pi} \left(\frac{56}{27\pi^2} - \frac{4}{3} \right)$$

$$\langle x^2 \rangle = a^2 \left(\frac{1}{6} - \frac{1}{4\pi^2} + \frac{1}{6} - \frac{1}{16\pi^2} + \left(\frac{56}{27\pi^3} - \frac{4}{3\pi} \right) \cos 3\omega t \right)$$

$$\langle x^2 \rangle = a^2 \left(\frac{1}{3} - \frac{5}{16\pi^2} + \left(\frac{56}{27\pi^3} - \frac{4}{3\pi} \right) \cos 3\omega t \right)$$

$$\sigma_x = \sqrt{\langle x \rangle^2 - \langle x^2 \rangle}$$

$$\begin{aligned}
\sigma_x &= \frac{a}{2} \sqrt{\frac{1}{3} - \frac{5}{4\pi^2} - \frac{1024}{81\pi^4} \cos^2(3\omega t)} \\
\langle H^2 \rangle &= \langle \Psi | \hat{H}^2 \Psi \rangle = \left\langle \frac{1}{\sqrt{2}} [\Psi_1 + \Psi_2] \left| \frac{1}{\sqrt{2}} [\hat{H}^2 \Psi_1 + \hat{H}^2 \Psi_2] \right. \right\rangle \\
\langle H^2 \rangle &= \frac{1}{2} (\langle \Psi_1 | \hat{H}^2 \Psi_1 \rangle + \langle \Psi_1 | \hat{H}^2 \Psi_2 \rangle + \langle \Psi_2 | \hat{H}^2 \Psi_1 \rangle + \langle \Psi_2 | \hat{H}^2 \Psi_2 \rangle) \\
\langle H^2 \rangle &= \frac{1}{2} (E_1^2 (\langle \Psi_1 | \Psi_1 \rangle + \langle \Psi_2 | \Psi_1 \rangle) + E_2^2 (\langle \Psi_1 | \Psi_2 \rangle + \langle \Psi_2 | \Psi_2 \rangle)) \\
\langle H^2 \rangle &= \frac{1}{2} (E_1^2 + E_2^2) = \frac{17}{2} E_1^2 \\
\sigma_H &= \sqrt{\langle H \rangle^2 - \langle H^2 \rangle} \\
\sigma_H &= \frac{3}{2} E_1 \\
D_t \langle x \rangle &= \frac{32\hbar^2}{3\pi^2} \sin(3\omega t)
\end{aligned}$$

Now, from the Energy Time uncertainty principle,

$$\begin{aligned}
\sigma_H^2 \sigma_x^2 &\geq \frac{\hbar^2}{4} (D_t \langle x \rangle)^2 \\
\frac{9}{4} E_1^2 \frac{a^2}{2} \left(\frac{1}{3} - \frac{5}{4\pi^2} - \frac{1024}{81\pi^4} \cos^2(3\omega t) \right) &\geq \frac{\hbar^2}{4} \left(\frac{1024\hbar^2}{9\pi^4} \sin^2(3\omega t) \right) \\
\frac{9}{4} \frac{\hbar^4 \pi^4}{4m^2 a^4} \frac{a^2}{2} \left(\frac{1}{3} - \frac{5}{4\pi^2} - \frac{1024}{81\pi^4} \cos^2(3\omega t) \right) &\geq \frac{256\hbar^4}{9\pi^2} \sin^2(3\omega t) \\
\frac{1}{3} - \frac{5}{4\pi^2} &\geq \frac{1024}{81\pi^4} (\cos^2(3\omega t) + \sin^2(3\omega t)) \\
\frac{1}{3} - \frac{5}{4\pi^2} &\geq \frac{1024}{81\pi^4}
\end{aligned}$$

Therefore, the uncertainty principle holds.

Problem 3.21

Given expectation values:

$$\begin{aligned}
\langle x \rangle &= \frac{\hbar l t}{m} \implies \frac{d\langle x \rangle}{dt} = \frac{\hbar l}{m} \\
\langle x^2 \rangle &= \frac{1}{4a} + \frac{\hbar^2 a t^2}{m^2} + \frac{\hbar^2 l^2 t^2}{m^2} \\
\sigma_x^2 &= \frac{1}{4a} + \frac{\hbar^2 a t^2}{m^2} \implies \sigma_x = \sqrt{\frac{1}{4a} + \frac{\hbar^2 a t^2}{m^2}} \\
\langle p^2 \rangle &= \hbar^2 (l^2 + a) \\
\langle p^4 \rangle &= \hbar^4 (l^4 + 6l^2 a + 3a^2)
\end{aligned}$$

Compute Hamiltonian expectation values:

$$\begin{aligned}
\langle H \rangle &= \frac{\langle p^2 \rangle}{2m} = \frac{\hbar^2 (l^2 + a)}{2m} \\
\langle H^2 \rangle &= \frac{\langle p^4 \rangle}{4m^2} = \frac{\hbar^4 (l^4 + 6l^2 a + 3a^2)}{4m^2}
\end{aligned}$$

$$\sigma_H^2 = \frac{\hbar^4}{4m^2} [(l^4 + 6l^2a + 3a^2) - (l^4 + 2l^2a + a^2)] = \frac{\hbar^4 a}{2m^2} (2l^2 + a)$$

$$\Delta E = \sigma_H = \frac{\hbar^2}{m} \sqrt{a \left(l^2 + \frac{a}{2} \right)}$$

$$\Delta t = \frac{\sigma_x}{|d\langle x \rangle / dt|} = \frac{m}{\hbar l} \sqrt{\frac{1}{4a} + \frac{\hbar^2 a t^2}{m^2}}$$

$$\Delta E \Delta t = \left[\frac{\hbar^2}{m} \sqrt{a \left(l^2 + \frac{a}{2} \right)} \right] \cdot \left[\frac{m}{\hbar l} \sqrt{\frac{1}{4a} + \frac{\hbar^2 a t^2}{m^2}} \right]$$

$$\Delta E \Delta t = \frac{\hbar}{l} \sqrt{\left(l^2 + \frac{a}{2} \right) \left(\frac{1}{4} + \frac{\hbar^2 a^2 t^2}{m^2} \right)}$$

$$\Delta E \Delta t = \frac{\hbar}{2} \sqrt{1 + \frac{a}{2l^2}} \sqrt{1 + \frac{4\hbar^2 a^2 t^2}{m^2}}$$

Since $a > 0$, $l^2 > 0$, and $t^2 \geq 0$, both radicals are ≥ 1 :

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

Thus, the energy-time uncertainty principle is satisfied for all t .

Problem 3.22

From the Energy-Time uncertainty principle,

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

$$\text{where } \Delta E = \sigma_E, \Delta t = \frac{\sigma_Q}{|D_t \langle Q \rangle|}$$

For $Q = x$, $D_t \langle x \rangle = \frac{1}{m} \langle p \rangle$, then

$$\begin{aligned} \frac{\sigma_E \sigma_x}{(1/m) |\langle p \rangle|} &\geq \frac{\hbar}{2} \\ \therefore \sigma_x \sigma_E &\geq \frac{\hbar}{2m} |\langle p \rangle| \end{aligned}$$

Problem 3.23

If $|\alpha\rangle$ is normalized vector, then the projection operator is given by

$$\hat{P} = |\alpha\rangle \langle \alpha|$$

Consider $\hat{P}^2$

$$\hat{P}^2 = \hat{P} \hat{P} = (|\alpha\rangle \langle \alpha|) (|\alpha\rangle \langle \alpha|) = |\alpha\rangle \underbrace{(\langle \alpha | \alpha \rangle)}_1 \langle \alpha|$$

$$\therefore \hat{P}^2 = |\alpha\rangle \langle \alpha| = \hat{P}$$

Therefore, the projection operator is idempotent.

Problem 3.24

Given that the operator $\hat{Q}$ is hermitian, its matrix elements in an orthonormal basis $|e_n\rangle$ are given by

$$\begin{aligned}\hat{Q} &= 1\hat{Q}1 = \left(\sum_{n=1} |e_n\rangle\langle e_n|\right) \hat{Q} \left(\sum_{m=1} |e_m\rangle\langle e_m|\right) \\ \hat{Q} &= \sum_{n,m=1} |e_n\rangle\langle e_n|\hat{Q}|e_m\rangle\langle e_m| = \sum_{n,m=1} Q_{nm}|e_n\rangle\langle e_m| \\ Q_{nm} &= \langle e_n|\hat{Q}|e_m\rangle = \langle e_n|\hat{Q}e_m\rangle\end{aligned}$$

And Q_{mn} is given by,

$$Q_{mn} = \langle e_m|\hat{Q}|e_n\rangle = \langle e_m|\hat{Q}e_n\rangle$$

Since $\hat{Q}$ is hermitian, we can write Q_{mn} as

$$\begin{aligned}Q_{mn} &= \langle \hat{Q}e_n|e_m\rangle^* = \langle e_n|\hat{Q}e_m\rangle^* = Q_{nm}^* \\ \therefore Q_{mn} &= Q_{nm}^*\end{aligned}$$

Problem 3.25

The Hamiltonian for a certain two level system is given by:

$$\hat{H} = \epsilon(|1\rangle\langle 1| - |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1|)$$

where $|1\rangle$ and $|2\rangle$ are orthonormal and ϵ is constant with the dimensions of energy. The components of the matrix H is given by:

$$\begin{aligned}H_{nm} &= \langle n|\hat{H}|m\rangle \\ H_{11} &= \epsilon\langle 1|(|1\rangle\langle 1| - |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1|)|1\rangle \\ H_{11} &= \epsilon\langle 1|(|1\rangle + |2\rangle) = \epsilon \\ H_{12} &= \epsilon\langle 1|(|1\rangle\langle 1| - |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1|)|2\rangle \\ H_{12} &= \epsilon\langle 1|(|1\rangle - |2\rangle) = \epsilon \\ H_{21} &= \epsilon\langle 2|(|1\rangle\langle 1| - |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1|)|1\rangle \\ H_{21} &= \epsilon\langle 2|(|1\rangle + |2\rangle) = \epsilon \\ H_{22} &= \epsilon\langle 2|(|1\rangle\langle 1| - |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1|)|2\rangle \\ H_{22} &= \epsilon\langle 2|(|1\rangle - |2\rangle) = -\epsilon \\ \therefore H &= \epsilon \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\end{aligned}$$

The eigenvalue equation for the Hamiltonian is:

$$\begin{aligned}\hat{H}|\psi\rangle &= E|\psi\rangle \\ \epsilon \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} |\psi\rangle &= E|\psi\rangle\end{aligned}$$

The eigenvalues E are given by the solution to $\det(\hat{H} - E\hat{I}) = 0$

$$\begin{aligned}\det \begin{pmatrix} \epsilon - E & \epsilon \\ \epsilon & -\epsilon - E \end{pmatrix} &= 0 \implies E^2 - 2\epsilon^2 = 0 \\ \therefore E &= \pm\sqrt{2}\epsilon\end{aligned}$$

The eigenvalues are therefore $\pm\sqrt{2}\epsilon$. Now, finding the eigenvectors:

$$\hat{H}|\psi\rangle = E|\psi\rangle \implies \epsilon(|1\rangle\langle 1| - |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1|)|\psi\rangle = \pm\sqrt{2}\epsilon|\psi\rangle$$

$$\epsilon(|1\rangle\langle 1| - |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1| \mp \sqrt{2})|\psi\rangle = 0$$

$$(|1\rangle + |2\rangle)\langle 1|\psi\rangle + (|1\rangle - |2\rangle)\langle 2|\psi\rangle = \pm\sqrt{2}|\psi\rangle$$

Let $\langle 1|\psi\rangle = \psi_1$ and $\langle 2|\psi\rangle = \psi_2$, then the eigenvectors are:

$$\therefore |\psi\rangle = \mp \frac{1}{\sqrt{2}}[(\psi_1 + \psi_2)|1\rangle + (\psi_1 - \psi_2)|2\rangle]$$

Problem 3.26

A three-dimensional vector space spanned by an orthonormal basis $|1\rangle, |2\rangle, |3\rangle$. The kets $|\alpha\rangle$ and $|\beta\rangle$ are given by:

$$|\alpha\rangle = i|1\rangle - 2|2\rangle - i|3\rangle \quad |\beta\rangle = i|1\rangle + 2|3\rangle$$

(a) The duals of $|\alpha\rangle$ and $|\beta\rangle$ are their hermitian conjugates,

$$\langle\alpha| = |\alpha\rangle^\dagger = (i|1\rangle - 2|2\rangle - i|3\rangle)^\dagger$$

$$\therefore \langle\alpha| = -i\langle 1| - 2\langle 2| + i\langle 3|$$

$$\langle\beta| = |\beta\rangle^\dagger = (i|1\rangle + 2|3\rangle)^\dagger$$

$$\therefore \langle\beta| = -i\langle 1| + 2\langle 3|$$

(b)

$$\langle\alpha|\beta\rangle = (-i\langle 1| - 2\langle 2| + i\langle 3|)(i|1\rangle + 2|3\rangle)$$

$$\langle\alpha|\beta\rangle = 1 + 2i$$

$$\langle\beta|\alpha\rangle = (-i\langle 1| + 2\langle 3|)(i|1\rangle - 2|2\rangle - i|3\rangle)$$

$$\langle\beta|\alpha\rangle = 1 - 2i$$

Notice that indeed $\langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle^*$.

(c) Given that the operator $\hat{A} = |\alpha\rangle\langle\beta|$. The matrix elements of A are given by,

$$A_{nm} = \langle n|\hat{A}|m\rangle = \langle n|\alpha\rangle\langle m|\beta\rangle$$

$$A_{nm} = (i\delta_{n1} - 2\delta_{n2} - i\delta_{n3})(-i\delta_{m1} + 2\delta_{m3})$$

$$\therefore A = \begin{pmatrix} 1 & 0 & 2i \\ -2-i & 0 & -4 \\ -1 & 0 & -2i \end{pmatrix}$$

No it is not hermitian, since $A^\dagger \neq A$.

Problem 3.27

$\hat{Q}$ is an operator with a complete set of discrete eigenvalues q_n on an orthonormal basis $\{|e_n\rangle\}$

$$\hat{Q}|e_n\rangle = q_n|e_n\rangle$$

(a)

$$\begin{aligned}\hat{Q} &= 1\hat{Q} = \left(\sum_{n=1} |e_n\rangle\langle e_n| \right) \hat{Q} \\ \hat{Q} &= \sum_{n=1} \hat{Q}|e_n\rangle\langle e_n| = \sum_{n=1} q_n|e_n\rangle\langle e_n|\end{aligned}$$

Therefore, any operator can be written as its spectral decomposition as follows:

$$\hat{Q} = \sum_{n=1} q_n|e_n\rangle\langle e_n|$$

(b) Given that for a function f and operator $\hat{Q}$ with eigenvalues q_n ,

$$f(\hat{Q}) = \sum_{n=1} f(q_n)|e_n\rangle\langle e_n|$$

In the case of $f(\hat{Q}) = e^{\hat{Q}}$, $f(q_n) = e^{q_n}$

$$\begin{aligned}e^{\hat{Q}} &= \sum_{n=1} e^{q_n}|e_n\rangle\langle e_n| \\ e^{\hat{Q}} &= \sum_{n=1} \sum_{m=0} \frac{q_n^m}{m!}|e_n\rangle\langle e_n| \\ e^{\hat{Q}} &= \sum_{n=1} \sum_{m=0} \frac{1}{m!} \hat{Q}^m|e_n\rangle\langle e_n| = \sum_{m=0} \frac{\hat{Q}^m}{m!} \underbrace{\sum_{n=1} |e_n\rangle\langle e_n|}_1 \\ \therefore e^{\hat{Q}} &= \sum_{m=0} \frac{\hat{Q}^m}{m!} = 1 + \hat{Q} + \frac{\hat{Q}^2}{2!} + \frac{\hat{Q}^3}{3!} + \dots\end{aligned}$$

Problem 3.28

Let $\hat{D} = D_x$, then

(a) $(\sin \hat{D})x^5$

$$\begin{aligned}\sin x &= 1 - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \\ \sin \hat{D} &= 1 - \frac{\hat{D}^3}{3!} + \frac{\hat{D}^5}{5!} + \dots \\ (\sin \hat{D})x^5 &= \left(1 - \frac{D_x^3}{3!} + \frac{D_x^5}{5!} + \dots \right) x^5 = x^5 - 10x^2 + 1\end{aligned}$$

(b) $\left(\frac{1}{1-\hat{D}/2} \right) \cos x$

$$\begin{aligned}\frac{1}{1-x} &= 1 + x + x^2 + x^3 + \dots \\ \frac{1}{1-\hat{D}/2} &= 1 + \frac{\hat{D}}{2} + \frac{\hat{D}^2}{4} + \frac{\hat{D}^3}{8} + \dots \\ \left(\frac{1}{1-\hat{D}/2} \right) \cos x &= \left(1 + \frac{D_x}{2} + \frac{D_x^2}{4} + \frac{D_x^3}{8} + \dots \right) \cos x \\ \left(\frac{1}{1-\hat{D}/2} \right) \cos x &= \cos x - \frac{1}{2} \sin x - \frac{1}{4} \cos x + \frac{1}{8} \sin x + \dots\end{aligned}$$

Problem 3.29

$\hat{A}$ and $\hat{B}$ are non commuting operators and they commute with their commutator $\hat{C} = [\hat{A}, \hat{B}]$.
(a)

$$\begin{aligned} [\hat{A}^n, \hat{B}] &= \hat{A}^n \hat{B} - \hat{B} \hat{A}^n = \hat{A}^{n-1}(\hat{A}\hat{B}) - (\hat{B}\hat{A})\hat{A}^{n-1} \\ [\hat{A}^n, \hat{B}] &= \hat{A}^{n-1}(\hat{C} - \hat{B}\hat{A}) - (\hat{A}\hat{B} - \hat{C})\hat{A}^{n-1} = 2\hat{A}^{n-1}\hat{C} + \hat{A}(\hat{A}^{n-2}\hat{B} - \hat{B}\hat{A}^{n-2})\hat{A} \\ [\hat{A}^n, \hat{B}] &= 2\hat{A}^{n-1}\hat{C} + \hat{A}[\hat{A}^{n-2}, \hat{B}]\hat{A} \end{aligned}$$

We have obtained a recursive formula for $[\hat{A}^n, \hat{B}]$. Substituting the formula for $[\hat{A}^{n-2}, \hat{B}]$ into this,

$$\begin{aligned} [\hat{A}^n, \hat{B}] &= 2\hat{A}^{n-1}\hat{C} + \hat{A}(2\hat{A}^{n-3}\hat{C} + \hat{A}[\hat{A}^{n-4}, \hat{B}]\hat{A})\hat{A} \\ [\hat{A}^n, \hat{B}] &= 4\hat{A}^{n-1}\hat{C} + \hat{A}^2[\hat{A}^{n-4}, \hat{B}]\hat{A}^2 \end{aligned}$$

For an odd n , perform the above steps $\frac{n-1}{2}$ times until we obtain

$$[\hat{A}^n, \hat{B}] = (n-1)\hat{A}^{n-1}\hat{C} + \hat{A}^{\frac{n-1}{2}} \underbrace{[\hat{A}, \hat{B}]}_{\hat{C}} \hat{A}^{\frac{n-1}{2}}$$

$$[\hat{A}^n, \hat{B}] = n\hat{A}^{n-1}\hat{C}$$

For an even n , perform the steps $n/2$ times until we obtain

$$\begin{aligned} [\hat{A}^n, \hat{B}] &= n\hat{A}^{n-1}\hat{C} + \hat{A}^{n/2}[\hat{A}^0, \hat{B}]\hat{A}^{n/2} \\ \therefore [\hat{A}^n, \hat{B}] &= n\hat{A}^{n-1}\hat{C} \end{aligned}$$

This can also be proved using induction.

(b)

$$\begin{aligned} [e^{\lambda\hat{A}}, \hat{B}] &= e^{\lambda\hat{A}}\hat{B} - \hat{B}e^{\lambda\hat{A}} = \left(\sum_{n=0}^{\infty} \frac{(\lambda\hat{A})^n}{n!}\right)\hat{B} - \hat{B}\left(\sum_{n=0}^{\infty} \frac{(\lambda\hat{A})^n}{n!}\right) \\ [e^{\lambda\hat{A}}, \hat{B}] &= \lambda(\hat{A}\hat{B} - \hat{B}\hat{A}) + \frac{\lambda^2}{2!}(\hat{A}^2\hat{B} - \hat{B}\hat{A}^2) + \frac{\lambda^3}{3!}(\hat{A}^3\hat{B} - \hat{B}\hat{A}^3) + \dots \\ [e^{\lambda\hat{A}}, \hat{B}] &= \sum_{n=1}^{\infty} \frac{\lambda^n}{n!}[\hat{A}^n, \hat{B}] = \sum_{n=1}^{\infty} \frac{\lambda^n}{n!}n\hat{A}^{n-1}\hat{C} \quad (\because \text{From (a)}) \\ [e^{\lambda\hat{A}}, \hat{B}] &= \lambda \left(\sum_{n=0}^{\infty} \frac{(\lambda\hat{A})^n}{n!}\right)\hat{C} \\ \therefore [e^{\lambda\hat{A}}, \hat{C}] &= \lambda e^{\lambda\hat{A}}\hat{C} \end{aligned}$$

(c) Let $\hat{f}(\lambda) = e^{\lambda(\hat{A}+\hat{B})}$ and $\hat{g}(\lambda) = e^{\lambda\hat{A}}e^{\lambda\hat{B}}e^{-\lambda^2\hat{C}/2}$. Notice that $\hat{f}(0) = \hat{g}(0)$. Taking their derivatives,

$$\begin{aligned} D_\lambda \hat{f} &= (\hat{A} + \hat{B})e^{\lambda(\hat{A}+\hat{B})} = (\hat{A} + \hat{B})\hat{f} \\ D_\lambda \hat{g} &= \hat{A}e^{\lambda\hat{A}}e^{\lambda\hat{B}}e^{-\lambda^2\hat{C}/2} + \hat{B}e^{\lambda\hat{A}}e^{\lambda\hat{B}}e^{-\lambda^2\hat{C}/2} - \lambda\hat{C}e^{\lambda\hat{A}}e^{\lambda\hat{B}}e^{-\lambda^2\hat{C}/2} \\ D_\lambda \hat{g} &= \hat{A}\hat{g} + \hat{B}\hat{g} - (\lambda e^{\lambda\hat{A}}\hat{C})e^{\lambda\hat{B}}e^{-\lambda^2\hat{C}/2} = \hat{A}\hat{g} + \hat{B}\hat{g} - [e^{\lambda\hat{A}}, \hat{C}]e^{\lambda\hat{B}}e^{-\lambda^2\hat{C}/2} \\ D_\lambda \hat{g} &= \hat{A}\hat{g} + \hat{B}\hat{g} - \hat{C}\hat{g} + \hat{C}\hat{g} = (\hat{A} + \hat{B})\hat{g} \end{aligned}$$

See that both $\hat{f}$ and $\hat{g}$ satisfy the same differential equation. Therefore, since they are equal at $\lambda = 0$, they must be equal for all λ . For $\lambda = 1$,

$$\therefore e^{\hat{A}+\hat{B}} = e^{\hat{A}}e^{\hat{B}}e^{-\hat{C}/2}$$

Problem 3.30

Let the position space wave function be $\Psi(x, t)$ and the "energy-space" wave function be $c_n(t)$, then

$$c_n(t) = \langle n | \mathcal{S}(t) \rangle = \langle n | 1 | \mathcal{S}(t) \rangle = \langle n | \left(\int_{-\infty}^{+\infty} |x\rangle \langle x| dx \right) | \mathcal{S}(t) \rangle$$

$$c_n(t) = \int_{-\infty}^{+\infty} \langle n | x \rangle \langle x | \mathcal{S}(t) \rangle dx = \int_{-\infty}^{+\infty} \langle x | n \rangle^* \Psi(x, t) dx$$

$\langle x | n \rangle$ is the energy eigenfunction in position space,

$$\hat{H} |n\rangle = E_n |n\rangle \iff \langle x | \hat{H} |n\rangle = E_n \langle x | n \rangle$$

$$\hat{H} \langle x | n \rangle = E_n \langle x | n \rangle \implies \hat{H} \psi_n(x) = E_n \psi_n(x)$$

$$\therefore c_n(t) = \int_{-\infty}^{+\infty} \psi_n^*(x) \Psi(x, t) dx$$

Problem 3.31

Let $|1\rangle = 1, |2\rangle = x, |3\rangle = x^2, |4\rangle = x^3$ and let $|1'\rangle, |2'\rangle, |3'\rangle, |4'\rangle$ be the orthonormalized basis. Defining the inner product on $(-1, 1)$ as follows,

$$\langle f | g \rangle = \int_{-1}^{+1} f g dx$$

By Gram-Schmidt orthonormalization,

$$|1'\rangle = \frac{|1\rangle}{||1\rangle|} = \frac{1}{\sqrt{\int_{-1}^{+1} 1 dx}}$$

$$\therefore |1'\rangle = \frac{1}{\sqrt{2}}$$

$$|2'\rangle = \frac{|2\rangle - \langle 1' | 2 \rangle |1'\rangle}{||2\rangle - \langle 1' | 2 \rangle |1'\rangle|}$$

$$\langle 1' | 2 \rangle = \int_{-1}^{+1} \frac{x}{\sqrt{2}} dx = \frac{x^2}{2\sqrt{2}} \Big|_{-1}^{+1} = 0$$

$$\therefore |2'\rangle = \sqrt{\frac{3}{2}} x$$

$$|3'\rangle = \frac{|3\rangle - \langle 2' | 3 \rangle |2'\rangle - \langle 1' | 3 \rangle |1'\rangle}{||3\rangle - \langle 2' | 3 \rangle |2'\rangle - \langle 1' | 3 \rangle |1'\rangle|}$$

$$\langle 2' | 3 \rangle = \int_{-1}^{+1} \sqrt{\frac{3}{2}} x^3 dx = 0$$

$$\langle 1' | 3 \rangle = \int_{-1}^{+1} \frac{x^2}{\sqrt{2}} dx = \frac{x^3}{3\sqrt{2}} \Big|_{-1}^{+1} = \frac{\sqrt{2}}{3}$$

$$\therefore |3'\rangle = \sqrt{\frac{45}{8}} \left(x^3 - \frac{1}{3} \right)$$

$$|4'\rangle = \frac{|4\rangle - \langle 3' | 4 \rangle |3'\rangle - \langle 2' | 4 \rangle |2'\rangle - \langle 1' | 4 \rangle |1'\rangle}{||4\rangle - \langle 3' | 4 \rangle |3'\rangle - \langle 2' | 4 \rangle |2'\rangle - \langle 1' | 4 \rangle |1'\rangle|}$$

$$\begin{aligned}
\langle 3'|4\rangle &= \sqrt{\frac{45}{8}} \int_{-1}^{+1} x^5 - \frac{1}{3}x^3 dx = 0 \\
\langle 2'|4\rangle &= \sqrt{\frac{3}{2}} \int_{-1}^{+1} x^4 dx = \sqrt{\frac{3}{2}} \frac{x^5}{5} \Big|_{-1}^{+1} = \frac{\sqrt{6}}{5} \\
\langle 1'|4\rangle &= \int_{-1}^{+1} \frac{x^3}{\sqrt{2}} dx = 0 \\
\therefore |4'\rangle &= \sqrt{\frac{175}{8}} \left(x^3 - \frac{3}{5}x \right)
\end{aligned}$$

Problem 3.32

A anti/skew-hermitian operator is an operator that is equal to the negative of its hermitian, i.e.

$$\hat{Q} = -\hat{Q}^\dagger$$

(a) The expectation value of the skew-hermitian operator is,

$$\begin{aligned}
\langle Q \rangle &= \langle \psi | \hat{Q} \psi \rangle = \langle -\hat{Q} \psi | \psi \rangle \implies \langle \psi | \hat{Q} \psi \rangle = -\langle \psi | \hat{Q} \psi \rangle^* \\
\text{Re}\langle Q \rangle &= -\text{Re}\langle Q \rangle \implies \text{Re}\langle Q \rangle = 0
\end{aligned}$$

Therefore, since it has no real part, the expectation value $\langle Q \rangle$ must be purely imaginary.

(b) The eigenvalue equation for a skew-hermitian operator is:

$$\hat{Q}|Q\rangle = q|Q\rangle$$

Multiply both sides with the bra $\langle Q|$,

$$\langle Q | \hat{Q} | Q \rangle = q \langle Q | Q \rangle \implies \langle Q | \hat{Q} Q \rangle = q \langle Q | Q \rangle$$

Exploiting the skew-hermiticity of $\hat{Q}$,

$$\begin{aligned}
-\langle \hat{Q} Q | Q \rangle &= q \langle Q | Q \rangle \implies -\langle Q | \hat{Q} Q \rangle^* = q \langle Q | Q \rangle \\
-q^* \langle Q | Q \rangle &= q \langle Q | Q \rangle \implies q = -q^*
\end{aligned}$$

Therefore, the eigenvalues q of a skew-hermitian operator $\hat{Q}$ are imaginary.

(c) Consider two distinct eigenvalues q, q' with eigenkets $|Q\rangle, |Q'\rangle$

$$\hat{Q}|Q\rangle = q|Q\rangle \quad \hat{Q}|Q'\rangle = q'|Q'\rangle$$

Multiplying the first with $\langle Q'|$ and the second with $\langle Q|$,

$$\langle Q' | \hat{Q} Q \rangle = q \langle Q' | Q \rangle \quad \langle Q | \hat{Q} Q' \rangle = q' \langle Q | Q' \rangle$$

By the skew-hermiticity of $\hat{Q}$ on the first equation,

$$\begin{aligned}
-\langle \hat{Q} Q' | Q \rangle &= q \langle Q' | Q \rangle \implies -\langle Q | \hat{Q} Q' \rangle^* = q \langle Q' | Q \rangle \\
(q')^* \langle Q' | Q \rangle &= q \langle Q' | Q \rangle
\end{aligned}$$

Since q and q' are distinct eigenvalues, the above equation is valid if and only if $\langle Q' | Q \rangle = 0$. Therefore, the eigenvectors of distinct eigenvalues of a skew-hermitian operator $\hat{Q}$ are orthogonal.

(d) Consider two hermitian operators $\hat{P}$ and $\hat{R}$. Their commutator is

$$[\hat{P}, \hat{R}] = \hat{P}\hat{R} - \hat{R}\hat{P}$$

Consider the hermitian adjoint of the commutator,

$$[\hat{P}, \hat{R}]^\dagger = (\hat{P}\hat{R} - \hat{R}\hat{P})^\dagger = \hat{R}^\dagger \hat{P}^\dagger - \hat{P}^\dagger \hat{R}^\dagger$$

But $\hat{P}$ and $\hat{R}$ are hermitian, so

$$[\hat{P}, \hat{R}]^\dagger = \hat{R}\hat{P} - \hat{P}\hat{R} = -[\hat{P}, \hat{R}]$$

Therefore, the commutator of two hermitian operators is skew-hermitian.

Now consider two skew-hermitian operators $\hat{A}, \hat{B}$ with commutator $[\hat{A}, \hat{B}]$. The hermitian adjoint of the commutator is

$$[\hat{A}, \hat{B}]^\dagger = (\hat{A}\hat{B} - \hat{B}\hat{A})^\dagger = \hat{B}^\dagger \hat{A}^\dagger - \hat{A}^\dagger \hat{B}^\dagger$$

Since $\hat{A}, \hat{B}$ are skew-hermitian,

$$[\hat{A}, \hat{B}] = (-\hat{B})(-\hat{A}) - (-\hat{A})(-\hat{B}) = \hat{B}\hat{A} - \hat{A}\hat{B} = -[\hat{A}, \hat{B}]$$

Therefore, the commutator of two skew-hermitian operators is skew-hermitian.

(e) Let $\hat{Z}$ be an operator with complex eigenvalues. The eigenvalues equation of $\hat{Z}$ is

$$\hat{Z}|f\rangle = z|f\rangle$$

Since the eigenvalue is complex, we write it as $z = x + iy$, where x, y are purely real.

$$\hat{Z}|f\rangle = x|f\rangle + iy|f\rangle$$

Let x and iy be eigenvalues of a hermitian and skew-hermitian operator $\hat{X}$ and $\hat{Y}$ with the same eigenvectors, then

$$\begin{aligned}\hat{X}|f\rangle &= x|f\rangle & \hat{Y}|f\rangle &= iy|f\rangle \\ \hat{Z}|f\rangle &= \hat{X}|f\rangle + \hat{Y}|f\rangle = (\hat{X} + \hat{Y})|f\rangle \\ \therefore \hat{Z} &= \hat{X} + \hat{Y}\end{aligned}$$

Therefore, any operator can be written as a sum of hermitian and skew-hermitian operators.

Consider $\hat{Z}^\dagger$,

$$\hat{Z}^\dagger = (\hat{X} + \hat{Y})^\dagger = \hat{X}^\dagger + \hat{Y}^\dagger = \hat{X} - \hat{Y}$$

Then, $\hat{X}$ and $\hat{Y}$ in terms of $\hat{Z}$ and $\hat{Z}^\dagger$ is

$$\hat{X} = \frac{1}{2}(\hat{Z} + \hat{Z}^\dagger) \quad \hat{Y} = \frac{1}{2}(\hat{Z} - \hat{Z}^\dagger)$$

Problem 3.33

Given that an observable A has normalized eigenstates ψ_1, ψ_2 corresponding to eigenvalues a_1, a_2 and observable B has normalized eigenstates ϕ_1, ϕ_2 corresponding to eigenvalues b_1, b_2 . Then eigenstates of A and B are related by:

$$\begin{aligned}\psi_1 &= \frac{3\phi_1 + 4\phi_2}{5} & \psi_2 &= \frac{4\phi_1 - 3\phi_2}{5} \\ \implies \phi_1 &= \frac{3\psi_1 + 4\psi_2}{5} & \phi_2 &= \frac{4\psi_1 - 3\psi_2}{5}\end{aligned}$$

Let us describe the state vector as following,

$$|S\rangle = \alpha_1|\psi_1\rangle + \alpha_2|\psi_2\rangle = \beta_1|\phi_1\rangle + \beta_2|\phi_2\rangle$$

$$\text{where } \alpha_1^2 + \alpha_2^2 = \beta_1^2 + \beta_2^2 = 1$$

From the given relations, we can derive that

$$\begin{aligned}\beta_1 &= \frac{3\alpha_1 + 4\alpha_2}{5} & \beta_2 &= \frac{4\alpha_1 - 3\alpha_2}{5} \\ \alpha_1 &= \frac{3\beta_1 + 4\beta_2}{5} & \alpha_2 &= \frac{4\beta_1 - 3\beta_2}{5} \\ |\mathcal{S}\rangle &= \left(\frac{3\beta_1 + 4\beta_2}{5}\right) |\psi_1\rangle + \left(\frac{4\beta_1 - 3\beta_2}{5}\right) |\psi_2\rangle = \left(\frac{3\alpha_1 + 4\alpha_2}{5}\right) |\phi_1\rangle + \left(\frac{4\alpha_1 - 3\alpha_2}{5}\right) |\phi_2\rangle\end{aligned}$$

(a) If A is measured as a_1 , then the state collapses to $|\psi_1\rangle$, then $\alpha_1 = 1$ and $\alpha_2 = 0$. The state of the system immediatly after this measurement will still be

$$|\mathcal{S}\rangle = |\psi_1\rangle = \frac{3}{5}|\phi_1\rangle + \frac{4}{5}|\phi_2\rangle$$

(b) If B is measured immediatly after, the state will collapse to $|\phi_1\rangle$ measuring b_1 with a probabily of 36% or it will collapse to $|\phi_2\rangle$ measuring b_2 with a probability of 64%.

(c) If b_1 is measured, $\beta_1 = 1$ and $\beta_2 = 0$, the state will be

$$|\mathcal{S}\rangle = |\phi_1\rangle = \frac{3}{5}|\psi_1\rangle + \frac{4}{5}|\psi_2\rangle$$

If A is measured immediatly after, the state will collapse to $|\psi_1\rangle$ measuring a_1 with a probabily of 36%.

If b_2 is measured, $\beta_1 = 0$ and $\beta_2 = 1$, the state will be

$$|\mathcal{S}\rangle = |\phi_2\rangle = \frac{4}{5}|\psi_1\rangle - \frac{3}{5}|\psi_2\rangle$$

If A is measured immediatly after, the state will collapse to $|\psi_1\rangle$ measuring a_1 with a probabily of 64%.